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# Quantum-Network-ready Source of Heralded Atom-Photon Entanglement

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# Abstract

Photon-pair sources play a central role in quantum optics and quantum information protocols, particularly for applications in quantum communication, quantum metrology, and foundational tests of quantum mechanics. To be useful in these contexts, sources must combine high efficiency and compatibility with optical fiber networks. This thesis presents the realization and detailed characterization of a high-performance photon-pair source based on a single $^{87}$Rb atom coupled to two crossed fiber Fabry-Pérot cavities. The atom, configured in a three-level ladder system, emits photon pairs sequentially into the two optical cavities via a cascaded decay process. The source achieves an in-fiber photon-pair emission efficiency of 22(1)%, with high single-photon purity and temporal correlation properties that are well understood through theoretical modeling.

Beyond its value as a photon-pair source, this system enables the heralded generation of atom-photon entanglement. The polarization of the upper photon is entangled with the spin of the atom, while the lower photon serves as a herald. This is achieved by employing a birefringent heralding cavity that enhances only specific atomic decay paths. Detecting the herald photon at the source enables tracking inefficiency and infidelity errors, improving the efficiency of generating atom-photon entanglement from 43(3)% to 68(3)% and the fidelity from 62(2)% to 86(2)%. Moreover, the detection of a herald photon enables the time-gated detection at the receiver with a reduced temporal window, improving the signal-to-noise ratio and making the entangled state more robust to photon loss and detector noise. In detector-noise-limited conditions, we measure an increased fidelity of the entangled state from 48(3)% to 75(3)%.

These results demonstrate the dual capabilities of the system: as a high-efficiency photon-pair source and as a heralded atom-photon entanglement interface. This makes it a promising building block for scalable and noise-resilient quantum networks.

Alla mia famiglia.
Il vostro supporto è le fondamenta del mio presente e futuro.

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# Contents

# 1. Introduction

The concept of correlations between distant events has fascinated scientists and philosophers for centuries. In classical physics, correlations were often attributed to common causes, following the principles laid out by thinkers such as Aristotle and later formalized in probability theory by Pierre-Simon Laplace [1]. The study of correlations played a central role in early scientific disciplines such as astronomy, where observed regularities in planetary positions and motions enabled predictive models of celestial mechanics [2, 3]. For example, the correlation between the orbital period of a planet and its distance from the Sun, later formalized in Kepler's laws, provided a powerful tool to anticipate planetary motion long before the underlying gravitational law was known. Similarly, in thermodynamics, statistical dependencies between variables like pressure, volume, and temperature guided the formulation of empirical laws [4] and, later, statistical mechanics [5, 6].

As physics progressed into the 20th century, a profoundly different form of correlation emerged, one that defied classical intuition. In classical physics, correlations arise from shared histories or common causes: two variables are statistically dependent because they are influenced by the same underlying factors. Once established, such correlations can persist across arbitrary distances without requiring further interaction. Quantum entanglement, however — as formalized by Albert Einstein, Boris Podolsky, and Nathan Rosen in their 1935 paper on what would later be known as the Einstein-Podolsky-Rosen (EPR) paradox [7], and shortly after by Erwin Schroedinger [8, 9] — goes beyond this classical picture. Entangled particles display correlations that cannot be accounted for by any prior shared information. Instead, measurements on one particle are linked to outcomes on the other in ways that exceed the limits of classical statistics, even when the particles are separated by vast distances. This phenomenon, which Einstein famously referred to as "spooky action at a distance" [10], challenged the very foundation of classical realism and locality.

The concept of entanglement has remained central to debates in quantum mechanics. A major breakthrough came in the 1960s, when John Bell formulated Bell's inequalities [11]: a mathematical criterion that distinguishes classical hidden variable theories from quantum mechanics. His work showed that the correlations predicted by quantum mechanics could not be explained by any local classical theory. Subsequent experiments by John Clauser [12], Alain Aspect [13, 14], and others [15] confirmed violations of Bell's inequalities, providing strong evidence for quantum nonlocality. More recently, loophole-free Bell tests have closed remaining experimental gaps, firmly establishing that nature behaves according to quantum, not classical, principles [16, 17]. These findings not only validate the nonlocal character of quantum mechanics but also open the door to technologies that use entanglement as a key resource.

Indeed, entanglement has emerged as a cornerstone of modern quantum technologies, playing a fundamental role in the development of quantum information science [18]. Entanglement is central to both quantum communication and quantum computing. Unlike classical bits, which take values of 0 or 1, quantum computers use qubits that can exist in superpositions of both states [19]. The power of quantum computing arises from superposition and entanglement: while superposition enables parallel computation, entanglement creates strong correlations between qubits, enhancing computational efficiency [20, 21, 22]. Algorithms like Shor's for factorization and Grover's for search leverage entanglement to achieve speedups unattainable by classical systems [23, 24]. In the context of quantum communication, flying qubits are used for quantum key distribution (QKD) [25], a method that enables two distant parties to generate a shared, secret cryptographic key with security guaranteed by the laws of quantum mechanics [26, 27]. Unlike classical encryption schemes, which rely on computational assumptions [28], QKD is fundamentally secure against any eavesdropping attempt, even by adversaries with unlimited computational power.

Quantum networks, differently, use entanglement to link quantum processors. Unlike classical communication networks, where information is encoded in independent bits that can be copied and transmitted at will, quantum networks rely on the coherent distribution of entangled qubits [29, 30]. In principle, they can distribute entanglement across large distances, potentially extending the range and robustness of secure communication significantly [31]. A central element in realizing long-distance scalable networks is the quantum repeater, which allows for entanglement distribution over long distances by mitigating photon loss through entanglement swapping and purification techniques [32, 33, 34, 35]. Beyond secure communication, quantum networks promise advances in distributed quantum computing [36, 37, 38, 39], precision-enhanced distributed sensing [40, 41], and foundational tests of quantum nonlocality over macroscopic scales [42]. Initial demonstrations of small-scale quantum networks consisting of two or three nodes [43, 44, 45, 46, 47, 48, 49, 50], as well as basic quantum gates operating between separate modules [51, 52], have already been achieved experimentally. However, despite significant theoretical and experimental progress in quantum networks in the past years, the practical realization of large-scale quantum information protocols still faces major experimental challenges.

Among the building blocks required for scalable quantum networks is the ability to generate high-quality photons in a controlled and efficient manner. In particular, sources of correlated photon pairs have proven indispensable across quantum science and technology. Historically, they have been used for foundational experiments such as early tests of Bell's inequalities [53, 13], and they now enable applied technologies including quantum cryptography [54], photonic quantum computing [55], and quantum metrology [56]. A key feature of such sources is their ability to herald the presence of a single photon through detection of its partner [57]. To meet the demands of quantum networks, these photons must not only be emitted with high efficiency and purity, but also be highly indistinguishable and be compatible with fiber-based optical infrastructure. This has motivated the development of photon-pair sources based on different physical systems, including atomic ensembles [58, 59], nonlinear crystals [60, 61], quantum dots [62, 63], and single molecules [64]. Yet, each platform presents trade-offs between emission efficiency, photon-number purity, mode matching, and indistinguishability. The realization of a source that simultaneously fulfills all these requirements remains a major technical challenge.

This thesis presents the implementation of a novel photon-pair source based on cascaded emission from a single atom coupled to two optical fiber cavities [65], achieving an in-fiber photon pair emission efficiency of 22(1)% with high single photon purity. This architecture enables independent control over both photons and provides a promising route toward efficient, pure, and indistinguishable photon pairs — key ingredients for future quantum network applications.

Another fundamental building block for scalable quantum network architectures is the ability to generate high-fidelity entanglement between a matter qubit and a photonic qubit[29]. Matter qubits, such as trapped atoms or ions [66, 67], serve as long-lived quantum memories [68], allowing for the storage and manipulation of quantum information. Photonic qubits, on the other hand, function as fast carriers of quantum information, enabling the transmission of entanglement between distant nodes [69]. Quantum systems are inherently sensitive to environmental disturbances, and any uncontrolled interaction with the surroundings can introduce decoherence or loss, degrading the fidelity of the desired quantum state. In isolated systems, such as those used in quantum computing where stationary qubits can be well-shielded from external noise, these errors can be kept relatively low [70]. In contrast, quantum communication relies on photonic qubits propagating through open systems and optical channels. These systems are assembled from many components, such as sources, cavities, fibers, and detectors, each of which introduces potential sources of imperfection [71].

Because these errors are difficult to eliminate completely, a powerful strategy is to identify and pre-certify the successful quantum events—a process known as heralding. Heralding enables one to discard failed or ambiguous attempts and retain only those in which the quantum state was generated and transmitted with high fidelity. For example, in quantum communication protocols, a herald signal can indicate the successful generation [72] or arrival [73, 74] of a photon, thereby improving the reliability of long-distance entanglement distribution [75]. While there is a broad body of research on heralded quantum states of light [76, 77, 78, 79, 80, 81, 82] and heralded quantum memories [83, 84], relatively little attention has been devoted to the heralded generation of atom-photon entanglement [85].

In conventional architectures, the herald signal, typically generated at the photon-receiving node, confirms the success of an entanglement distribution attempt. However, heralding at the photon-sending node, where the atom-photon entangled state is created, offers several distinct advantages. First, it enables real-time feedback, allowing the sender to abort and reset upon failure without waiting for round-trip communication delays. Second, it avoids attenuation of the heralding signal over long optical links, ensuring higher detection efficiencies and a better signal-to-noise ratio. Third, and perhaps most crucially, the heralding event precisely timestamps the creation of the entangled state. This timing information can be shared with the receiver, allowing precise temporal gating of detectors and suppression of background noise — an essential feature for high-fidelity quantum state transmission over long distances.

Building on the cascaded emission process developed for efficient photon-pair generation, this thesis presents the development and characterization of a novel heralded atom-photon entanglement source, where the detection of one photon acts as a herald for the successful creation of an entangled atom-photon pair [86]. These results mark a step forward in the realization of scalable quantum networks, demonstrating the potential of fiber-coupled cavity QED systems as robust quantum network nodes.

This thesis is organized as follows: Chapter 2 introduces the theoretical framework, covering cavity quantum electrodynamics, the dynamics of a driven three-level atom in a cavity, open quantum systems, and Hong-Ou-Mandel visibility as a measure of photon indistinguishability. Chapter 3 describes the experimental setup, including high-finesse fiber cavities, the magneto-optical trap, single-atom trapping, microwave control, and state detection. Chapter 4 presents the generation of photon pairs from single atoms, addressing both the theory and experimental challenges such as atomic photoionization, as well as photon indistinguishability and strong-coupling effects. Chapter 5 describes the heralded generation of atom-photon entanglement, outlining the concept, implementation, and results. Chapter 6 concludes the thesis with a summary and outlook on future developments toward scalable quantum networks.

# 2. Theoretical Backgroud

## 2.1. Basics about cavity quantum electrodynamics

The interaction between an atom and light within a cavity is described by cavity quantum electrodynamics (cQED). A foundational theoretical model in this field is the Jaynes-Cummings model, which describes a two-level atom interacting with a single quantized mode of an optical cavity. While it is an approximation, it provides the theoretical groundwork for modeling more complex atom-cavity systems, including those explored in this work. Therefore, it serves as an optimal starting point. Setting $\hbar = 1$, the Jaynes-Cummings Hamiltonian describing this system is

$$\hat{H} = \omega_a \hat{\sigma}^\dagger \hat{\sigma} + \omega_c \hat{a}^\dagger \hat{a} + g(\hat{a}\hat{\sigma}^\dagger + \hat{a}^\dagger \hat{\sigma}) \tag{2.1}$$

where $\omega_a$ and $\omega_c$ are the angular frequencies associated with, respectively, the atom and the cavity field energy. Here, $\hat{a}$ and $\hat{a}^\dagger$ are the annihilation and creation operators of the cavity mode, whereas $\hat{\sigma}$ and $\hat{\sigma}^\dagger$ are the operators which describe the atomic transition from the excited state to the ground state and vice-versa, respectively. The first two terms on the right side of the equation are the atom's and the cavity field's energy, whereas the coherent exchange of a system excitation between the two of them is expressed by the coupling rate $g$. This arises from the coupling between the electric dipole moment $\boldsymbol{d}$ and the cavity electric field $\boldsymbol{E}$ and is indeed defined as [66]

$$g = \frac{\boldsymbol{d} \cdot \boldsymbol{E}}{\hbar} \tag{2.2}$$

Considering the normalized cavity mode function $u(\boldsymbol{r})$, this expression can be rewritten as [87, 66]

$$g(\boldsymbol{r}) = \sqrt{\frac{\omega_c \left|\boldsymbol{d}\right|^2}{2\hbar\epsilon_0 V}} u(\boldsymbol{r}) = g_0 \cdot u(\boldsymbol{r}) \tag{2.3}$$

where $V$ is the cavity mode volume and $g_0$ is the maximum coupling rate. Equation 2.1 describes only the coherent dynamics of the system. However, for an accurate description, dissipative channels that take into account the system's coupling to the environment must be considered.

The atomic spontaneous decay in free space is a significant dissipative channel. The relevant parameter of this process is the population decay rate, defined as [88, 89]

$$2\gamma = \frac{\omega_a^2 \left|\boldsymbol{d}\right|^2}{3\pi\epsilon_0\hbar c^2} \tag{2.4}$$

This is equivalent to the atomic natural linewidth. It depends on the atomic transition since it is determined by $\omega_a$ and $|\boldsymbol{d}|^2$.

Analogously, one can define the cavity decay rate $\kappa$ as the rate at which the cavity electric field's amplitude decays out of the resonator. The photon-loss rate is given by $2\kappa$ and it is equivalent to the cavity linewidth.

To observe the coherent dynamics described by the Jaynes-Cummings model, the rate of the coherent exchange of energy between the atom and the cavity must be bigger than the dissipation rates, namely $g \gg \kappa, \gamma$. This condition is referred to as the strong coupling regime. In this regime fascinating cQED effects can be observed, but it is often hard to realize experimentally, especially for the kind of apparatus used in this work. The parameters adopted for this experimental work lie in a regime of intermediate coupling, where $g \approx \kappa \gg \gamma$. This choice is often advantageous because it allows having a coherent exchange of energy between the atom and the cavity and, at the same time, a fast photonic out-coupling. In many applications within cQED, it is useful to define a parameter that quantifies the system's ability to coherently exchange energy compared to dissipation. This parameter is known as the cooperativity $C$. In this work, we define it as [90]

$$C = \frac{g^2}{2\kappa\gamma} \tag{2.5}$$

It is important to note that various alternative definitions of cooperativity exist in the literature. The fact that the coupling rate is inversely proportional to the square root of the cavity mode volume has motivated the manufacturing of resonators with small mode volumes in order to obtain higher cooperativity. In this work $C \approx 1$.

The presence of the cavity around an atom enhances the atomic decay rate into the cavity mode. This is known as Purcell effect and the polarization decay rate $\gamma_P$ is related to the cooperativity as [91, 92]

$$\gamma_P = 2C\gamma \tag{2.6}$$

The resulting total decay rate is therefore

$$\gamma' = \gamma_P + \gamma = (2C+1)\gamma \tag{2.7}$$

Another important quantity of a cQED system is the probability with which an atom emits a single photon into the cavity. This is related to the cooperativity as [66]

$$\eta_c = \frac{\gamma_P}{\gamma + \gamma_P} = \frac{2C}{2C+1} \tag{2.8}$$

To derive an expression for the photon emission efficiency into an optical fiber, two additional loss factors must be considered: the outcoupling efficiency and the cavity-fiber mode matching. The first arises from the intentionally unbalanced reflectivities of the cavity mirrors, ensuring that photons are more likely to be outcoupled in the desired direction. The second results from the imperfect spatial mode matching between the cavity and a single-mode fiber. The fraction of photons collected by the cavity and successfully coupled into a single-mode fiber is given by [66]

$$\eta_f = \frac{2C}{2C+1} \cdot \frac{\kappa_{out}}{\kappa} \cdot \mu_{cf} \tag{2.9}$$

where $\kappa_{out}$ is the decay rate associated with the outcoupling mirror and $\mu_{cf}$ is the cavity-fiber mode matching.

## 2.2. Driven three-level atom in a cavity

The Jaynes-Cummings model describes the interaction between the cavity electric field and a two-level atom. This is an approximation since, in reality, atoms have multiple levels and, therefore, multiple transitions that can interact with the electric field of the cavity or of an external laser. The goal of this section is to provide a formal treatment of the scenario in which a three-level atom in a resonator is externally driven by a laser. This constitutes, indeed, the foundation for the theory describing the more complicated scenarios of this work.

For a three-level atom, we define the states as $|g\rangle$, $|e\rangle$ and $|u\rangle$ and introduce the operators $\sigma_{ge} = |g\rangle\langle e|$ and $\sigma_{ue} = |u\rangle\langle e|$, associated with the atomic transitions $|e\rangle \to |g\rangle$ and $|e\rangle \to |u\rangle$, respectively. Figure 2.1a and 2.1b illustrate the considered atom-cavity system and the relevant energy levels. In this scenario, the states $|g\rangle$ and $|u\rangle$ remain uncoupled, meaning no direct transition occurs between them. As a result, the system forms a $\Lambda$-type configuration.

The interaction with the external laser is modeled as an oscillating electric field with angular frequency $\omega_L$ and amplitude $\Omega_{ue}$ driving the transition $|u\rangle \leftrightarrow |e\rangle$ [93]

$$\hat{H}_L(t) = \Omega_{ue}(e^{i\omega_L t}\hat{\sigma}_{ue} + e^{-i\omega_L t}\hat{\sigma}_{ue}^\dagger) \tag{2.10}$$

The complete Hamiltonian of this system is given by

$$\hat{H} = \omega_{ge}\hat{\sigma}_{ee} + (\omega_{ge} - \omega_{ue})\hat{\sigma}_{uu} + \omega_c\hat{a}^\dagger\hat{a} + g(\hat{a}^\dagger\hat{\sigma}_{ge} + \hat{a}\hat{\sigma}_{ge}^\dagger) + \Omega_{ue}(e^{i\omega_L t}\hat{\sigma}_{ue} + e^{-i\omega_L t}\hat{\sigma}_{ue}) \tag{2.11}$$

where $\hat{\sigma}_{ee} = |e\rangle\langle e|$, $\hat{\sigma}_{uu} = |u\rangle\langle u|$, and $\omega_{ge}$ and $\omega_{ue}$ are the angular frequencies associated with the energies of the atomic transitions $|g\rangle - |e\rangle$ and $|u\rangle - |e\rangle$, respectively. The zero-point of energy is chosen to be at state $|g\rangle$. The electric field amplitude $\Omega_{ue}$ is called

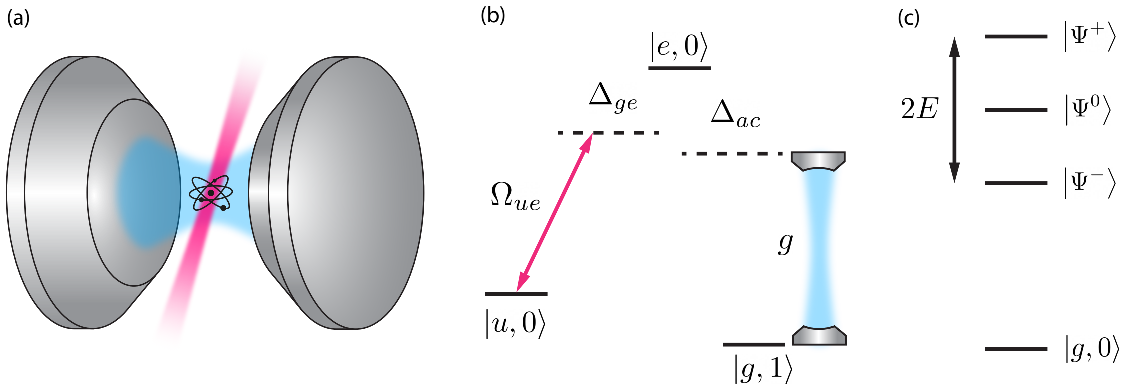


**Figure 2.1.:** (a) An atom within a Fabri-Perot interferometer driven by an external laser beam coming from the side (magenta). The blue area represents the cavity spatial mode. (b) The atomic level scheme, including the cavity (blue), the external drive (magenta), and the detunings. (c) The eigenstates of the coupled system for 0 and 1 system excitations.

Rabi frequency and quantifies the frequency at which the atom state population oscillates between $|u\rangle$ and $|e\rangle$ [94].

When dealing with an external laser driving an atom in a cavity, the natural Hamiltonian is time-dependent because the laser oscillates at a frequency $\omega_L$. However, solving time-dependent Hamiltonians is often tricky. Indeed, the oscillatory terms make both analytical and numerical calculations more complicated. Moving to the interaction picture, applying the rotating wave approximation and transforming this Hamiltonian to a rotating frame gives an effective time-independent Hamiltonian that reads:

$$\hat{H} = (\Delta_{ge} - \Delta_{ac})\hat{\sigma}_{uu} - \Delta_{ac}\hat{\sigma}_{ee} + g(\hat{\sigma}_{ge}\hat{a}^\dagger + \hat{\sigma}_{ge}^\dagger\hat{a}) + \Omega_{ue}(\hat{\sigma}_{ue}^\dagger + \hat{\sigma}_{ue}) \tag{2.12}$$

where $\Delta_{ge} = \omega_L - \omega_{ue}$ is the laser detuning from the atomic resonance $|u\rangle - |e\rangle$ and $\Delta_{ac} = \omega_c - \omega_{ge}$ is the cavity detuning from the atomic resonance $|g\rangle - |e\rangle$. When the cavity and driving detuning are the same, i.e. $\Delta_{ac} = \Delta_{ge} = \Delta$, it is easy to get an analytical solution for the stationary Schrödinger equation. If we choose the states $|g,1\rangle$, $|u,0\rangle$ and $|e,0\rangle$ as a basis for the important scenario of having a single excitation in the system, the eigenenergies of the system are given by [95]:

$$E^0 = 0 \tag{2.13}$$

$$E^\pm = \frac{\Delta}{2} \pm \sqrt{g^2 + \Omega_{ue}^2 + \Delta^2/4} \tag{2.14}$$

This form of coupling results in a triplet of states. Whereas one eigenstate remains at the zero point of energy, the other two experience a shift given by ref Eq. 2.14. The corresponding eigenstates of the system are

$$\left|\Psi^0\right\rangle = \cos\theta\,|u,0\rangle - \sin\theta\,|g,1\rangle \tag{2.15}$$
$$\left|\Psi^+\right\rangle = \sin\phi\sin\theta\,|u,0\rangle + \sin\phi\cos\theta\,|g,1\rangle + \cos\phi\,|e,0\rangle \tag{2.16}$$
$$\left|\Psi^-\right\rangle = \cos\phi\sin\theta\,|u,0\rangle + \cos\phi\cos\theta\,|g,1\rangle - \sin\phi\,|e,0\rangle \tag{2.17}$$

where the mixing angles $\theta$ and $\phi$ are defined by the trigonometric equations

$$\tan\theta = \frac{\Omega_{ue}}{g} \quad \text{and} \quad \tan\phi = \frac{\sqrt{g^2+\Omega_{ue}^2}}{\sqrt{g^2+\Omega_{ue}^2+\Delta^2/4}+\Delta/2} \tag{2.18}$$

Differently from the state $|\Psi^\pm\rangle$, the state $\left|\Psi^0\right\rangle$ is a superposition of solely the states $|u,0\rangle$ and $|g,1\rangle$, without any contribution from the excited state $|e,0\rangle$. Therefore, this is also defined as a dark state since an atom in this state can not be excited and, consequently, can not spontaneously decay in free space.

## 2.3. Open quantum systems

Until now, we have treated the atom-cavity system as an isolated system, ignoring interactions with the surrounding environment. However, as previously discussed, all physical systems inevitably couple to their environment, leading to dissipative processes. When these effects are significant, the system is referred to as an open quantum system. To account for this, we describe the system's state using the density matrix $\rho$, whose evolution is governed by the Master equation [96]:

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H,\rho] + \mathcal{L}[\rho]. \tag{2.19}$$

Here, $\mathcal{L}[\rho]$ represents the Liouvillian superoperator, which captures the dissipative dynamics induced by environmental coupling. It is defined as [96]

$$\mathcal{L}[\rho] = \sum_n \frac{1}{2}\left[2\hat{C}_n\rho(t)\hat{C}_n^\dagger - \rho(t)\hat{C}_n^\dagger\hat{C}_n - \hat{C}_n^\dagger\hat{C}_n\rho(t)\right] \tag{2.20}$$

where the $\hat{C}_n = \sqrt{\Gamma_n}\hat{A}_n$ are the collapse operators, $\hat{A}_n$ are the operators through which the system couples to the environment, and $\Gamma_n$ are the corresponding rates. This formalism allows us to model the decoherence mechanisms such as spontaneous emission and cavity losses. Indeed, the relevant collapse operators for a system like the one described in section 2.2 are

$$\hat{C}_1 = \sqrt{2\kappa}\hat{a} \tag{2.21}$$
$$\hat{C}_2 = \sqrt{2\gamma_{ge}}\,|g\rangle\langle e| \tag{2.22}$$
$$\hat{C}_3 = \sqrt{2\gamma_{ue}}\,|u\rangle\langle e| \tag{2.23}$$

Here, $\hat{C}_1$ represents photon loss from the cavity, whereas $\hat{C}_2$ and $\hat{C}_3$ describe the spontaneous emission from the excited state $|e\rangle$ to $|g\rangle$ and $|u\rangle$, respectively. These transitions occur with polarization decay rates $\gamma_{ge}$ and $\gamma_{ue}$.

This serves as a foundational example, providing the theoretical framework for the analysis and simulations conducted in this work. In particular, equation 2.19 will be useful later in this work to model the photon-pair emission from a single atom coupled to two Fabry-Pérot resonators (Chapter 4). Throughout the theoretical investigation in this thesis, the master equation is solved numerically using QuTip [97].

## 2.4. Principles of photon indistinguishability

Having discussed the interaction between single atoms and optical cavities, as well as the tools to model open quantum systems, we now turn to an essential requirement for photonic quantum information protocols: photon indistinguishability. This concept plays a central role in quantum communication and photonic quantum computing, where protocols often rely on the interference of single photons emitted from independent sources.

A key example is the Bell-state measurement using linear optics [98], which is essential for many quantum networking tasks based on entanglement. These include quantum teleportation of photonic qubits [76], quantum repeaters [99], and the heralded entanglement of remote quantum memories [44]. The success of such measurements fundamentally relies on the indistinguishability of the interfering photons. Two photons are said to be indistinguishable if they are identical in all their quantum properties, such as frequency, polarization, arrival time, temporal shape, and spatial mode. Any residual distinguishability introduces errors that reduce the fidelity of the resulting quantum states.

The Hong-Ou-Mandel (HOM) interference experiment provides a direct way to quantify photon indistinguishability. The setup involves sending two photons into a 50:50 beam splitter from different input ports and measuring coincidences at the two output ports. If the photons are completely indistinguishable, they will always bunch into the same output mode due to quantum interference, leading to a complete suppression of coincidence counts.

With reference to Figure 2.2a, the initial state of two identical photons entering a 50:50 beam splitter is:

$$|1\rangle_A |1\rangle_B \tag{2.24}$$

where A and B label the two input modes. The action of the beam splitter transforms this state into:

$$\frac{1}{\sqrt{2}} \left( |2\rangle_C |0\rangle_D + |0\rangle_C |2\rangle_D \right) \tag{2.25}$$

where C and D are the output modes. This means both photons always exit in the same mode, leading to zero coincidence counts. However, if the photons are partially

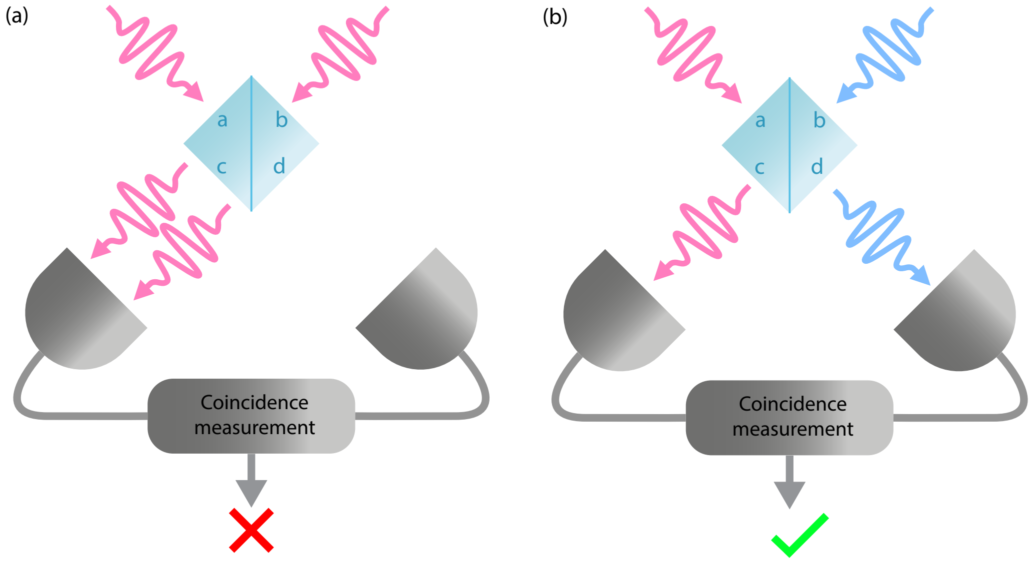


**Figure 2.2.:** Depiction of the HOM interference experiment. (a) shows one of the two possible outcomes when two indistinguishable photons at ports a and b interfere in a 50/50 beamsplitter. In this scenario, both photons exit the beam splitter from port c or d, leading to vanishing detector coincidences. (b) shows the case where the photons impinging on ports a and b are distinguishable (illustrated in two different colors here). The different photon paths don't interfere and there are detector coincidences.

distinguishable, the probability of coincidence detection is nonzero (see Figure 2.2b). The degree of indistinguishability can be quantified using the HOM visibility defined as:

$$V_{HOM} = \frac{C_{max} - C_{min}}{C_{max}} \tag{2.26}$$

where $C_{max}$ is the coincidence count rate when the photons are fully distinguishable (e.g., arriving at different times) and $C_{min}$ is the coincidence count rate at perfect overlap (i.e., complete quantum interference). For perfectly indistinguishable photons, we expect $C_{min} = 0$, leading to $V_{HOM} = 1$. Any deviation from this ideal value indicates partial photon distinguishability. Specifically, in the case of pulsed single-photon sources, differences in the temporal profile of two photons lead to a two-photon distinguishability. To characterize the photon-indistinguishability of the photon source investigated in this work, it is useful to introduce some theoretical tools. It can be shown that the normalized second-order correlation function at zero delay in the two-photon interference scenario $g^{(2)}_{HOM}(0)$ (that indicates the probability of detecting a coincidence count between two single-photon sources, i.e. a cross-correlation) is connected to the first- and second-order correlation functions of the single-photon source ($g^{(1)}(0)$ and $g^{(2)}(0)$, respectively) as the following [100]

$$g^{(2)}_{HOM}(0) = \frac{1}{2}\left(g^{(2)}(0) + 1 - \left|g^{(1)}(0)\right|^2\right) \tag{2.27}$$

As $g^{(2)}(0) = 0$ characterizes an ideal single-photon source, indicating perfect anti-bunching and the absence of multi-photon events, this condition must also hold for each individual photon in a photon-pair source where the two photons are emitted into distinct output modes. In such cases, each output channel should still exhibit a single-photon character to ensure high interference visibility. Under this assumption, the HOM visibility of each channel is directly related to $g^{(1)}(0)$ [101]:

$$V_{HOM} = \left|g^{(1)}(0)\right|^2 = \frac{2\int_0^T \int_0^T dtd\tau \left|\left\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\right\rangle\right|^2}{\left(\int_0^T \left\langle \hat{a}^\dagger(t)\hat{a}(t)\right\rangle dt\right)^2} \tag{2.28}$$

where $\hat{a}$ and $\hat{a}^\dagger$ are the annihilation and creation operators of the studied optical mode and $T$ is the time-window width over which the photon is detected.

# 3. Experimental setup

The system used in this work is a single $^{87}$Rb atom coupled to two crossed fiber Fabry-Perot resonators in an ultra-high vacuum (UHV) chamber. After being trapped and positioned in the cavities mode center, the atom can be manipulated in its internal electronic degrees of freedom by means of several lasers and microwave radiation, targeting different atomic transitions. The experimental setup, originally constructed in 2019, is described in detail in [102], with complementary components covered in [103]. This chapter presents the cavities system used in the previous works and outlines the upgrades and innovations implemented during the course of this PhD work. It covers the atom loading process using light-induced atom desorption (LIAD) for the magneto-optical trap (MOT), the trapping and precise positioning of atoms at the center of the cavities with a 852 nm standing wave in combination with a galvo-scanner, the internal state manipulation, and the atomic state readout technique.

## 3.1. The cavities system

The key element of the experimental setup is a system of two orthogonally arranged optical fiber cavities, illustrated in Fig. 3.1. These cavities are distinguished by their lengths: 76 µm for the 'short' cavity (SC), and 161 µm for the 'long' cavity (LC). This distinction in length is the origin of their common names. The small transverse size of optical fibers is crucial to enabling this crossed-cavity configuration. Each fiber is held in place by a self-made holder consisting of different elements, as illustrated in detail in Fig 3.1a.

Light is coupled into and out of each cavity via a single-mode (SM) fiber. With reference to Fig.3.1b, these are labeled as LC-SM and SC-SM for the long and the short cavity, respectively. This single-sided design is achieved through differential mirror coatings: the multi-mode (MM) fiber surfaces are coated with a highly reflective dielectric coating (10 ppm transmission), while the opposing single-mode fiber surfaces are coated with a dialectric mirror coating with lower reflectivity (340 ppm transmission).

These cavity fibers are fabricated using a $CO_2$ laser machining technique initially developed by Jakob Reichel, David Hunger, and colleagues [104, 90, 105, 106]. This method has since become a widely spread approach for constructing experiments in cavity quantum electrodynamics [107, 108, 109, 110, 111, 112, 113, 114, 115, 116]. A dedicated $CO_2$ laser machining setup was built by Gerhard Rempe's group and was used to produce the four fibers used in the current work. By focusing the $CO_2$ laser on the fiber tip, spherical and, notably, elliptical fiber surfaces can be manufactured. Further details on this process are available in [102]. In particular, the fiber surfaces of LC are almost spherical, whereas

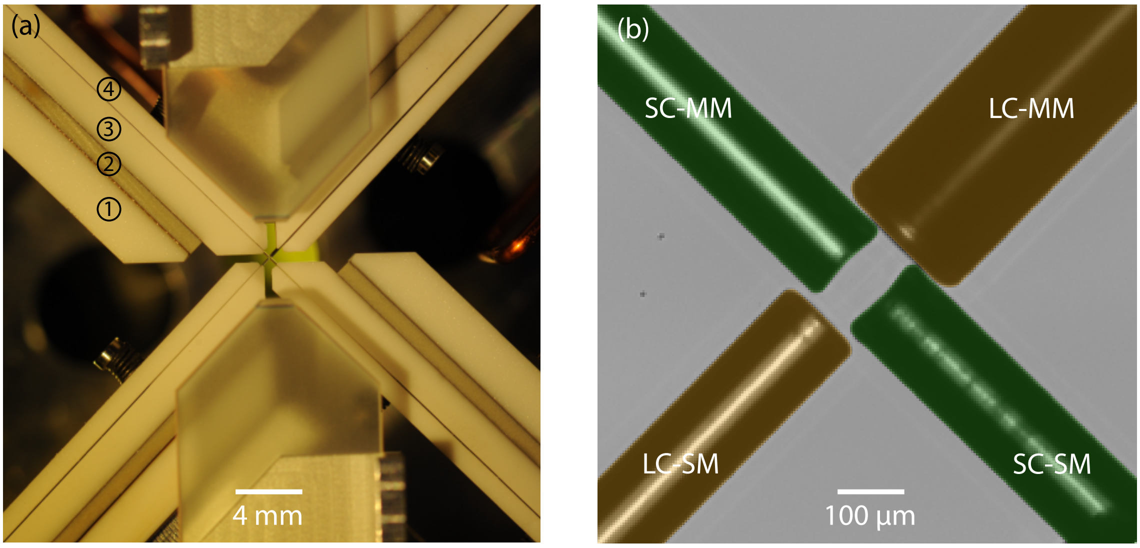


**Figure 3.1.:** (a) A vertical top view of the construction holding the crossed fiber cavities in place. The holder of each fiber consists of 4 elements, each of which is labeled in the picture: ① Base element for mounting the fiber holder to downstream components; ② Shear piezo actuator for active stabilization of the resonator length; ③ Mount in which a V-groove is fabricated - where the fiber is positioned and mechanically clamped; ④ Lid to clamp the fiber within the V-groove mount. A zoom of the center of the picture is shown in (b). The orange-colored cavity has a length of 161 µm and it is composed of a single-mode fiber (LC-SM) with a low reflective mirror coating and a multi-mode fiber (LC-MM) with a highly reflective mirror coating. The green-colored cavity is 75.7 µm long and is composed of a single- (SC-SM) and multi- (SC-MM) mode fiber with reflective coatings analogous to LC-SM and LC-MM, respectively. This photo has been colored in post-production.

those of SC are strongly elliptical. This leads to a cavity eigenmodes splitting [117], i.e. cavity birefringence, as shown in Fig. 3.2.

The fibers themselves have a 125 µm cladding diameter, comparable to a human hair. While standard optical fibers are often coated with acrylate polymers for protection – materials incompatible with ultra-high vacuum – the fibers used here are coated with copper to ensure both UHV compatibility and mechanical robustness. The details about the fiber types and coatings are shown in Table 3.1.

| **Fiber** | **Fiber type** | **Coating** ⌀ (µm) | **Cladding** ⌀ (µm) | **Core** ⌀ (µm) |
|---|---|---|---|---|
| SC/LC-SM | Cu800 | $165 \pm 10$ | $125 \pm 1$ | $6.0 \pm 0.5$ |
| SC-MM | Cu50-125 | $165 \pm 10$ | $125 \pm 1$ | 50 |
| LC-MM | Cu50-125 | $260 \pm 10$ | $200 \pm 2$ | 50 |

**Table 3.1.:** Fibers adopted for the crossed fiber cavities setup, produced by the company *IVG Fiber Ltd.*

The single-sided cavity design dictates specific fiber types and mirror coatings requirements. Multi-mode fibers are preferred at the highly reflective mirror for two reasons: their larger core diameter (Table 3.1) facilitates efficient and stable light coupling, beneficial for cavity transmission measurements, and the CO2 laser machining produces better concave surface structures [118].

The CO2 laser machining process can cause fiber edge reflow, resulting in a concave mirror area smaller than the fiber diameter [118]. To overcome this, larger cladding diameter fibers can be used to increase the effective mirror area, which is particularly important for the LC-MM fiber in the crossed-cavity configuration. This is due to both the cavity length and the cavity mode waist shifted towards the LC-SM fiber, hence making the cavity mode radius at the MM fiber mirror large. This design has been chosen in order to optimize the fiber-cavity mode matching. An alternative technique to obtain a larger concave mirror area is the CO2 dot milling technique [119], which employs a series of weak, precisely targeted CO2 pulses to sculpt the fiber surface. This method produces mirrors with lower roughness and more uniform structures.

## 3.2. Cavity parameters

The parameters for the long and short cavity used for this work are summarized in Table 3.2.

The concave surfaces of the four fiber mirrors were measured using a *Bruker Contour-GT* white light interferometer, shown in Fig. 3.2. In order to precisely determine their radii of curvature (ROCs), a spherical fit (not shown) was applied to a 5 μm region around the mirror center, a size chosen to match the cavity mode waist on the mirror surface.

The free spectral range (FSR) is measured by simultaneously holding two lasers resonant with the cavity and determining their frequency difference. The FSR is then determined as [120]:

$$\mathrm{FSR} = \Delta\nu / n \tag{3.1}$$

where $\Delta\nu$ is the lasers' frequency difference and $n$ is an integer number, quantifying the number of longitudinal modes (cavity resonances) between the two laser frequencies. The cavity length is then inferred from the FSR as:

$$L = \frac{c}{2\,\mathrm{FSR}} \tag{3.2}$$

where $c$ is the speed of light in vacuum. The ROCs values can be used to calculate the cavity mode geometrical parameters, such as the waist at different positions and the mode volume. Assuming a Gaussian mode, the waist $w(z)$ along the cavity axis ($z$-axis) can be calculated from the ROCs of the mirrors $R_1$ and $R_2$ and the cavity length [121, 122]:

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2} \tag{3.3}$$

$$w_0^2 = \frac{2L}{k}\sqrt{\frac{(R_1-L)(R_2-L)(R_1+R_2-L)}{L(R_1+R_2-2L)^2}} \tag{3.4}$$

where $k = \omega/c = 2\pi/\lambda$ is the wavenumber and $z_R = k\omega_0^2/2$ is the Rayleigh-length. The distance $z_1$ between the waist radius $w_0$ and the mirror 1 is calculated as [121, 122]:

$$z_1 = \frac{L(R_2-L)}{R_1+R_2-2L} \tag{3.5}$$

The mode volume is calculated according to [90]

$$V = \frac{\pi}{4}w_0^2 L \tag{3.6}$$

which leads to the theoretical maximum atom-photon coupling rate $g_0$, calculated with Eq. 2.3. This value is provided for the two transitions employed in Chapter 4, namely $5P_{3/2}$, $F' = 2$, $m_F = \pm 1 \leftrightarrow 5D_{5/2}$, $F'' = 3$, $m_F = 0$ for the long cavity and $5S_{1/2}$, $F = 2$, $m_F = \pm 1 \leftrightarrow 5P_{3/2}$, $F' = 2$, $m_F = \pm 1$ for the short cavity.

Due to ROCs of the cavity mirrors, the cavity modes are asymmetric around the cavity center, which is ultimately the atom's position. This intentional asymmetry shifts the cavity waist radius towards the OC mirror, enhancing fiber-cavity mode matching. Assuming a mode field diameter of 6.0 µm (Table 3.1) within the single-mode fiber, one can predict the theoretical mode matching values for both cavities, reported in Table 3.2.

Transmission spectra, shown in Fig. 3.2, were recorded for both cavities with the probe field polarization aligned to the cavity's polarization eigenmodes. This prevents simultaneous excitation of multiple modes, which would broaden the spectra, assuming non-degenerate polarization eigenmodes. Analysis of these spectra yields the cavity linewidth ($2\kappa$), field decay rate ($\kappa$), and the frequency splitting of the polarization eigenmodes. These quantities are used to determine the cavity finesse as [120]

$$\mathcal{F} = \frac{2\kappa}{\mathrm{FSR}} \tag{3.7}$$

In addition to the transmission losses given by the coatings, the mirrors manifest undesired parasitic losses, which are attributed to mirror absorption.

| Parameter | Long Cavity | Short Cavity |
|---|---|---|
| Radius of curvature of OC[1] | 340 µm | 115 µm and 283 µm |
| Radius of curvature of HR[2] | 170 µm | 96 µm and 167 µm |
| Cavity length | 161 µm | 75.7 µm |
| Free spectral range | 0.93061 THz | 1.98043 THz |
| Waist radius | 3.5 µm | 3.5 µm and 4.4 µm |
| Waist radius pos. from OC | 7.7 µm | 25.8 µm and 22.7 µm |
| Mode waist at OC | 3.6 µm | 3.9 µm and 4.6 µm |
| Mode waist at HR | 11.3 µm | 5.9 µm and 5.2 µm |
| Mode waist at cavity center | 6.2 µm | 3.5 µm and 4.4 µm |
| Mode volume ($\lambda = 780$ nm) | $10200 \cdot \lambda^3$ | $1900 \cdot \lambda^3$ |
| Mode matching | 0.97 | 0.82 |
| $g(3'' \leftrightarrow 2')/(2\pi)$ at cavity center[3] | 7 MHz | |
| $g(2' \leftrightarrow 2)/(2\pi)$ at cavity center[4] | | 44 MHz |
| Transmission through OC | 340 ppm | 340 ppm |
| Transmission through HR | 10 ppm | 10 ppm |
| Parasitic losses | 80 ppm | 50 ppm |
| Total losses | 430 ppm | 400 ppm |
| $\kappa/(2\pi)$ | 31.7 MHz | 61.5(3) MHz |
| Linewidth | 63.4 MHz | 122.9(6) MHz |
| Finesse | 14089 | 16531 |
| Outcoupling ratio | 0.79 | 0.85 |
| Eigenmodes freq. splitting | 16 MHz | 505.0(5) MHz |

**Table 3.2.:** Parameters of the cavities used in this work.
[1] OC stands for outcoupling mirror, located on the SM fiber tip.
[2] HR stands for highly reflective mirror, located on the MM fiber tip.
[3] Predicted coupling strength on the transition $5P_{3/2}$, $F' = 2$, $m_F = \pm 1 \leftrightarrow 5D_{5/2}$, $F'' = 3$, $m_F = 0$.
[4] Predicted coupling strength on the transition $5S_{1/2}$, $F = 2$, $m_F = \pm 1 \leftrightarrow 5P_{3/2}$, $F' = 2$, $m_F = \pm 1$.

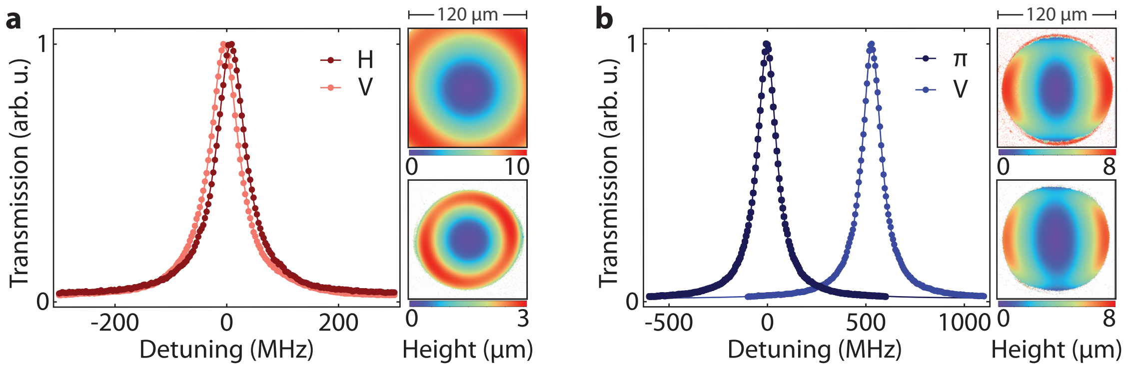


**Figure 3.2.:** Transmission spectra were recorded for both the long (a) and short (b) cavity. For the long cavity, the probe field was polarized to match the cavity eigenmodes, revealing a small residual polarization mode splitting. Contour images of the mirrors – for the MM and SM fibers – are shown alongside the spectrum. In contrast, the short cavity exhibited a large polarization mode splitting, a result of the strong ellipticity of its mirror surfaces. The error bars are smaller than the data point markers. This figure first appeared in [84].

The overall losses are

$$\mathcal{L} = \mathcal{L}_{OC} + \mathcal{L}_{HR} + \mathcal{L}_{parasitic} \tag{3.8}$$

where $\mathcal{L}_{OC}$, $\mathcal{L}_{HR}$ and $\mathcal{L}_{parasitic}$ are the losses, respectively, of the OC mirror, HR mirror and from parasitic effects. The related cavity field decay rate can be expressed as a sum of field decay rates that originate from each cavity loss channel:

$$\kappa = (\mathcal{L}_{OC} + \mathcal{L}_{HR} + \mathcal{L}_{parasitic})\frac{c}{4L} = \kappa_{OC} + \kappa_{HR} + \kappa_{parasitic} \tag{3.9}$$

By measuring the field decay rate $\kappa$ - Fig. 3.2 - and using the known coatings losses, we can calculate $\mathcal{L}_{parasitic}$ from Eq. 3.9.

## 3.3. The atom

The atom used in this work is $^{87}$Rb, one of two naturally occurring isotopes (with a relative abundance of around 27%, being $^{85}$Rb the other one) of the metal rubidium. It is solid at room temperature and it is quasi-stable with a half-life of $5 \cdot 10^{10}$ years [89]. The simplicity of its electronic structure, the ready availability of diode lasers at the relevant wavelenghts and the moderate temperatures required to vaporize it make rubidium a popular choice for ultracold atoms experiments. Because it has only one electron in its outermost shell, rubidium exhibits a comparatively simple energy level structure, facilitating cooling, precise control and manipulation. Indeed, it was famously used to realize the first Bose-Einstein condensate [123] in 1995, which was later awarded the Nobel Prize in Physics in 2001 [124].

The electronic ground state of $^{87}$Rb is the $5S_{1/2}$ state. It exhibits a hyperfine splitting of about 6.8 GHz due to the coupling of the total angular momentum of the electron to the angular momentum of the nucleus. Especially important for cooling and trapping is the D2-line, the $5S_{1/2}$ - $5P_{3/2}$ transition, originating from the fine structure of the atom due to spin-orbit coupling and its transition wavelength in vacuum is 780.2 nm. Important for this work is also the upper excited state $5D_{5/2}$, since the cascaded decay from this state is used to produce a photon-pair - see Section 4. If the atom is in the excited state $5D_{5/2}$, it will decay either in $5P_{3/2}$ or in $6P_{3/2}$, emitting a photon with a wavelength of 776 nm or 5.23 µm.

The level scheme is depicted in Fig. 3.3. In the experiment, a magnetic guiding field is applied. This lifts the degeneracy of the magnetic sub-levels and leads to a Zeeman splitting $\Delta E_{|F,m_F\rangle}$ on the order of a hundred kHz, according to [125]

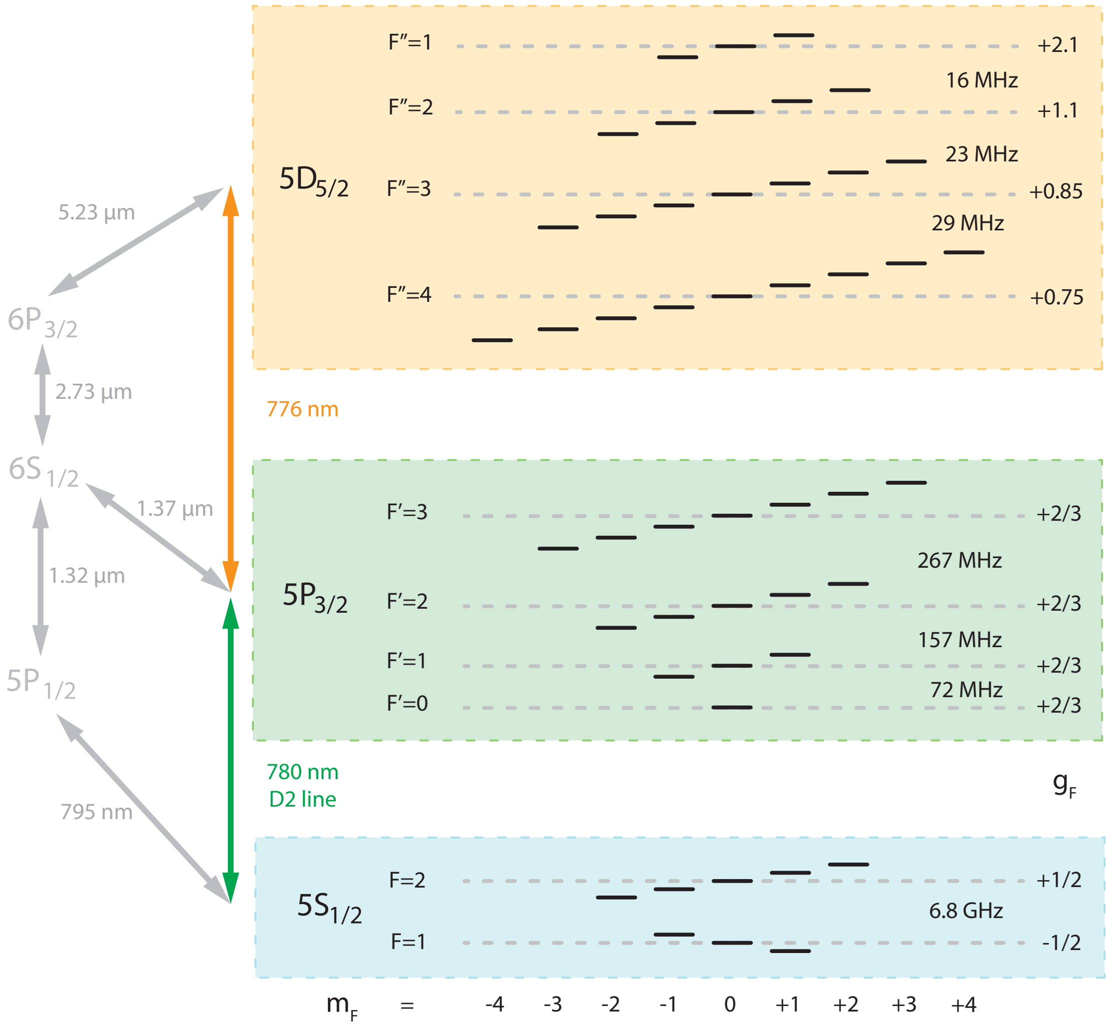


**Figure 3.3.:** Level scheme of $^{87}$Rb, depicting the fine-structure splitting into the states $5S_{1/2}$, $5P_{3/2}$ and $5D_{5/2}$, their hyperfine splitting and the Zeeman splitting of each of these. The right side of the scheme reports the Landé factor $g_F$ of each hyperfine state.

$$\Delta E_{|F,m_F\rangle} = \mu_B \, g_F \, m_F \, B_z \tag{3.10}$$

where $\mu_B$ is the Bohr magneton, $g_F$ is the considered level's Landé factor, $m_F$ is the magnetic sublevel and $B_z$ is the magnetic field along the quantization axis.

## 3.4. Magneto-optical trap

The experiments require trapping single atoms at the center of the fiber cavities. For this purpose, the experimental sequence begins by trapping a cloud of atoms in a magneto-optical trap (MOT). This is required in order to have a localized reservoir of ultra-cold atoms, from which single atoms can then be loaded into the cavities' center. The working principle of a MOT is well explained in [126, 127].

The first step is obtaining Rubidium vapor in the chamber. In previous works, the loading of the MOT began by letting current flow through a Rubidium dispenser placed in the vacuum chamber next to the cavities setup [103, 102]. The current would stay on for the whole experiment duration. This would heat the dispenser up to 500 °C, causing the evaporation of a significant quantity of $^{87}$Rb, which would eventually fill the chamber. Even though this simple yet efficient technique provides sufficient vapor pressure for these experiments, it presents some drawbacks.

First of all, this procedure can indeed render high background pressures in the vacuum chamber, potentially leading to shorter atomic lifetimes in the trap because of collisions with the hot atoms. Even though this effect could be mitigated by pulsing the current in the dispenser, we have observed that the switching of the dispenser is slow, leading to MOT loading times of several seconds. Secondly, we have observed that the MOT size fluctuates in time, probably due to the dispenser's fluctuating sputtering rate, causing fluctuations in the loading probability of single atoms. Thirdly, and most importantly, the dispenser heats the elements surrounding the cavity over the experiment duration. This causes a thermal expansion of the cavities and, therefore, a drift of the cavity length, which has to be actively compensated by applying a voltage to the shear piezo actuators (Fig. 3.1a). Over the experiment duration, the required compensation voltage can exceed the piezos' damage threshold, jeopardizing the setup integrity.

The light-induced atom desorption (LIAD) offers a solution to this problem. This technique has been shown to be suitable as an atomic source for large MOTs, and it is already largely employed in ultracold atoms experiments [128, 129, 130, 131, 132, 133]. It can be seen as the atomic analogue of the photoelectric effect. A light source is pulsed for the length of time needed to load the MOT (typically between a few seconds up to tens of seconds), after which the light is turned off and the partial pressure drops back to a lower value. The LIAD is most effective when the light source has a short wavelength, typically in the ultraviolet (UV). In this work, a UV diode *ILH-XQ01-S410* with emitting wavelength 410 nm illuminating a chamber side-viewport has been implemented - see Fig. 3.4. Section 3.6 offers a quantitative comparison of the thermal effects induced by different heat sources,

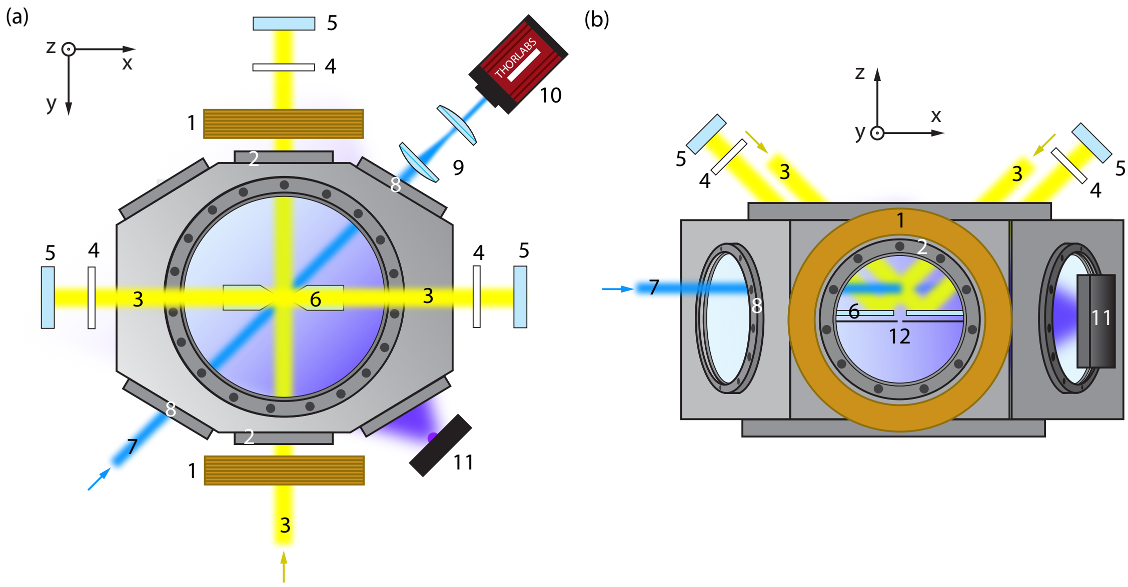


**Figure 3.4.:** Top (a) and side (b) view of the vacuum chamber in which all the elements used for realizing and characterizing the MOT are displayed. A more detailed description is provided in the main text. (1) The MOT coils. (2) The two viewports aligned along the symmetry axis of the MOT coils. (3) The three MOT beams, aligned in three orthogonal directions. (4) $\lambda/4$ waveplates. (5) $0°$ mirrors. (6) Chamber mirrors reflecting the MOT beams coming diagonally from the top. The structure holding these mirrors is not shown for simplicity. (7) Absorption imaging beam. (8) The two viewports through which the imaging beam is transmitted. (9) Imaging telescope. (10) Imaging camera. (11) UV diode. (12) Fiber cavity. The structure holding the fibers is not shown for simplicity. These drawings are not in scale and are shown here to give a qualitative idea to the reader of the setup used.

such as the dispenser, the UV light during the MOT loading procedure and the atom trap laser.

Fig. 3.4 shows a top- and a side-view of the chamber with all the elements employed for the experimental realization of the MOT, described in the following. The MOT coils (1) are mounted outside the vacuum chamber and are spaced 21.7 cm apart to accommodate the chamber's width. Their symmetry axis passes through two CF40-viewports (2). The MOT itself is formed using three retro-reflected optical beams (3), each with a diameter of 12 mm. One beam is aligned horizontally along the MOT coils' symmetry axis and is retro-reflected (5) on the other side of the chamber and its polarization is rotated by a $\lambda/4$ waveplate (4). The other two beams enter through the top viewport, reflect off in-vacuum mirrors (6), their polarization is rotated by a $\lambda/4$ waveplate (4) and are then eventually retro-reflected outside the chamber (5). The $\lambda/4$ waveplates in front of each retro-reflecting mirror have the role to preserve the circular polarization of the MOT beams to allow the magneto-optical trapping of the atoms and to perform Sisyphus cooling [134]. With this configuration, the beams intersect 1 cm above the plane of the horizontal cavities. A detailed description of the MOT coils, the MOT coils' circuit and the water cooling circuit is provided in [103].

The MOT loading procedure consists of two parts: the loading phase and the molasses phase. During the loading phase the MOT coils, the MOT beams (consisting of a cooling laser and of a repumping laser) and the UV light (11) are switched on simultaneously. During this time, the atomic cloud is cooled down and can reach a temperature close to the $^{87}$Rb $D_2$-line Doppler temperature of 146 µK [89]. Directly after that, an optical molasses is applied, which can bring the atoms to a temperature much below the Doppler limit. The detuning of the MOT beams increases, whereas the optical power decreases. At the end, the MOT coils, the MOT beams, and the UV light are switched off. The parameters used for the experiments performed in this work are reported in Table 3.3.

| | |
|---|---|
| MOT coil current | 23.5 A |
| UV diode current | 140 mA |
| MOT beams' diameter | 12 mm |
| **Loading phase** | |
| Duration | 450 ms |
| Cooling laser detuning from $5S_{1/2}$, $F = 2 \leftrightarrow 5P_{3/2}$, $F' = 3$ | -13 MHz |
| Repumping laser detuning from $5S_{1/2}$, $F = 1 \leftrightarrow 5P_{3/2}$, $F' = 2$ | 0 MHz |
| Cooling laser power | 20 mW |
| Repumping laser power | 250 µW |
| **Molasses phase** | |
| Duration | 1 ms |
| Cooling laser detuning from $5S_{1/2}$, $F = 2 \leftrightarrow 5P_{3/2}$, $F' = 3$ | -31 MHz |
| Repumping laser detuning from $5S_{1/2}$, $F = 1 \leftrightarrow 5P_{3/2}$, $F' = 2$ | -20 MHz |
| Cooling laser ramp-down power | 1.7 mW |
| Repumping laser power | 250 µW |

**Table 3.3.:** Parameters used for the MOT loading phase. The UV diode and the MOT coils are permanently on the whole time. The cooling and repumping laser power is measured behind a polarizing beam splitter, which separates the horizontal MOT beam from the top-diagonal ones.

In order to characterize the obtained MOT, absorption imaging of the atomic cloud is performed by using a 10 mm diameter beam (7), which propagates through two opposing CF40-viewport (8) and is then imaged via an optical telescope (9) with a magnification factor of 0.5 onto the CMOS-camera (10) *Thorlabs CS135MUN*. By measuring the attenuation of the beam propagating through the MOT it is possible to derive the number of atoms in the MOT, as explained in [103]. Furthermore, the number of atoms versus the loading time can be measured - see Fig. 3.5a - theoretically modeled as:

$$N(t) = R \cdot \tau \left(1 - e^{-t/\tau}\right) \tag{3.11}$$

where $R$ is the loading rate and $\tau$ is the $1/e$ lifetime. By fitting Eq. 3.11 to the measured data yields R = 1324(70) Hz and $\tau = 25(2)$. For long loading times, Eq. 3.11 converges towards the steady-state atom number $R \cdot \tau = 3.2 \cdot 10^4$. The number of atoms in the MOT is investigated also as a function of the UV-diode current, see Fig. 3.5b.

Another key parameter characterizing an atomic cloud is its temperature, which can be determined using time-of-flight (TOF) measurements. In this method, absorption images are taken at varying times after the MOT phase ends, allowing the cloud to expand as a result of its finite temperature. By fitting a two-dimensional Gaussian function to each image, the waist size $w$ of the cloud is extracted. Fig. 3.5c shows the evolution of the waist as a function of the TOF, which is modeled by:

$$w(t) = \sqrt{w^2(0) + \sigma^2 t^2} \tag{3.12}$$

$$\sigma^2 = \frac{k_B T}{m} \tag{3.13}$$

where $\sigma^2$ is the variance of the Maxwell-Boltzmann distribution for particles with mass $m$ moving with an average temperature $T$. Fitting Eq. 3.12 to the measured data yields a mean temperature $T = 100(8)\,\mu\text{K}$ after the cooling phase (blue curve) and $T = 23(3)\,\mu\text{K}$ after the molasses phase (green curve), which is significantly below the Doppler limit ($T = 146\,\mu\text{K}$). Given the steady-state atom number and the atomic cloud size, the atom cloud density is estimated to be around $10^{10}$ atoms/cm$^3$.

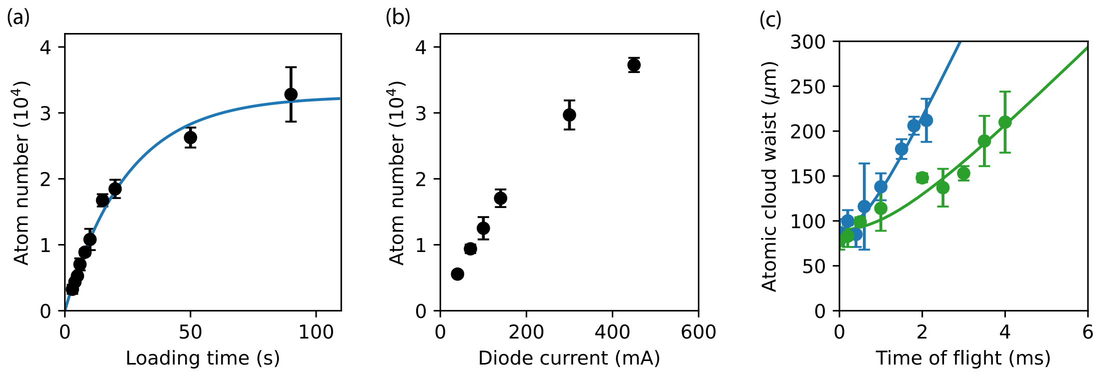


**Figure 3.5.:** (a) The number of atoms in the atomic cloud as a function of the loading time for a diode current of 140 mA. The solid line represents the fit of Eq. 3.11 to the data. (b) The number of atoms in the atomic cloud after 20 s of loading as a function of the diode current. (c) Atomic cloud size as a function of the time of flight after the loading phase (blue data points) and after the molasses phase (green data points). The solid lines represent the fit of Eq. 3.12 to the data. The beams' powers and detunings and the coils' current used for these measurements are reported in Table 3.3.

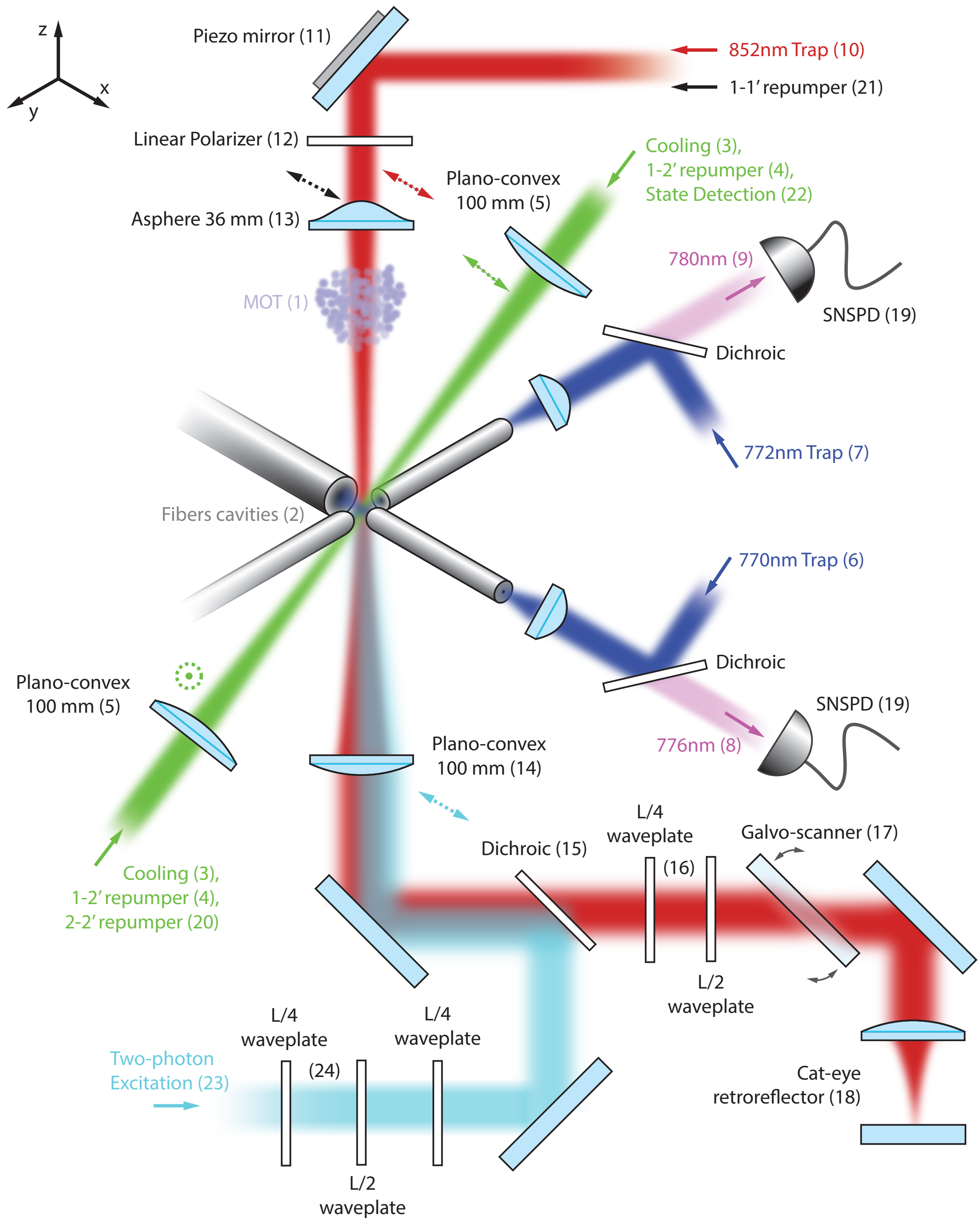


**Figure 3.6.:** Overview of experimental setup used in this work. According to the reference frame, the z, the x, and the y axes are, respectively, the vertical, the Long Cavity, and the Short Cavity axes. The dashed arrows and the dashed circle indicate the laser beams' polarization. Each main beam path is represented with a different color, and its function is explained in the main text.

## 3.5. Trapping and positioning

After creating an ultra-cold atomic cloud, the next step is to trap a single atom at the center of the two fiber cavities. Fig. 3.6 shows a qualitative illustration of the laser beams and optical elements used for performing all the experiments described in this work and will serve as a reference for the following descriptions. The MOT (1) is located about 10 mm above the cavity plane (xy). After the sub-Doppler cooling phase, the trapping potential is released, and the atoms fall towards the cavity plane. Laser beams are switched on for cooling and trapping purposes once the free-falling atomic cloud reaches the cavities' crossing point (2).

A cooling (3) and a repumping (4) laser are shone diagonally onto the crossing point in a counter-propagating configuration. The counter-propagating beams have orthogonal polarizations in a lin⊥lin configuration, such that the atoms are cooled to sub-Doppler temperatures via Sisyphus-cooling. The beams are focused onto the atoms via 100 mm plano-convex lenses (5) and have a diameter of 20 µm at the atom's position.

The trapping of the atoms at the cavities' crossing point is provided by three lasers that cross in three directions (x, y and z). Two trapping beams are provided by the intracavity light fields. A 770 nm (6) and a 772 nm (7) laser are coupled via the OC mirror to the long (x-axis) and the short (y-axis) cavity, respectively. Each of these lasers self-interferes in the cavity, forming a standing wave. Since they are blue detuned from the $D_2$- and the $D_1$-line [89], they create two orthogonal repulsive standing waves, which confine the atom at their nodes - i.e. interference minima. Moreover, the reflection of these lasers at the OC mirror in combination with the shear piezo actuators - see Fig. 3.1a -is used to stabilize the cavity length. In practice, the laser is first locked to the cavity using the PDH-lock technique [135], and the cavity length is then indirectly stabilized by locking the laser frequency to an optical frequency comb, which acts as a stable frequency reference [102]. The laser frequency is carefully chosen so that each cavity is resonant with the atomic transition relevant for the performed experiments. This is known as the double resonance condition. In this work, the long cavity resonates with the $5P_{3/2} - 5D_{5/2}$ transition at 776nm (8) and the short cavity with the $D_2$-line at 780nm (9).

The atom confinement along the vertical axis (z-axis) is provided by a 852 nm laser (10), which is shone from the top to the cavities' crossing point. Its precise alignment to this point is ensured by a piezo mirror (11) *Tholarbs POLARIS-K1S2P* that can steer the beam in the x- and y-direction. This laser beam is linearly polarized (12) along the long cavity axis and subsequently focused to a diameter of about 20 µm with a 36.5 mm custom lens (13). It is then transmitted through the chamber and collimated by a 100 mm plano-convex lens (14) to a mode diameter of 1 cm. It is subsequently transmitted through a dichroic element (15), a set of retardation waveplates (16), a refractive substrate (17) mounted on a galvo scanner and, eventually, it is retroreflected in a cat-eye configuration (18). Since it is far red-detuned from the atom's $D_2$- and the $D_1$-line, this laser forms a trapping standing wave, where the atoms are confined in its antinodes - i.e. interference maxima.

Once the cooling and trapping beams have been switched on, a sequence takes place to position the atom at the location of maximum cavity-coupling. Here, a short description

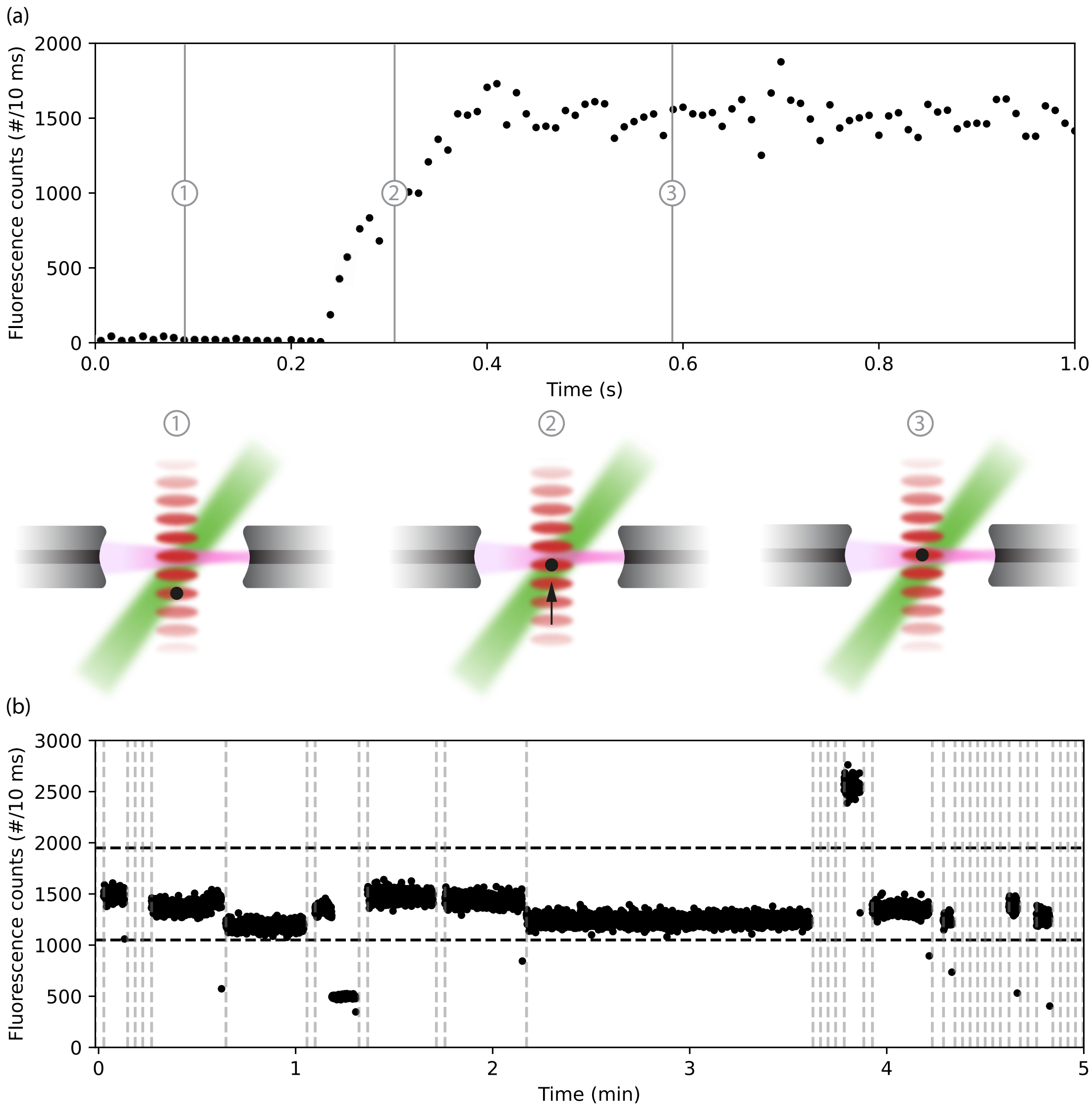


**Figure 3.7.:** (a) shows a typical trace in which fluorescence counts are recorded during the atom positioning procedure as a function of time. The MOT release sets the time = 0. Around time 0.1 s, an atom is trapped in the 852 nm trap and is not coupled to the cavity (1). Around time 0.2 s, the positioning procedure begins, in which the galvo is swept in one direction. During this time, the counts are recorded (2). Once the maximum coupling position is reached (3), the positioning sequence is interrupted, and the atom is kept in place while being cooled. (b) Fluorescence counts recorded over a 5-minute experimental run. Typically, multiple atoms are sequentially trapped, with each trapping attempt indicated by a vertical dashed line. Only atoms producing fluorescence signals within the horizontal dashed lines are post-selected for further analysis.

of this procedure is provided; a detailed description can be found in Tobias Frank's PhD thesis [136]. Fluorescence counts coming from the atom during the cooling phase are collected by the short cavity and detected with superconducting nanowire single-photon detectors (SNSPDs, 19). Meanwhile, the angle of the 5 mm thick substrate (17) is scanned by a galvo scanner. This modifies the beam's refraction angle, effectively changing the optical path length. This changes the standing wave interference pattern, scanning the antinodes' position and, therefore, the atom's position along the vertical axis. A closed live-loop consisting of the FPGA, the SNSPDs and the galvo scanner optimizes the number of fluorescence counts collected from the atom, ensuring the positioning of the atom at the maximum cavity-coupling. The loop restarts the MOT-loading phase if no atom is found after 1 s. Fig. 3.7a shows a trace during the positioning sequence, where the fluorescence counts increase in time until the optimum position is found. The loading of a single atom in this work has a success probability of about 30%. Fig. 3.7b reports a typical experimental trace with several subsequent atom trapping attempts.

## 3.6. Thermal stability of the setup

Over the course of this PhD, several heat sources have been observed to affect the setup stability over the measurement times. Here we characterize the effect of the heating induced by the 852 nm trap laser, the dispenser and the UV diode. For this purpose, the SC is locked on resonance with the 772 nm laser for the whole experiment duration. The cavity length is chosen such that at the beginning of the experiment it is resonant with a second laser of wavelength 780 nm, with which the cavity is probed. In order to track potential temperature changes of the vacuum chamber, a temperature sensor (*PT100*) has been put in contact with the chamber's external walls.

Fig. 3.8a shows the voltage applied to the cavity-length stabilizing piezo while heating the setup with different heat sources. For each curve of this measurement, the setup starts from a thermalized condition at rest. Whereas the dispenser and the trap are turned on continuously, the UV diode is switched on every 2 s for 0.5 s, based on realistic working conditions for the experiments performed in this PhD. For this measurement, the dispenser and the UV LED are driven with a current of 3.6 A and 140 mA, respectively - these values are chosen such that the atom loading rate in both cases is comparable. While all sources induce a change in the voltage, the dispenser clearly shows the biggest impact. The adoption of a background gas loading mechanism involving the LIAD instead of the dispenser activation is indeed beneficial for the system stability. A remarkable feature is the change in the slope in each curve. One possible explanation for this effect is that different components in the chamber, which expand with different rates and direction, reach their equilibrium at different times.

Surprisingly, a further instability lies in the cavity spectrum. By probing the cavity with the 780 nm light, two peculiar effects were observed while activating the dispenser. While the probe reflected signal tuned onto the starting double resonance condition changes substantially during the dispenser activation (first increasing and later decreasing, see Fig. 3.8b), a new probe resonance peak appears, increasing its separation from the initial resonance, as shown in Fig. 3.8c. Fig. 3.8d shows the cavity spectrum when probing the SC

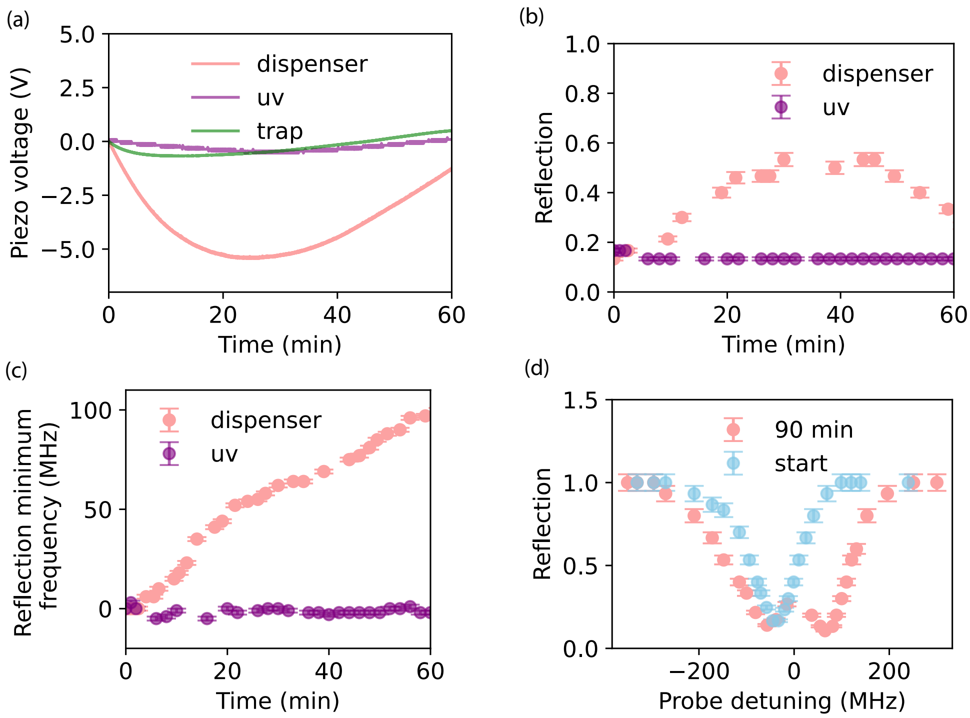


**Figure 3.8.:** Characterization of heating effects induced by the dispenser, the UV diode and the 852 nm atomic trap. (a) Locking voltage applied to the SC piezo while heating with different heat sources. (b) Reflection of the 780 nm probe. Its frequency is kept constant at the cavity double resonance at time 0. (c) Frequency of the reflection spectrum minimum. (d) Cavity spectrum in reflection at the beginning of the measurement and after 90 minutes. The latter appears broader and exhibits two peaks, spaced by about 120 MHz.

before activating the dispenser (blue dots) and after 90 minutes (red dots), which appears broader and exhibits two peaks. This observation remains sofar unexplained. Heating the setup with the trap or the UV light yields an unchanged cavity spectrum, confirming the positive impact of the LIAD in the setup stability.

Although remarkable thermal effects have been observed in the cavity, a constant temperature of 24.0(6)°C was measured on the outer chamber walls.

## 3.7. AC Stark shifts and standing wave polarization

The 852 nm standing wave induces strong AC Stark shifts on the atomic energy levels $|nL_J, F, m_F\rangle$, where $n$ is the main quantum number, $L$ is the orbital angular momentum, $J$ is the total angular momentum, $F$ is the hyperfine quantum number and $m_F$ labels the Zeeman state. The applied electric field is denoted as $\mathbf{E}(t) = \frac{1}{2}|E|\hat{\boldsymbol{\epsilon}}e^{i\omega t} + c.c.$ where $|E|$ is the field amplitude, $\hat{\boldsymbol{\epsilon}}$ is the polarization vector of unit length, and $\omega$ is the angular frequency. The AC Start shift can be expressed as [137]:

$$\Delta E_{nL_J,F,m_F} = -\frac{|E|^2}{4}\left[\alpha^{(s)}_{nL_J,F} - \alpha^{(v)}_{nL_J,F}\Im(\epsilon_x^*\epsilon_y)\frac{m_F}{F} - \alpha^{(t)}_{nL_J,F}\frac{(1-3|\epsilon_z|^2)}{2}\frac{(3m_F^2 - F(F+1))}{F(2F-1)}\right] \tag{3.14}$$

where $\alpha^{(s)}_{nL_J,F}(\omega)$, $\alpha^{(v)}_{nL_J,F}(\omega)$, and $\alpha^{(t)}_{nL_J,F}(\omega)$ are, respectively, the scalar, the vectorial and the tensorial dynamic polarizability of the hyperfine level. Eq. 3.14 leads to a shift of energy levels that is independent (scalar), linearly dependent (vector), or quadratically (tensor) dependent on the Zeeman quantum number $m_F$. The dynamic polarizabilities can be calculated by summing the contribution of the couplings to different state as in the following:

$$\alpha^{(s)}_{nL_J,F} = \frac{1}{\sqrt{3(2J+1)}}\alpha^{(0)}_{nL_J} \tag{3.15}$$

$$\alpha^{(v)}_{nL_J,F} = (-1)^{J+I+F}\sqrt{\frac{2F(2F+1)}{F+1}}\begin{Bmatrix} F & 1 & F \\ J & I & J \end{Bmatrix}\alpha^{(1)}_{nL_J} \tag{3.16}$$

$$\alpha^{(t)}_{nL_J,F} = (-1)^{J+I+F}\sqrt{\frac{2F(2F+1)(2F-1)}{3(F+1)(2F+3)}}\begin{Bmatrix} F & 2 & F \\ J & I & J \end{Bmatrix}\alpha^{(2)}_{nL_J} \tag{3.17}$$

$$\tag{3.18}$$

where $\alpha^{(K)}_{nL_J}$ is frequency dependent and is defined as:

$$\alpha^{(K)}_{nL_J}(\omega) = (-1)^{K+J+1}\sqrt{2K+1}\sum_{n',L',J'}(-1)^{J'}\begin{Bmatrix} 1 & K & 1 \\ J & J' & J \end{Bmatrix}\times$$
$$\times (2J+1)|\langle nL_J||er||n'L'_{J'}\rangle|^2\frac{1}{\hbar}\left(\frac{1}{\omega_{n'L'_{J'}\leftrightarrow nL_J}-\omega} + \frac{(-1)^K}{\omega_{n'L'_{J'}\leftrightarrow nL_J}+\omega}\right) \tag{3.19}$$

Here, $I = 3/2$ is the nuclear angular momentum, $\omega_{n'L'_{J'}\leftrightarrow nL_J}$ is the angular frequency of the transition between the state $|nL_J\rangle$ and $|n'L'_{J'}\rangle$, $\langle nL_J||er||n'L'_{J'}\rangle$ denotes the reduced dipole moment matrix element of the transition between these two states, and the curly brakets denote the Wigner-6j-symbol. The polarizabilities calculated with Equations 3.15-3.19 can be found in Table 3.4. From 3.14 it follows that linearly polarized light, i.e. $\Im(\epsilon_x^*\epsilon_y) = 0$ does not cause vector AC Stark shifts, even in case $\alpha^{(v)}_{nL_J,F} \neq 0$.

From Eq. 3.19, it follows that the tensor polarizability of the ground state $5S_{1/2}$ ($n = 5$, $L = 0$, $J = 1/2$) is zero. Eq. 3.14 shows that if the trap is elliptical, it acts as a virtual magnetic field along its direction of propagation, since the induced AC Stark shifts lift the Zeeman states' degeneracy.

| **Dynamic polarizabilities - $\lambda = 852$ nm** | |
|---|---|
| $\alpha^{(s)}_{5S_{1/2}}$ | 2065 a.u. |
| $\alpha^{(v)}_{5S_{1/2},F=1}$ | 170 a.u. |
| $\alpha^{(v)}_{5S_{1/2},F=2}$ | -341 a.u. |
| $\alpha^{(s)}_{5P_{3/2}}$ | -3845 a.u. |
| $\alpha^{(v)}_{5P_{3/2},F=1}$ | -2678 a.u. |
| $\alpha^{(t)}_{5P_{3/2},F=1}$ | 1198 a.u. |
| $\alpha^{(v)}_{5P_{3/2},F=2}$ | -5356 a.u. |
| $\alpha^{(t)}_{5P_{3/2},F=2}$ | 0 a.u. |
| $\alpha^{(v)}_{5P_{3/2},F=3}$ | -8035 a.u. |
| $\alpha^{(t)}_{5P_{3/2},F=3}$ | -2995 a.u. |

**Table 3.4.:** Calculated dynamic polarizabilities for the wavelength of the red detuned trap used in this work. The scalar polarizabilities are reported only once, since they do not depend on the specific hyperfine state. The atomic unit for polarizability is 1 a.u. $= a_0^2 e^2/E_h = 1.658 \cdot 10^{-41}$ $\mathrm{C^2 m^2 J^{-1}}$, where $a_0$ is the Borh radius, $e$ is the electron charge, and $E_h$ is the Hartree energy.

Even though the trap is transmitted through a linear polarizer (Fig. 3.6, 12) , its polarization at the atom position is not perfectly linear, especially because of the rotations accumulated by the retro-reflected beam during its propagation. However, having a linearly polarized trap is of great importance for the experiments presented in Chapter 5. Firstly, it ensures that the effective magnetic field is preserved while the trap is switched off - see Section 4.3. Secondly, it makes the atomic qubit stored in a superposition of the states $\left|5S_{1/2}, F=2, m_F=\pm 1\right\rangle$ insensitive to trap power fluctuations.

From Eq. 3.14 we can derive an experimental method to measure the residual trap ellipticity. First of all, a static magnetic field is applied along the vertical direction, such that the quantization axis is aligned with the laser's longitudinal direction. The Zeeman splitting frequency between the states $\left|5S_{1/2}, F=2, m_F=\pm 1\right\rangle$ is given by:

$$\Delta\nu = \Delta\nu_0 + \alpha^{(v)}_{5S_{1/2},F=2} \Im(\epsilon_x^* \epsilon_y) \frac{4P}{h\pi c \varepsilon_0 w_0^2} \tag{3.20}$$

where $\Delta\nu_0$ is the Zeeman splitting frequency induced by the external magnetic field, $P$ is the total power of the trap beam, and $w_0$ is the beam waist radius at the atom position. This equation has been derived assuming the intensity at the center of the beam at its waist to be $I = 2P/\pi w_0^2 \times 4$, which is valid for Gaussian beams. The factor 4 comes from the fact that we are considering a standing wave.

Fig. 3.9a shows the measured Zeeman splitting frequency, obtained via microwave spectroscopy - see Section 3.10 - as a function of the 852 nm laser power. By fitting Eq. 3.20

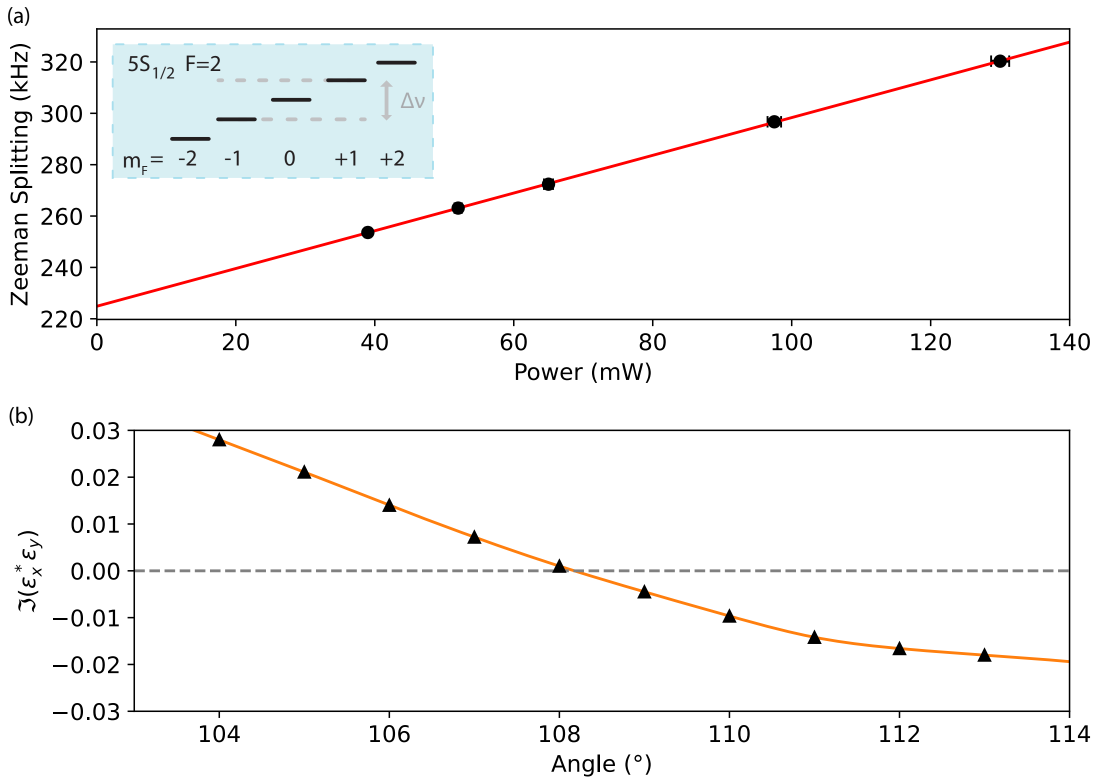


**Figure 3.9.:** Zeeman splitting induced by the trap ellipticity. (a) Frequency difference between the states $|5S_{1/2}, F = 2, m_F = \pm 1\rangle$ as a function of the trap power. The red line is the linear fit of Eq. 3.20 to the data. By using Eq. 3.21, one can calculate $\Im(\epsilon_x^* \epsilon_y) = 0.17995(7)$. (b) Derived degree of ellipticity as a function of the $\lambda/2$ retardation waveplate angle (Fig. 3.6, 16). By proper choice of this angle, it is possible to reach a degree of ellipticity $|\Im(\epsilon_x^* \epsilon_y)| \leq 10^{-4}$, obtained in this case for 108.75°. The orange line represents a smooth data interpolation. For both plots, the error bars are smaller than the data points.

to the data, it yields $\Delta\nu_0 = 224.9(2)$ kHz and a slope $a = 0.734(3)$ kHz/mW. The degree of ellipticity is determined as

$$\Im(\epsilon_x^* \epsilon_y) = \frac{hc\pi\varepsilon_0 w_0^2}{4\alpha^{(v)}_{5S_{1/2},F=2}} a \tag{3.21}$$

In order to cancel the polarization rotations accumulated by the laser, a $\lambda/4$ and a $\lambda/2$ waveplates (Fig. 3.6, 16) have been installed. By proper choice of the waveplates' angles, it is possible to obtain a perfect linear polarization, as shown in Fig. 3.9b.

## 3.8. Atomic-state preparation

The experiments with single atoms performed in this work require the preparation of the atom in the hyperfine states $5S_{1/2}, F = 1$, $5S_{1/2}, F = 2$, or in the Zeeman state $5S_{1/2}, F = 1, m_F = 0$.

Optical pumping (OP) is a technique used to prepare an atom in a specific ground state by repeatedly exciting it to an upper state from which it can decay into various ground states, with probabilities determined by the dipole matrix elements of the transitions. The excitation light is configured so that all ground states are coupled to an excited state, except for the desired target state. As a result, the target state does not absorb the excitation light and therefore does not contribute to fluorescence. Once the atom decays into this dark state, it remains there, completing the state preparation. In this work, the D2-line serves as the OP transition manifold, with laser light used to drive the necessary excitations. In total, three different laser beams - summarized in Table 3.5 - are utilized for the pumping schemes employed.

| Transition | Frequency | Short Notation | Polarization |
|---|---|---|---|
| $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F = 2$ | 384.227890 THz | $2 \leftrightarrow 2'$ | mixed |
| $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F = 2$ | 384.234720 THz | $1 \leftrightarrow 2'$ | mixed |
| $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F = 1$ | 384.234562 THz | $1 \leftrightarrow 1'$ | $\pi$ |

**Table 3.5.:** Laser beams utilized for the OP schemes. Here, $\pi$-polarization refers to linear polarization aligned with the LC axis.

Depending on the targeted initialization state, a distinct subset of these laser beams is used, as illustrated in Fig.s 3.10a, 3.10b, and 3.10c. To prepare the atom in the $5S_{1/2}, F = 1$ state, the laser beam $2 \leftrightarrow 2'$ is employed - see Fig. 3.6, 20. This beam depopulates the $F = 2$ level, effectively accumulating the atomic population into the $F = 1$ state.

Additionally, if the atom is to be prepared in the Zeeman state $5S_{1/2}, F = 1, m_F = 0$, laser beam $1 \leftrightarrow 1'$ is also applied. This beam is shone from the top of the chamber and it is $\pi$-polarized, since it is transmitted through the linear polarizer - see Fig. 3.6, 21. It takes advantage of the dipole selection rules, which forbid the transition between the states $5S_{1/2}, F = 1, m_F = 0$ and $5P_{3/2}, F = 1, m_F = 0$. As a result, $5S_{1/2}, F = 1, m_F = 0$ state remains dark, while all other states within this hyperfine manifold are bright. These beams combined depopulate the states $F = 2$ and $F = 1, m_F = \pm 1$, effectively accumulating population into the state $F = 1, m_F = 0$. Fig. 3.10d shows the fluorescence counts emitted by the atom in the SC during OP in $F = 1, m_F = 0$, which takes approximately $40\,\mu s$.

If the target state is $F = 2$, the laser beam $1 \leftrightarrow 2'$ is employed - see Fig. 3.6, 4 - which eventually depletes the population in the $F = 1$ level, leaving the atom population in the $F = 2$ level.

The precise quantification of the OP performance is tricky, since it depends on other tech-

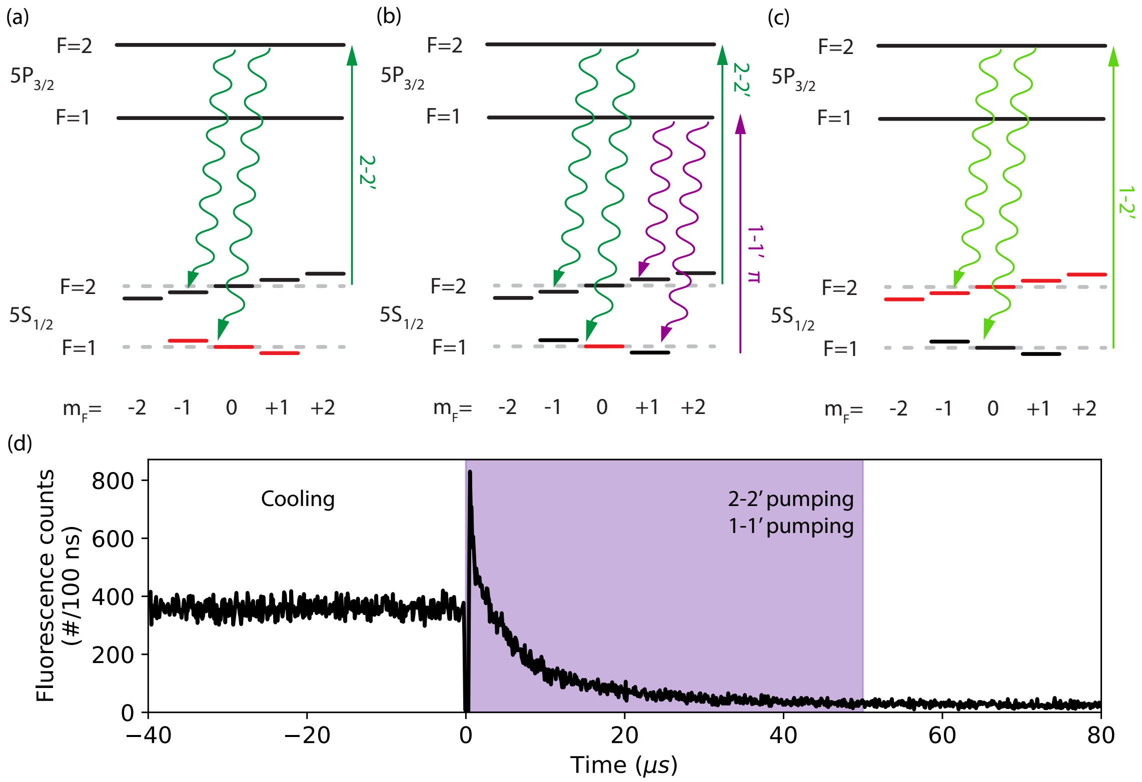


**Figure 3.10.:** Schemes utilized for OP in $F = 1$ (a), $F = 1, m_F = 0$ (b), and $F = 2$ (c). Each sketch shows the used lasers (straight arrows), the atomic decay (wavy arrows), and the targeted state (red lines). (d) Histogram of the fluorescence counts emitted by the atom in the SC during OP in $F = 1, m_F = 0$ with a duration of 50 µs. For this measurement, the SC is resonant with the $2 - 2'$ transition. Time 0 corresponds to the start of the OP procedure.

niques explained in the following sections, which are prone to error. We can conservatively estimate an OP probability in $F = 1$ and in $F = 2 \geq 99.5\%$ and in $F = 1, m_F = 0 \geq 97\%$.

## 3.9. Hyperfine state detection

Another fundamental feature of the utilized setup is the capability to detect whether the atom is in the hyperfine ground state $F = 1$ or $F = 2$. The used technique is cavity-enhanced fluorescence state detection (SD), and it is utilized for all the sequences involving the atomic qubit readout, e.g., for the atom-photon state tomography described in Chapter 5. Compared to other techniques, such as ionization-based SD [138] or free space fluorescence SD [139], this technique offers the main advantage of being inherently lossless - meaning that the atom is not lost after its state is read out - and very fast at the same time. Firstly, the fact that the atom remains trapped can potentially yield higher experimental repetition rates; secondly, it can be accomplished within time scales of µs, while having an SD fidelity $> 99\%$ [140]. This is of practical importance for quantum network protocols involving the atomic qubit readout after entanglement distribution over the network [141].

State readout based on fluorescence light detection distinguishes two atomic states by detecting either a high rate of scattered photons (identified as "bright" state) or no scattered photons ("dark" state) when the atom is state-selectively excited with a probe laser. When a cavity is coupled to the atom and is in the vicinity of the atomic resonance, it enhances the scattering of photons in the cavity optical mode. These are then detected by single-photon detectors, and based on the photon counts, it is inferred which atomic state is populated.

In our case, the SD beam enters the chamber along a diagonal axis - see Fig. 3.6, 22 - and excites the atom via the $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}$ transition. Fig. 3.11a shows the atomic levels in the presence of the AC-Stark shifts induced by the 852 nm trap and the detunings of the SC eigenmodes used. The SC eigenmode with polarization parallel to the LC-axis (abbreviated as SC $\pi$-mode) is $\sim$ 40 MHz red detuned to $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F = 2, m_F = \pm 2$, whereas the perpendicular SC eigenmode (abbreviated as SC V-mode, since the polarization is vertical) is $\sim$ 219 MHz blue detuned to $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F = 3, m_F = 0$. Even though having the cavity resonant with the $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}$ transition would enhance the photon emission and collection, this was not possible for the experiments presented in this thesis, for which the SC has to collect efficiently photons emitted in the $5P_{3/2}, F = 2, m_F = \pm 2 \rightarrow 5S_{1/2}, F = 2, m_F = \pm 2$ decay - see Chapter 4.

The spectrum of the $5P_{3/2}, F = 3$ level is measured by scanning the SD beam frequency at a low and constant power and recording the detected counts in the two SC eigenmodes. Thanks to a set of waveplates and a PBS, it is possible to map the photons emitted in the two modes onto two different SNSPDs. The result is shown in Fig. 3.11b. Here, two different spectra for the two eigenmodes can be clearly distinguished. This is mainly due to the fact that the coupling strength of the atom to each cavity depends on the Zeeman states considered in the $5S_{1/2}, F = 2$ and $5P_{3/2}, F = 3$ manifolds. As a consequence, each Zeeman level in $5P_{3/2}, F = 3$ gets shifted differently, due to the coupling of the atom to both modes. By using the theory presented in chapter 4, one can construct a Hamiltonian tailored for this scenario to simulate the expected spectra having as only free parameter the Rabi frequency of the driving laser - see Appendix C for the model and the parameters used. Fig. 3.11b depicts the net shift of each $5P_{3/2}, F = 3$ Zeeman state from this model, and the solid lines represent the predicted spectra. A comparison between the theoretical prediction and the measured data reveals a noticeable discrepancy in the width of the V-mode spectrum. This broadening is likely caused by the mechanical motion of the atom within the trap during the absorption and subsequent re-emission of laser light.

Since the cyclic absorption and emission of fluorescence photons heats the atoms, reducing their lifetime in the trap, a red detuning of 16 MHz to $\left|5S_{1/2}, F = 2\right\rangle \leftrightarrow \left|5P_{3/2}, F = 3, m_F = \pm 3\right\rangle$ is chosen for this work. Fig. 3.12a shows the photon number histogram measured during the cavity-enhanced fluorescence SD when the atom is in state $5^2S_{1/2}, F = 1$ (black bars) and in state $5^2S_{1/2}, F = 2$ (pink bars) for a SD beam with frequency 384.228109 THz, power of 170 nW and a duration of 7.5 µs. Both distributions are highly distinguishable, due to their small overlap. The dark state ($F = 1$) distribution is concentrated at the zero count number, whereas for the bright state ($F = 2$), the distribution yields a mean photon number of 10. The SD fidelity is quantified as [142]

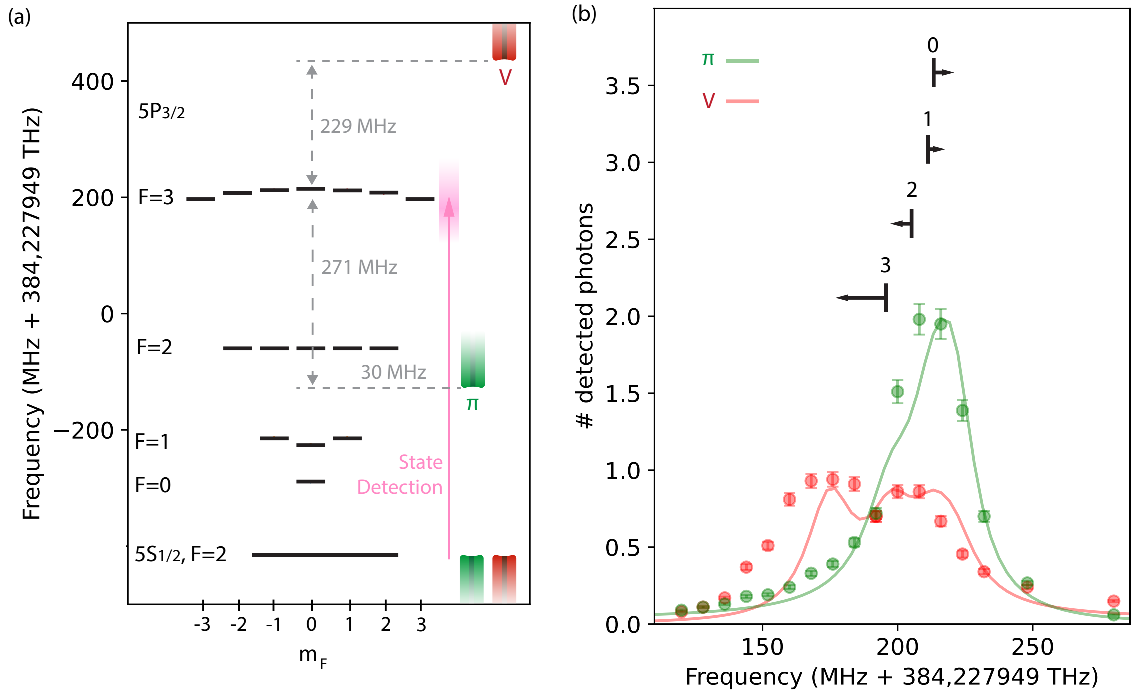


**Figure 3.11.:** (a) Calculated transition frequencies in $^{87}$Rb in the presence of a 852 nm standing wave trap and SC detuning for both eigenmodes. The SD beam is scanned over the cycling transition $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F = 3$. For this calculation, known $^{87}$Rb spectroscopy values from literature are assumed [89], and equations 3.15-3.19 are used to estimate the AC Stark shifts. A power of 130 mW and a waist of 10.5 µm are assumed. The numerical results can be found in Appendix A. (b) Measured (dots) and simulated (solid lines) spectra of the $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F = 3$ transition for photons emitted into the two cavity eigenmodes. Using a set of waveplates and a polarizing beam splitter (PBS), the two modes are mapped onto separate SNSPDs, allowing for independent detection. The observed spectral separation arises from mode-dependent coupling strengths, which vary with the atomic Zeeman sublevels in the $5S_{1/2}, F = 2$ and $5P_{3/2}, F = 3$ manifolds. This leads to differential AC Stark shifts of the Zeeman states induced by the SC. These are depicted in the upper part of the plot, where the labels represent the absolute value of $m_F$.

$$\text{Fidelity} = \min\left(\sum_{n=0}^{\text{threshold}} p_{F=1}(n),\, 1 - \sum_{n=0}^{\text{threshold}} p_{F=2}(n)\right) \tag{3.22}$$

where $p_{F=1}(n)$ and $p_{F=2}(n)$ are the probabilities of detecting $n$ photons when having the atom in $F = 1$ or $F = 2$, respectively. By setting the threshold to $n = 1$, the SD fidelity is maximized to the value 0.989(2). Fig. 3.12b show the fidelity and the lifetime of the atom in the trap as a function of the SD duration A possible approach for reaching a higher fidelity is implementing a SD sequence with duration conditioned on the photon counts, in which the SD beam is interrupted as soon as a photon number higher than the threshold is detected. This would allow for optimizing the SD frequency without thereby heating the atoms. This approach is currently being investigated in the laboratory.

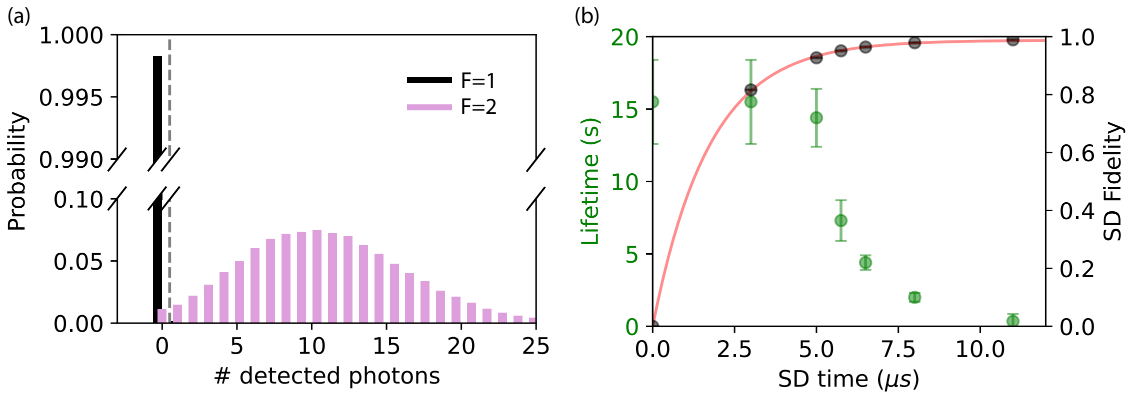


**Figure 3.12.:** (a) Probability distribution of detected photons during SD for 7.5 µs for an atom prepared in the state $F = 1$ (black bars) and $F = 2$ (pink bars). The vertical dashed line represents the threshold. For this measurement, an SD power of 170 nW and a frequency of 384.228109 THz are used. (b) Mean lifetime in the trap (green) and SD fidelity (black) as a function of the SD duration. Even though longer SD durations provide a higher number of photons detected, yielding a higher SD fidelity, the cycling absorption and emission of photons heat the atom, causing a shorter lifetime in the trap - here the SD is performed with a repetition rate of 450 Hz. For this reason, the chosen working point is 7.5 µs, for which we measure a SD fidelity of 0.989(2). The SD fidelity data points can be well fitted with the function $F = p_{F=2}(n > 0) = F_{max}\,(1 - e^{-r \cdot t})$ (red solid line), which yields $r = 0.577(7)$ photon/µs and a maximum fidelity of $F_{max} = 0.988(2)$.

## 3.10. Atomic ground-state manipulation

In the ground state $5S_{1/2}$ there are a total of eight Zeeman states. The two hyperfine states $F = 1$ and $F = 2$ are separated by about 6.8 GHz, and the Zeeman states of a hyperfine state are coupled to those of the other one by the magnetic dipole moment. Consequently, it is possible to drive transitions between $F = 1$ and $F = 2$ by shining an MW magnetic field resonant with the atomic transitions. This is accomplished by employing a square self-resonant loop antenna with circumference $c/6.8$ GHz $= 4.4$ cm. This antenna is placed in the vacuum chamber about 1 cm away from the atom location (distance from the antenna's front side) and consists of a copper wire coated with a *Kapton* layer to guarantee the electrical insulation.

To generate short MW pulses with sufficient power, the setup employs a MW signal circuit, illustrated in Fig. 3.13a. The circuit begins with a MW signal generator (1) *Rohde & Schwarz SMA 100B*, which is locked to a highly stable 10 MHz reference signal. The signal produced by the source is then mixed with a RF signal in the range of 30 MHz (provided by the experimental control hardware) by a *Polyphase Microwave SSB 4080A* (2). This allows us to rapidly change the frequency of the MW signal sent to the atom, and to have control over the MW phase, both necessary features for the experiment performed in Chapter 5. Rectangular MW pulses lasting several microseconds are produced using an electronic switch (3) *Mini-Circuits ZFSWA2-63DR+*. These pulses are then amplified by two cascaded amplifiers: a +26 dB gain amplifier (4) *Minicircuits ZVA-183-S+* and a high-power amplifier (5), *Microwave Amps Ltd AM53-6.5-7-37-37*, providing a gain of +37 dB and a maximum output power of +37 dBm. Due to possible impedance mismatches at the MW antenna and between the circuit components, some power reflection is expected. To protect the amplifier and the source from damage caused by reflected signals, MW circulators (6) are placed immediately after the two. The amplified signal is then routed to the antenna (7).

In order to find out the ground state transition frequencies, MW spectroscopy measurements are performed. In the presence of an external magnetic field, the Zeeman states lift their degeneracy and split according to Eq. 3.20. Fig. 3.13b depicts the typical level structure of the $5S_{1/2}$ manifold. Here, each arrow represents the allowed magnetic dipole moment transitions, and the colors represent their frequency. Due to the equal but opposite shift of the levels $F = 1, m_F = 1(-1)$ and $F = 2, m_F = 1(-1)$, the transitions $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 1(-1)$ and $F = 1, m_F = 1(-1) \leftrightarrow F = 2, m_F = 0$ are degenerate (this is represented by the choice of the same color for these transitions).

The experimental sequence of a MW spectroscopy measurement starts by cooling the atom for 500 µs. Subsequently, the atom is optically pumped in the $F = 1$ hyperfine state or in the $F = 1, m_F = 0$ Zeeman state - see section 3.8. A MW pulse of duration 450 µs is applied, transferring the population from the initial state into $F = 2$ if close to resonance. Eventually, the hyperfine atomic population is measured via cavity-enhanced fluorescence SD for 7.5 µs - see section 3.9. By scanning the RF-signal frequency in the mixer, we can measure the population transferred in $F = 2$ as a function of the MW-signal frequency applied to the atom. Fig. 3.13c and d show the resulting MW spectrum for an atom prepared in the $F = 1$ hyperfine state or in the $F = 1, m_F = 0$ Zeeman state, respectively. In the first case, seven spectral peaks are visible, which correspond to the nine ground-state transitions, two of which are degenerate. In the second case, only three peaks are visible, which correspond to the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$, $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = \pm 1$ transitions.

From the background level of the spectrum, one can infer that optical pumping into the $F = 1$ state achieves an efficiency of 99.7(3)%, while pumping into the specific Zeeman sublevel $F = 1, m_F = 0$ reaches about 97(1)%. By fitting Lorentzian functions to the spectral peaks, a mean Zeeman splitting of 99.5(8) kHz is extracted, which corresponds to a magnetic field strength of 142(1) mG, as calculated using Eq. 3.20.

In the absence of a static magnetic field, there is no Zeeman splitting, and only one peak is present. This property is used to cancel residual environmental magnetic fields by adjusting compensating fields until only one peak remains. Once this condition is achieved, a well-defined static magnetic field is applied along the LC axis. This is done by setting precisely the current in three pairs of Helmholtz coils, positioned along the vertical, the LC, and the KC axis. The current is provided by a self-made, highly stable current generator.

Once the ground-state frequencies are known, it is possible to coherently manipulate the atomic population in the ground state. By tuning the MW on resonance with any transition, we can drive Rabi oscillations [143] between two ground state, and the evolution of the population in the Zeeman state of $F = 2$ can be modeled as [144]

$$P_2 = P_1 \cdot \frac{1}{2}\left(1 - \cos(2\pi\Omega t) \cdot e^{-\chi t}\right) \tag{3.23}$$

where $P_1$ is the initial population after OP in the state $|F = 1, m_F = 0\rangle$, $\Omega$ is the Rabi

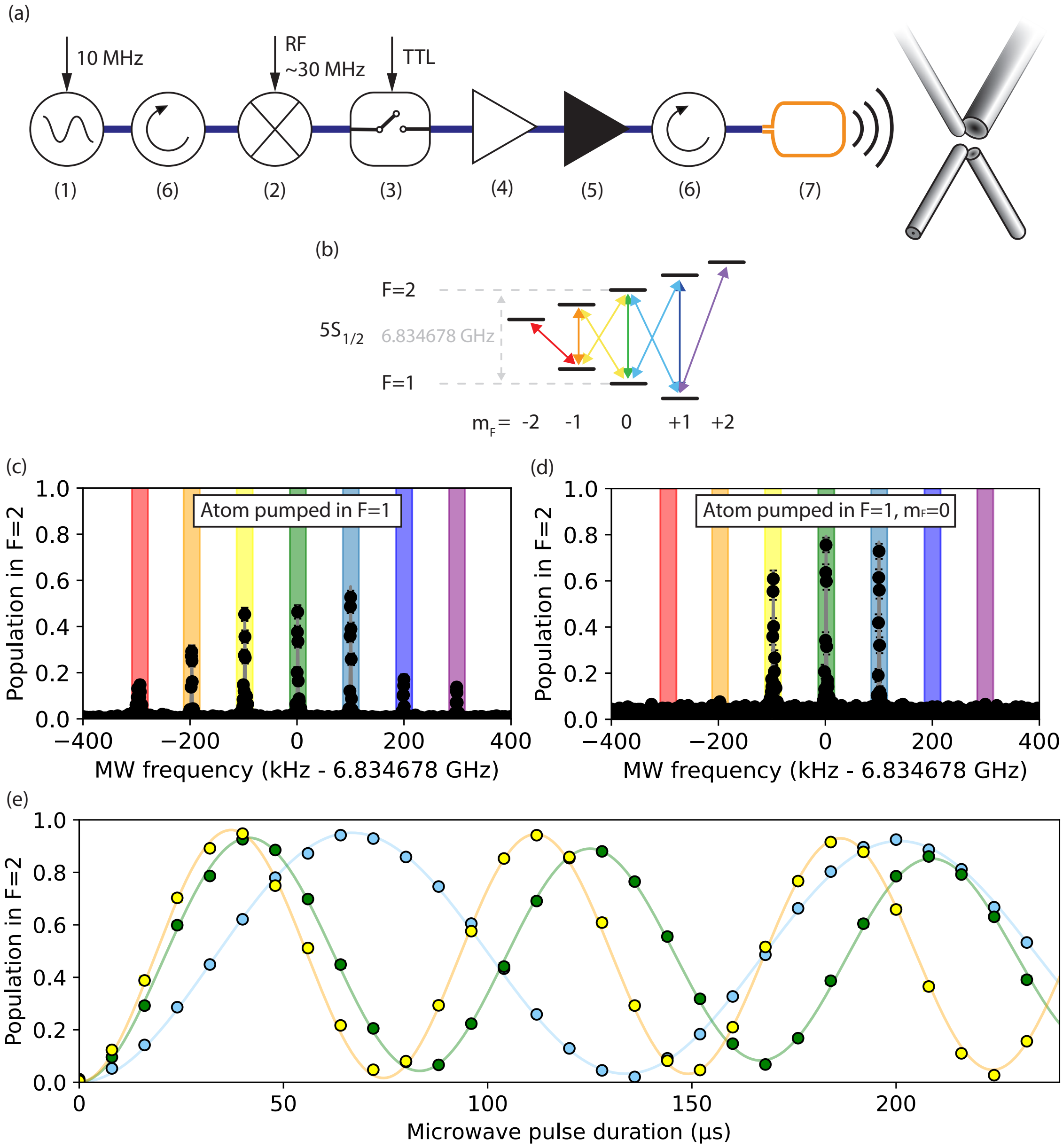


**Figure 3.13.:** (a) MW signal circuit used for the MW spectroscopy and the Rabi oscillation measurements. The elements employed are labeled and described in the main text. These include the following: (1) MW source, (2) signal mixer, (3) switch, (4-5) amplifiers, (6) circulators, and (7) the MW antenna. (b) Energy structure of $5S_{1/2}$ in the presence of a static magnetic field. The arrows indicate the magnetic dipole moment ground state transitions. Each color corresponds to a different frequency. Plots (c) and (d) show the resulting MW spectra when the atom is initialized in $F = 1$ and in $F = 1, m_F = 0$, respectively. The colored windows indicate the transitions depicted in (b). When pumping the atom in $F = 1, m_F = 0$, only three spectral peaks are visible, corresponding to the transitions $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = -1$ (yellow), $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ (green), and $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 1$ (light blue). Plot (e) shows the Rabi oscillations for these three transitions. The solid lines represent the fit of Eq. 3.23 to the data points. Here, the same color labeling as for (b), (c), and (d) is used.

frequency, which is the oscillation frequency and is determined by the transition's dipole moment and by the magnetic field strength, and $\chi$ is the damping rate. The sinusoidal term of Eq. 3.23 describes the coherent transfer of population between the two states, whereas the exponential factor represents the dissipative mechanism that damps the oscillations and brings the system into a steady state.

The experimental sequence of a Rabi oscillation measurement is very similar to the one used for the MW spectroscopy, with the exception that the MW frequency is kept constant and the MW pulse duration is scanned. Fig. 3.13c shows the resulting Rabi oscillations for an atom prepared in $F = 1, m_F = 0$ and transferred to the states $F = 2, m_F = -1$ (yellow), $F = 2, m_F = 0$ (green) or $F = 2, m_F = 1$ (light blue). Fitting Eq. 3.23 to the data yields a mean initial population of 0.983(7), Rabi frequencies of $\Omega_{0\leftrightarrow-1}$ = 13.41(1) kHz, $\Omega_{0\leftrightarrow0}$ = 11.97(1) kHz, and $\Omega_{0\leftrightarrow+1}$ = 7.46(1) kHz and damping rates of $\chi_{0\leftrightarrow-1}$ = 460(70) Hz, $\chi_{0\leftrightarrow0}$ = 1.14(8) kHz, and $\chi_{0\leftrightarrow-1}$ = 520(70) Hz. Driving these transitions coherently is essential for the experiment performed in chapter 5. The state evolution's coherence is mainly limited by magnetic field fluctuations and atomic temperature.

# 4. Cavity mediated photon-pairs with single atoms

The contents of this chapter were first published in [65]:

**Two-cavity-mediated photon-pair emission by one atom**
Gianvito Chiarella, Tobias Frank, Pau Farrera, and Gerhard Rempe
*Optica Quantum* **2**, 346–350 (2024).

This chapter explores the generation of photon pairs from a single atom coupled to two optical cavities. The main advantage of this source with respect to other sources lies in the individual control of each photon through the parameters of the respective cavities. This chapter presents a comprehensive investigation of this process, combining theoretical modeling with experimental implementation. We first introduce a theoretical model capturing the dynamics of the atom–cavity system. The model incorporates the relevant atomic-level structure and captures the influence of relevant parameters such as detuning, laser intensity, and cavity coupling strengths.

The experimental sequence is then outlined, including state preparation, photon detection, and a discussion of ionization effects that may limit performance. Key results are presented, including the measured efficiency as a function of powers and detunings, the photons' detection probability as a function of time, and their temporal cross-correlation. Some results presented here are different from those reported in [65], due to some improvements that have been implemented after the publication. In particular, a more accurate calibration of the excitation pulses' power and a better positioning of the atom in the cavities have led to higher emission probability. However, the sequences and experiments executed are the same.

The chapter concludes with a theoretical investigation of the photon indistinguishability and of the strong coupling regime, in which the system exhibits dynamics that is analogous to stimulated Raman adiabatic passage (STIRAP) [145].

## 4.1. Effective Hamiltonian for photon-pair generation

We start by considering a single neutral atom coupled to two independent optical cavities, as depicted in Fig. 4.1a. The atom has three energy levels in a ladder configuration (ground $|g\rangle$, intermediate $|i\rangle$, and excited $|e\rangle$) supporting the two transitions that are coupled to the optical cavities - see Fig. 4.1b. Here, $g_u(g_l)$ is the coupling rate of the cavity that couples to the upper $|i\rangle - |e\rangle$ (lower $|g\rangle - |i\rangle$) transition, $\Delta_u$ ($\Delta_l$) is the cavity detuning to

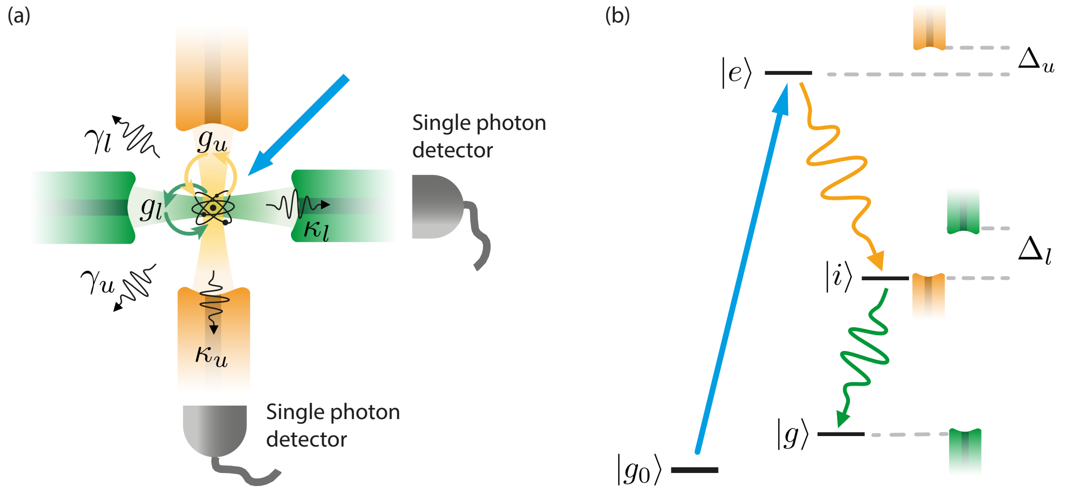


**Figure 4.1.:** (a) Simple illustration of the system employed for the photon-pair production. A single atom is trapped at the intersection of two optical cavities. It couples to the two optical modes with rates $g_u$ and $g_l$, and it decays in free space with polarization decay rates $\gamma_u$ and $\gamma_l$. Photons emitted in the cavities are outcoupled with rates $\kappa_u$ and $\kappa_l$, and are eventually detected by single photon detectors. (b) Level structure utilized. The atom is initialized in the ground state $|g_0\rangle$ and externally driven to $|e\rangle$. The atom undergoes a cavity-mediated decay in the ladder $|e\rangle \rightarrow |i\rangle \rightarrow |g\rangle$, emitting a photon in each resonator.

the upper (lower) atomic transition, $\hat{a}_u$ ($\hat{a}_l$) and $\hat{a}_u^\dagger$ ($\hat{a}_l^\dagger$) are the annihilation and creation operators for the upper (lower) cavity mode, and $\hat{\sigma}_{ie}$ and $\hat{\sigma}_{gi}$ are the atomic operators which describe the transition from $|e\rangle$ to $|i\rangle$ and from $|i\rangle$ to $|g\rangle$, respectively.

The atom is initialized in an additional ground state $|g_0\rangle$, and a drive laser with Rabi frequency $\Omega_D$ and detuning to the transition $\Delta_D$ couples this state to the excited state. The atomic operator that describes the transition from $|g_0\rangle$ to $|e\rangle$ is $\hat{\sigma}_{g_0 e}^\dagger$.

The system's natural Hamiltonian is time-dependent and therefore quite unhandy. By following the derivation in Appendix B, one gets an effective time-independent Hamiltonian given by:

$$\begin{aligned}\hat{H} = g_u(\hat{a}_u^\dagger \hat{\sigma}_{ie} + \hat{a}_u \hat{\sigma}_{ie}^\dagger) + g_l(\hat{a}_l^\dagger \hat{\sigma}_{gi} + \hat{a}_l \hat{\sigma}_{gi}^\dagger) + \Delta_u \hat{a}_u^\dagger \hat{a}_u + \Delta_l \hat{a}_l^\dagger \hat{a}_u + \\ + \Omega_D(\hat{\sigma}_{g_0 e}^\dagger + \hat{\sigma}_{g_0 e}) - \Delta_D \hat{\sigma}_{g_0 e}^\dagger \hat{\sigma}_{g_0 e} - \Delta_D \hat{a}_u^\dagger \hat{a}_u \quad (4.1)\end{aligned}$$

The first four terms describe the coherent dynamics of the ladder system exchanging energy with the two optical modes, whereas the last three terms model the interaction of the atom with the external drive. The dynamics of the system are simulated throughout this work with the Master Equation 2.19, in which the only assumed dissipative processes are the atomic decay in free space with polarization decay rate $\gamma_u$ ($\gamma_l$) for the state $|e\rangle$ ($|i\rangle$) and the cavity decay with rate $\kappa_u$ ($\kappa_l$) for the upper (lower) mode.

## 4.2. Two-photon excitation

The photon-pair production is based on the cascaded emission of the atom from the higher excited state $5D_{5/2}, F = 3$. Since the electrical dipole moment transition between the ground state $5S_{1/2}$ and the excited state $5D_{5/2}$ is prohibited, a non-degenerate two-photon excitation has been used in order to populate the atom into this state. This is based on the simultaneous absorption of two photons coming from two lasers at different wavelengths, whose energy sum equals the energy difference between the states $5S_{1/2}, F = 1$ and $5D_{5/2}, F = 3$ [146, 147]. For this purpose, a 780 nm and a 776 nm laser with a photon detuning of 1 GHz from $5P_{3/2}, F = 2$ - see Fig. 4.2a - are shone on the atom simultaneously from the bottom of the chamber along the vertical direction. These lasers share the same beam path (Fig. 3.6, 23) and are focused onto the atom to a diameter of about 20 µm.

For the experiments performed in Chapter 5, it is important to make the two-photon excitation Zeeman-state-selective by properly calibrating the polarization of the lasers. In particular, the excitation beams are carefully calibrated by means of a set of waveplates (Fig. 3.6, 24) to obtain a linear polarization aligned with the LC-axis at the atom position ($\pi$-polarization). The calibration procedure takes place in two steps and is illustrated in Fig. 4.2b. First, a linear polarizer is inserted between the aspherical lens and the linear polarizer. The 852 nm trap power is monitored with a power meter on the other side of the chamber. At this point, the newly inserted polarizer's angle is adjusted such that the transmitted power is minimized. Since the other polarizer is parallel to the LC axis, the newly inserted polarizer is aligned perpendicularly to the LC axis. In the second step, the power meter is placed behind this polarizer in order to monitor the power of the 776 nm and 780 nm beams, and the set of waveplates is adjusted such that the transmitted power is minimized. At this point, the excitation lasers' polarization is aligned along the LC-axis and the newly inserted polarizer is removed.

## 4.3. Photoionization

The fine atomic levels chosen to realize the atomic ladder are $5D_{5/2} \rightarrow 5P_{3/2} \rightarrow 5S_{1/2}$. The $5D$ state has a binding energy of 0.99 eV, therefore absorbing a 852 nm trap photon with energy $h\nu$=1.45 eV is sufficient to ionize the atom (see Fig. 4.3a), leading to its loss from the trap. From Fig. 4.3a it is possible to see that the choice of $|e\rangle = 4D$ or $|e\rangle = 6S$ would solve this problem, since they have a binding energy of 1.8 eV and 1.7 eV (double check these values), both $> 1.45$ eV. However, the cavity mirror coatings are unsuitable for photons emitted during the decay to $4D \rightarrow 5P$ or $6S \rightarrow 5P$. These have indeed wavelength in the telecom regime, whereas the coatings preserve their highly-reflective properties only over a span of about 80 nm around 790 nm.

While photoionization is undesirable for the photon-pair experiments, it was initially exploited to perform atomic loss spectroscopy of the $5D_{5/2}$ level (Fig. 4.3b). In this measurement the atom's mean lifetime is monitored as a function of the two-photon detuning. This technique allows to identify coarsely the two-photon resonance, however for more precise measurements of the atomic level structure, such as the hyperfine spectroscopy

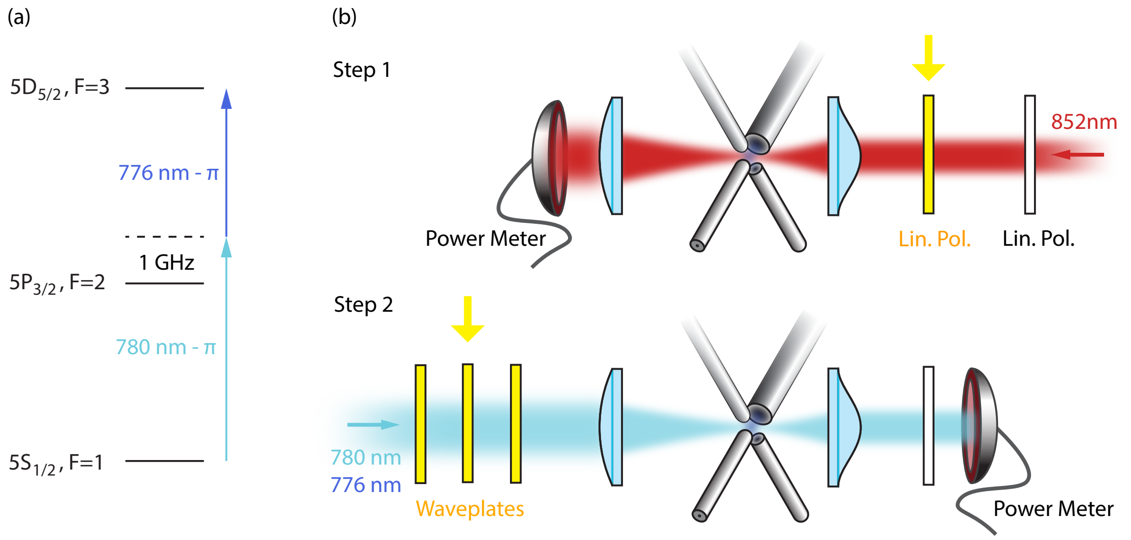


**Figure 4.2.:** (a) Sketch of the employed two-photon absorption process. A 780 nm and a 776 nm $\pi$-polarized laser excite resonantly the two-photon transition $5S_{1/2}, F=1 \leftrightarrow 5D_{5/2}, F=3$. (b) 780 nm and 776 nm lasers' polarization calibration procedure. Step 1: A linear polarizer is inserted and rotated such that the 852 nm laser power is minimized. Step 2: The waveplates set is inserted and rotated such that the excitation laser's power is minimized.

of $5D_{5/2}$, a different technique must be employed - see Section 4.4. Fig. 4.3c shows such atom loss effect when sending excitation pulses every 200ms as a function of the trap beam intensity. We can observe how the lifetime of the atom in the trap decreases when we increase the power of the trap during the atom excitation. For this measurement, the trap intensity is kept constant with intensity $3.36 \times 10^9$ W/m$^2$ for 200ms and is decreased to the values shown in the plot horizontal axis for 800ns during the atom excitation to $5D_{5/2}$. We use the theory described in [148] in order to fit the data:

$$P_{PI} = \eta \left(1 - e^{-\frac{\sigma F \lambda}{hc}}\right) \tag{4.2}$$

where $P_{PI}$ is the photoionization probability, $\eta$ is the excitation efficiency to the excited level, $\sigma$ is the photoionization cross section, $\lambda$ the light wavelength, and $F = \int_{-\infty}^{\infty} I\mathrm{d}t$ is the ionizing fluence related to the light intensity $I$. From the fit we obtain a photoionization cross section of $\sigma = 17(6)$ Mb which is compatible with the photoionization cross section given in the literature for light at 852 nm ($\sigma = 12$ Mb) [148]. In order to prevent the photoionization of the atom, the trap is switched off during the atomic excitation. However, since the atom in that time is not confined along the vertical direction, it can escape the cavities and eventually the cooling region. Because of this, longer switch-off times result in a shorter mean lifetime in the trap. These two concurring effects are summarized in Fig. 4.3d, where the atom loss rate - defined as the inverse of the mean atom lifetime in the trap - is measured as function of the trap-off duration. From this, we infer that a trap-off duration of 800 ns is a good compromise and is used in the following described experiments.

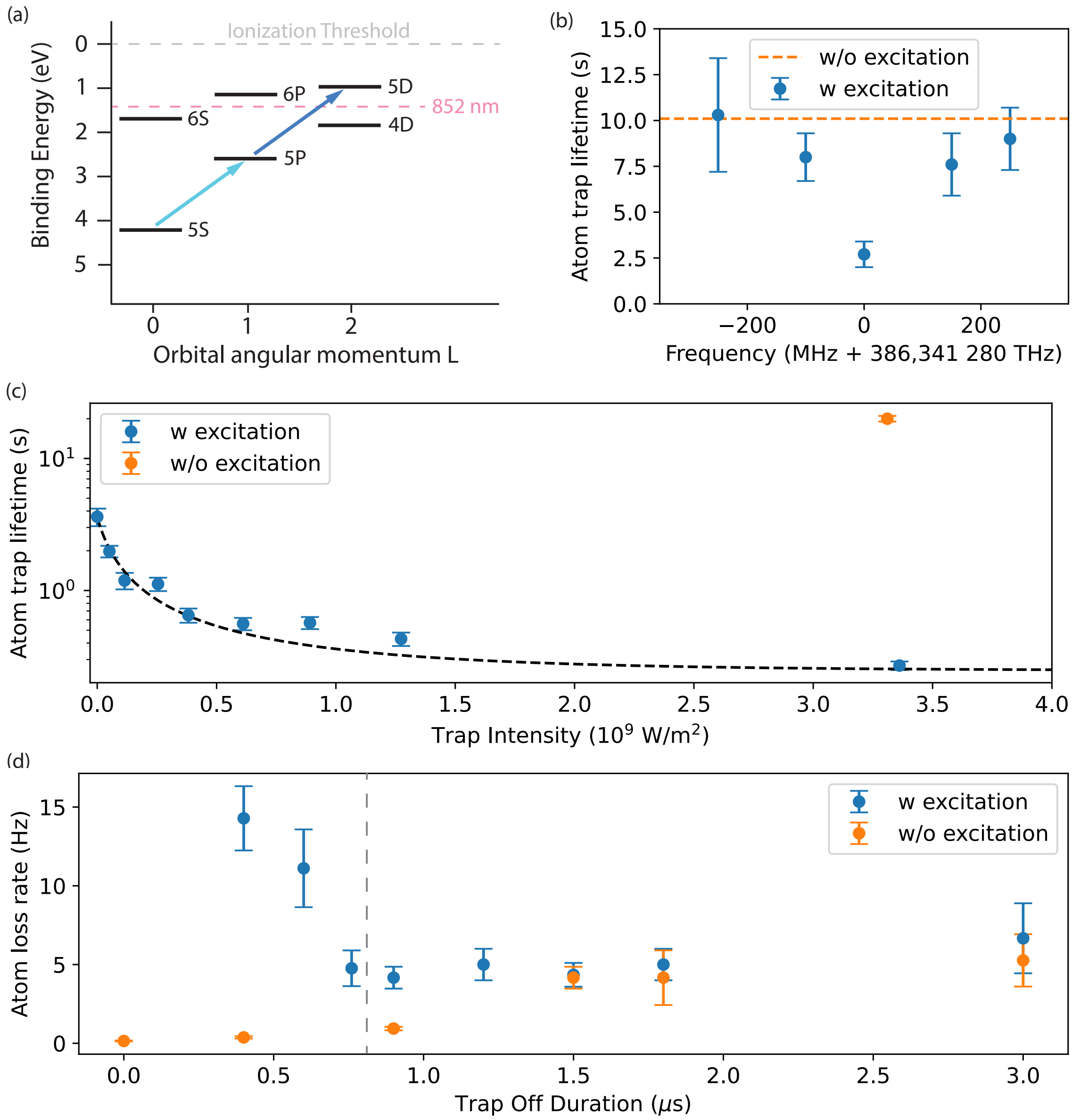


**Figure 4.3.:** (a) Levels binding energy in $Rb^{87}$. Fine and hyperfine structure are not shown. The red dashed line marks the binding energy which can be ionized by the absorption of a 852 nm photon. The atom is ionized by a 852 nm photon if it occupies the levels with binding energy above the dasehd line, such as 6P or 5D. (b) Atom loss spectroscopy of $5D_{5/2}$. For this measurement, the 780 nm excitation laser frequency is left unchanged, while the 776 nm excitation laser is scanned. Around 386,341280 THz the mean atom lifetime in the trap decreases noticeably, indicating the vicinity to the resonance. The dashed line marks the trapped atom lifetime with no excitation. (c) Trapped atom lifetime as a function of the trap beam intensity during the atom excitation (blue dots). The black dashed line shows a fit corresponding to the model of Eq. 4.2. The orange dot represents the trap atom lifetime with no atomic excitation. (d) Atom loss rate as a function of the trap-off duration, measured with (blue dots) and without (orange dots) atomic excitation. The vertical dashed line marks the switch-off duration of 800 ns chosen for the successive experiments. A repetition rate of 1 kHz has been used for this measurement.

## 4.4. Experimental realization

The photon-pair generation experiment starts by trapping the atom at the intersection of the two crossed cavities. The utilized experimental sequence is summarized in Fig. 4.4a. After cooling the atom for 5 ms, it is optically pumped in the state $|g_0\rangle = |5S_{1/2}, F = 1, m_F = 0\rangle$. The 852 nm trap is ramped down and switched of for 800 µs and the 2-photon excitation is applied to drive the atom in the state $|e\rangle = |5D_{5/2}, F = 3, m_F = 0\rangle$. This consists of two simultaneous laser pulses with a duration of 60 ns that couple the transitions $|g_0\rangle \leftrightarrow |5P_{3/2}, F = 2, m_F = 0\rangle$ and $|5P_{3/2}, F = 2, m_F = 0\rangle \leftrightarrow |e\rangle$ - see Fig. 4.4b for the precise pulse shape monitored with photodiodes (PDs).

These lasers have a photon detuning of 1 GHz from $|5P_{3/2}, F = 2, m_F = 0\rangle$ and are both $\pi$-polarized in order to drive selectively the $|5S_{1/2}, F = 1, m_F = 0\rangle \rightarrow |5D_{5/2}, F = 3, m_F = 0\rangle$ transition. The LC is resonant with the $|5D_{5/2}, F = 3\rangle \leftrightarrow |5P_{3/2}, F = 2\rangle$ transition at 776 nm. Since the cavity emission of photons with polarization parallel to the cavity-axis is forbidden, the LC mediates only the decays $|e\rangle \rightarrow |i\rangle = |5P_{3/2}, F = 2, m_F = \pm 1\rangle$. The KC $\pi$-mode is resonant with the $|5P_{3/2}, F = 2\rangle \leftrightarrow |5S_{1/2}, F = 2\rangle$ transition at 780 nm, mediating the decays $|i\rangle \rightarrow |g\rangle = |5S_{1/2}, F = 2, m_F = \pm 1\rangle$. After a delay of 800 ns, the trap is ramped up and the sequence is repeated. The atomic levels and the cavity configuration employed are depicted in Fig. 4.4c. Given this choice of atomic levels and transitions, the atom–cavity parameters for the upper cavity (LC) are $(g_u, \gamma_u, \kappa_u) = (6, 0.33, 31.7)$ $\times 2\pi$ MHz and those for the lower cavity (KC) are $(g_l, \gamma_l, \kappa_l) = (21.9, 3, 61.5)$ $\times 2\pi$ MHz. From these parameters and Eq. 2.5, we derive cooperativities of $C_u = 1.7$ for the upper transition and $C_l = 1.33$ for the lower one.

The characterization and optimization of our source start by performing a $5D_{5/2}$ spectroscopy by scanning one of the two excitation frequencies and keeping the other one and the cavities constant. The two-photon electric-dipole selection rules, summarized in Table 4.1, forbid the transitions $|5S_{1/2}, F = 1, m_F = 0\rangle \rightarrow |5D_{5/2}, F = 4, m_F = 0\rangle$ since $\Delta F > 2$ and $|5S_{1/2}, F = 1, m_F = 0\rangle \rightarrow |5D_{5/2}, F = 2, m_F = 0\rangle$ because they have opposite parity, defined as $(-1)^F$. Because both excitation lasers are $\pi$-polarized, only the transitions $|5S_{1/2}, F = 1, m_F = 0\rangle \rightarrow |5D_{5/2}, F = 1, m_F = 0\rangle$ and $|5S_{1/2}, F = 1, m_F = 0\rangle \rightarrow |5D_{5/2}, F = 3, m_F = 0\rangle$ are electric-dipole allowed.

The allowed and prohibited transitions are depicted in Fig. 4.5a. The resulting spectrum is shown in Fig. 4.5b, where it is possible to see resonance peaks corresponding only to the allowed transitions. The resonance peak which corresponds to state $F'' = 3$ shows a higher photon-detection efficiency compared to the resonance peak of state $F'' = 1$, making the state $|5D_{5/2}, F = 3, m_F = 0\rangle$ the optimal choice for an efficient source. This is to be expected from the electric-dipole coupling strengths of the $5P_{3/2} \leftrightarrow 5D5/2$ transitions, whose Clebsch-Gordan coefficients have been calculated and reported in Appendix D.

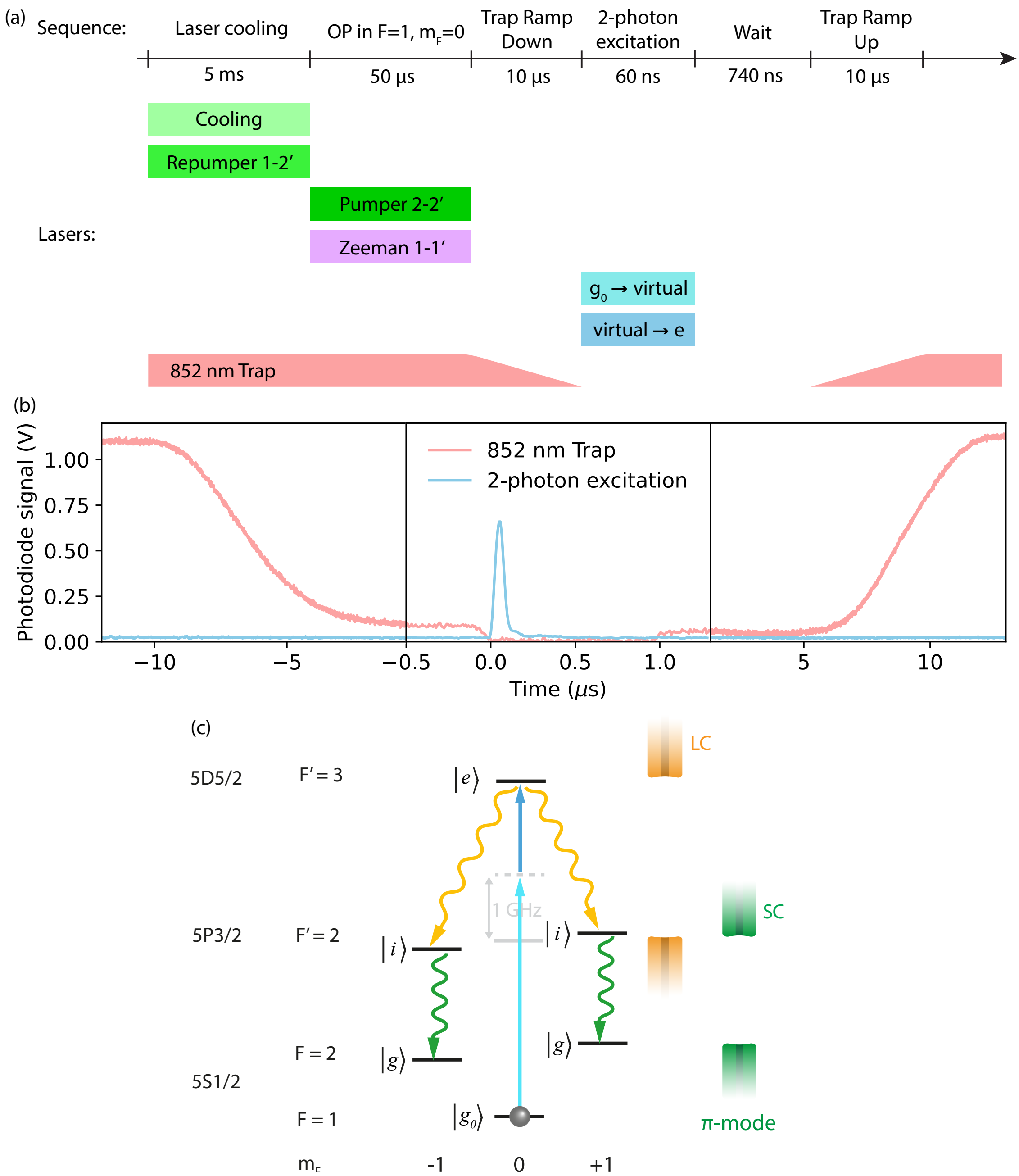


**Figure 4.4.:** (a) Experimental sequence used to generate photon-pairs from a single atom. (b) Signal recorded by PDs of the atomic excitation pulses (blue line) and of the 852 nm trap (red line) during trap ramp-down and switch-off. (c) Atomic levels and cavity configuration employed for the generation of photon-pairs. After having initialized the atom in the ground state $|g_0\rangle = |5S_{1/2}, F = 1, m_F = 0\rangle$, it is excited to $|e\rangle = |5D_{5/2}, F = 3, m_F = 0\rangle$. The atom undergoes a cavity-mediated decay in the ladder $|e\rangle \to |i\rangle = |5P_{3/2}, F = 2, m_F = -1\rangle \to |g\rangle = |5S_{1/2}, F = 2, m_F = -1\rangle$ or $|e\rangle \to |i\rangle = |5P_{3/2}, F = 2, m_F = 1\rangle \to |g\rangle = |5S_{1/2}, F = 2, m_F = 1\rangle$, emitting a photon in each resonator. Both decay paths are equally likely to happen.

| **General selection rules** | | |
|---|---|---|
| $\lvert\Delta F\rvert \leq 2$ | | |
| same parity | | |
| $F_i + F_f$ = integer | | |
| **Polarization selection rules** | | |
| Photon 1 | Photon 2 | Forbidden Transitions |
| $\sigma^+$ | $\sigma^-$ | $\Delta m \neq 0$ |
| $\sigma^+$ | $\pi$ | $\Delta F : 0 \leftrightarrow 0$ ; $\Delta m \neq 1$ |
| $\sigma^-$ | $\pi$ | $\Delta F : 0 \leftrightarrow 0$ ; $\Delta m \neq -1$ |
| $\pi$ | $\sigma$ | $\Delta F : 0 \leftrightarrow 0$ ; $\Delta m \neq \pm 1$ |
| $\pi$ | $\pi$ | $\Delta F : 0 \leftrightarrow 1$ ; $\Delta m \neq 0$ |
| $\sigma^+$ | $\sigma^+$ | $\Delta F : 0 \leftrightarrow 1$ , $0 \leftrightarrow 0$, $1/2 \leftrightarrow 1/2$ ;$\Delta m \neq 2$ |
| $\sigma^-$ | $\sigma^-$ | $\Delta F : 0 \leftrightarrow 1$ , $0 \leftrightarrow 0$, $1/2 \leftrightarrow 1/2$ ; $\Delta m \neq -2$ |

**Table 4.1.:** Two-photon electric-dipole selection rules. The notation $i$ and $f$ labels the initial and the final states, considered here to be respectively the states with lower and higher energy. Here $\Delta F = F_f - F_i$, $\Delta m = m_f - m_i$, $\pi$ is the light polarized along the quantization axis, $\sigma$ is the light linearly polarized orthogonally to the quantization axis, and $\sigma^+$ and $\sigma^-$ are the circularly polarized light. This table is adapted from [147].

Further characterizations of the source include the study of the photon emission dependence as a function of the cavity detunings. Fig. 4.5c shows the photon efficiency as a function of the drive beam and LC detuning. In this measurement, the detuning of the driving field and of the LC are changed by the same amount. Numerical simulations based on the master equation (Eq. 2.19), using the Hamiltonian in Eq. 4.1 and known dissipation channels, capture the overall trend of the measured data (dots), as shown by the simulated curves (solid lines), despite deviations in absolute values. A similar trend is expected when scanning the detuning of the SC. We study also the efficiency dependence on the detuning of the two cavities when they are changed by the same amount but with opposite sign, such that the two photon resonance is preserved. In this case we only simulate this situation and the result is plotted in Fig. 4.5d. The maximal efficiency is obtained for zero detuning.

Finally, we optimize the photon-pair generation efficiency by performing a two-dimensional scan over the powers of the two excitation beams. This measurement identifies the optimal power settings that maximize the emission probability, as shown in Fig. 4.5e. The resulting optimal values are then fixed for all subsequent experiments.

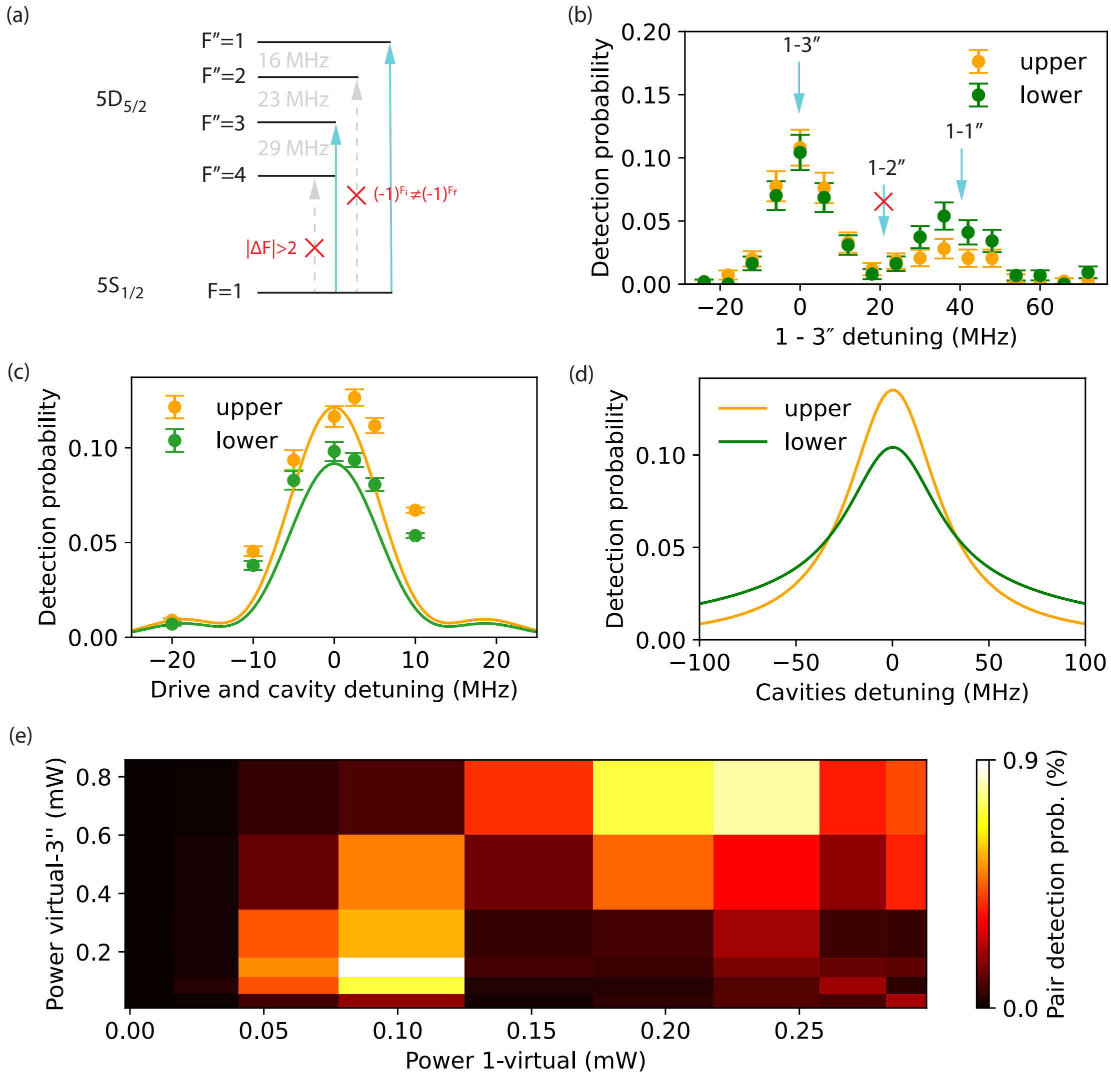


**Figure 4.5.:** Parameters calibration. (a) Sketch of the electric-dipole allowed and forbidden two-photon transitions according to the table 4.1. An atom in the state $5S_{1/2}, F = 1$ can be transferred only in the states $5D_{5/2}, F'' = 1$ or $F'' = 3$ via electric-dipole interaction. This can be seen in (b), where the detection probability of both photons is shown as a function of the drive detuning from the 1-3" transition. (c) The photons' detection probability is monitored as a function of the upper cavity and drive detuning, set to be equal for this measurement. The data points follow the trend of the simulated curves (solid lines). (d) Simulated detection probability as a function of the upper and lower cavity detunings, set to be equal but opposite to preserve the two-photon resonance. (e) The probability of detecting a photon pair as a function of both excitation beams' power. The photon-pair efficiency is optimized by the powers $P_{|g_0\rangle\to|virt\rangle} = 80\,\mu\mathrm{W}$ and $P_{|virt\rangle\to|e\rangle} = 150\,\mu\mathrm{W}$.

Fig. 4.6 shows the in-fiber photon emission efficiencies achieved. We observe that we generate photon pairs with an in-fiber efficiency of 22(1)% and single photons with in-fiber efficiencies of $\eta_u = 43(3)\%$ in the upper cavity and $\eta_l = 34(2)\%$ in the lower cavity. The generation efficiency of a photon pair is greater than the product of the single photons' efficiencies ($\eta_u \cdot \eta_l = 15(1)\ \%$), indicating that the photons are correlated. Another figure of merit is the probability that a photon is emitted into a cavity fiber once a photon was detected from other one cavity. This is especially important for characterizing the performance of the atom-photon entanglement described in chapter 5. We measure the probability that a photon is emitted into LC (KC) once a photon from KC (LC) is detected to be $\eta_{u|l} = 67(3)\%$ ($\eta_{l|u} = 52(2)\%$).

The inefficiencies include the cavity–fiber mode matching (measured $\eta_{mm}^{LC} = 0.94$ and $\eta_{mm}^{KC} = 0.81$, the cavity outcoupling efficiency ($\eta_{oc}^{LC} = 0.79$ and $\eta_{oc}^{KC} = 0.85$), and the emission into the cavity mode considering the finite cooperativity. This is calculated by using Eq. 2.8, which yields $\eta_C^u = 0.62$ and $\eta_C^l = 0.73$. The three aforementioned efficiencies are inherently constrained by the cavity design and cannot be optimized independently. For instance, the cavity-fiber mode matching efficiency $\eta_{mm}$ is maximized when the cavity mode waist coincides with the fiber mode field diameter at the outcoupling mirror. However, this configuration increases the mode waist at the cavity center—where the atom is trapped—thereby reducing the cooperativity and the photon emission efficiency into the cavity mode, $\eta_C$. This trade-off could be mitigated by employing fiber-integrated mode-matching optics, which enable both high mode-matching and a small mode waist at the cavity center [149]. Another limiting factor is the intracavity loss, mainly due to scattering and absorption at the fiber mirror surfaces (80 ppm and 50 ppm), and residual transmission through the high-reflectivity mirrors (10 ppm each). While increasing the transmission of the outcoupling mirrors (340 ppm) enhances the outcoupling efficiency, it simultaneously reduces the cooperativity. To address this, minimizing intracavity losses — e.g., by improving the surface quality of the fiber mirrors during fabrication — is essential. Moreover, the choice of stronger transitions would increase the cooperativity and therefore the emission efficiency into the cavity mode.

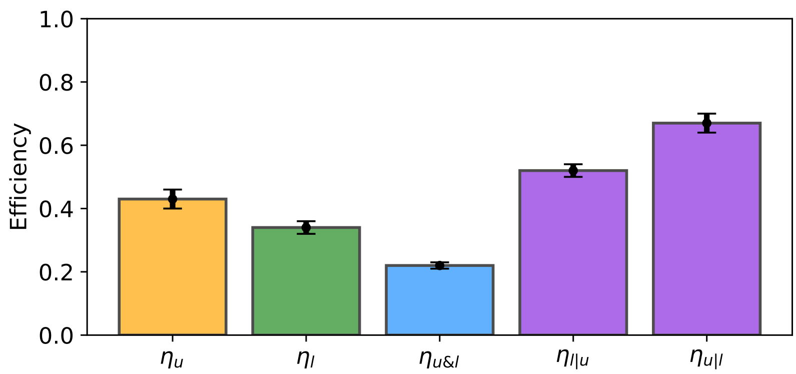


**Figure 4.6.:** In-fiber efficiency to generate the upper photon ($\eta_u$), the lower photon ($\eta_l$), the photon-pair ($\eta_{u\&l}$), the upper photon upon detection of the lower photon ($\eta_{u|l}$), and the lower photon upon detection of the lower photon ($\eta_{l|u}$).

We observe autocorrelation values of $g_u^{(2)}(0) = 0.05(1)$ and $g_l^{(2)}(0) = 0.02(1)$ for the photon fields in the upper and lower cavities, respectively, as shown in Fig. 4.7a. These low values indicate high single-photon purity in both modes. The residual deviation from ideal antibunching is primarily due to the occasional presence of multiple atoms in the cavities. Although we trap mainly single atoms, the probability of trapping more than one atom is non-negligible. Based on the measured $g^{(2)}(0)$ values, we estimate average an atom numbers of 1.04(1) in the upper cavity and 1.02(1) in the lower cavity.

Studying the temporal and correlation properties gives important insights into the behavior of our photon-pair source. Fig. 4.7b shows the time-resolved photon counts from both the upper and lower cavities following the two excitation pulses. Due to the significantly longer lifetime of the upper state $|e\rangle$ compared to the intermediate state $|i\rangle$, the photon durations are mainly set by the Purcell-enhanced decay of $|e\rangle$ (the free-space lifetime of $5D_{5/2}$ is $\tau_{|e\rangle} = 231(8)$ ns [150]). The observed photon lifetimes are $\tau_u = 60(2)$ ns and $\tau_l = 61(2)$ ns for the upper and lower cavities, respectively. Numerical simulations predict the same values ($\tau_u = 60$ ns), confirming the consistency of the model. The solid lines in Fig. 4.7b represent the simulated temporal shape. Alternatively, the Purcell-enhanced decay time can be estimated from the system's cooperativity and the free space decay time as $\tau_{|e\rangle\mathcal{P}} = \tau_{|e\rangle}/(2C_u + 1)$ (derived from Eq. 2.7), yielding a value of 52(3) ns. We attribute the difference between this estimate and the measured value to the finite lifetime of the intermediate state, which weakens the coherent dynamics between the excited and intermediate states. This reduces both the emission rate and the efficiency of photon generation in the upper cavity.

Fig. 4.7c shows the two-photon-coincidence histogram as a function of their delay time. We can observe a correlation peak with different rise and fall times . The rise time ($\tau = 2.7(2)$ ns) is determined by the optical lifetime of the upper cavity ($\tau = 2.5$ ns) while the fall time ($\tau = 7.6(4)$ ns) is given by the Purcell-enhanced decay from state $|i\rangle$ (in free space $\tau_{|i\rangle} = 26.2$ ns), matching well the expected value $\tau_{|i\rangle\mathcal{P}} = \tau_{|i\rangle}/(2C_l + 1) = 7.1(2)$ ns. Fig. 4.7d shows the same kind of histogram for photons detected in different trials. Since the photons are uncorrelated, the temporal width is determined by the duration of both upper and lower cavity single-photon wave-packets. Because the temporal width of the two-photon cross-correlation peak is significantly shorter, the two-photon emission is strongly temporally correlated.

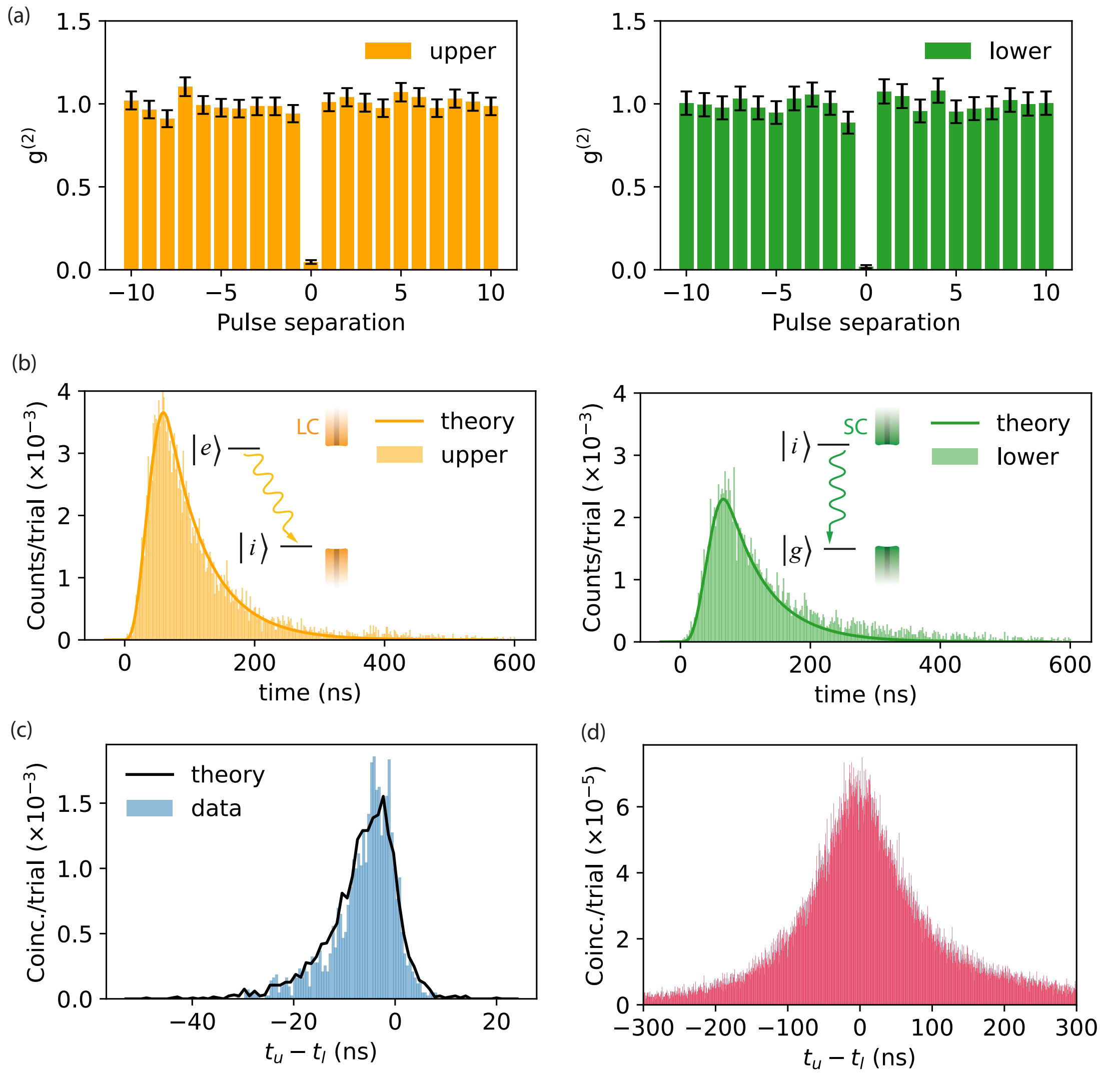


**Figure 4.7.:** Photon-pair properties. (a) Auto-correlation function for the photons emitted from the upper (orange) and the lower (green) cavities, where we observe autocorrelation values of $g_u^{(2)}(0) = 0.05(1)$ and $g_l^{(2)}(0) = 0.02(1)$, respectively, indicating high single-photon purity in both modes. (b) Time histograms showing the photon counts detected at the outputs of the upper (orange) and lower (green) cavities after exciting the atom to state $|e\rangle$ at time zero. The solid lines show the photons' temporal profile predicted by the theoretical simulation. (c) Coincidence time histogram between photons emitted in the same trial. The solid black line indicates the curve predicted by the theoretical model in a quantum Monte Carlo simulation. (d) Coincidence time histogram between photons emitted in different trials. For this data, the correlation peak has been arbitrarily centered to $t_u - t_l = 0$.

## 4.5. Photons' indistinguishability

The temporal width of the two-photon cross-correlation peak is significantly shorter than the duration of both upper and lower cavity single-photon wave packets, showing that the two-photon emission is temporally correlated. In other words, the emission time of one photon tells information about the emission time of the other photon. This implies that two upper (or lower) photons generated by two identical sources are not identical, since their temporal wavepackets (4.7c) are in general temporally displaced, unless they are synchronized by measuring the emission time of the other photon - which in turn would require implementing fast optics and electronics to comply with the nanosecond scale time shift.

To quantify the photon's degree of indistinguishability, we simulate the scenario in which two photons interfere on a beam splitter (Fig. 4.8a) and calculate the resulting Hong-Ou-Mandel (HOM) interference visibility for both cavity fields by using Eq. 2.28. Having two independent cavities coupled to the atom allows us, in principle, to independently tune the atom-cavity coupling parameters for each transition. By doing so, a set of parameters can be found for which the upper and lower cavity fields are generated with no temporal correlation. Fig. 4.8b shows the simulated $V_{HOM}$ of the upper photon as a function of the coupling rates of the cavities $g_u$ and $g_l$. The other atom-cavity parameters are set to the values used in the experiment and a pulse with a duration of 7 ns is considered.

A $V_{HOM}$ smaller than unity indicates that the photons are not perfectly indistinguishable. The interference visibility is determined by the temporal width of the two-photon cross-correlation compared to the temporal width of the single photon wave packets. In other words, if the temporal width of the cross-correlation between two photons emitted in the same trial is the same as the one between photons emitted in different trials, the emission of a photon does not tell any information about the other one. Since the emission rate in each cavity is controlled by the respective coupling strengths, the ratio between $g_u$ and $g_l$ is a critical factor. As shown in Fig.4.8b, for a fixed $g_u$, an increasing $g_l$ corresponds to a decreasing $V_{HOM}$. This is because the higher $g_l$, the faster the emission in the lower cavity will take place, giving rise to a temporal correlation between the two photons.

By properly tuning the coupling strengths, one can achieve a HOM interference visibility for the two fields close to unity, as indicated by the green cross in Fig. 4.8b. This point corresponds to the cavities couplings that provide high indistinguishability ($V_{HOM} \geq 95\%$) and photon-pair efficiency comparable to our measurements. Fig.4.8c and Fig.4.8d show the simulated intensity profile and the cross-correlation corresponding to this point.

## 4.6. Theoretical investigation of the strong coupling

Being able to independently tune the cavity parameters is useful both for manipulating the properties of the fields generated and for investigating the atom-field dynamics in different regimes. In particular, one can extend the study of this system to the regime of strong atom-cavity coupling (in which $g > (\kappa,\gamma)$).

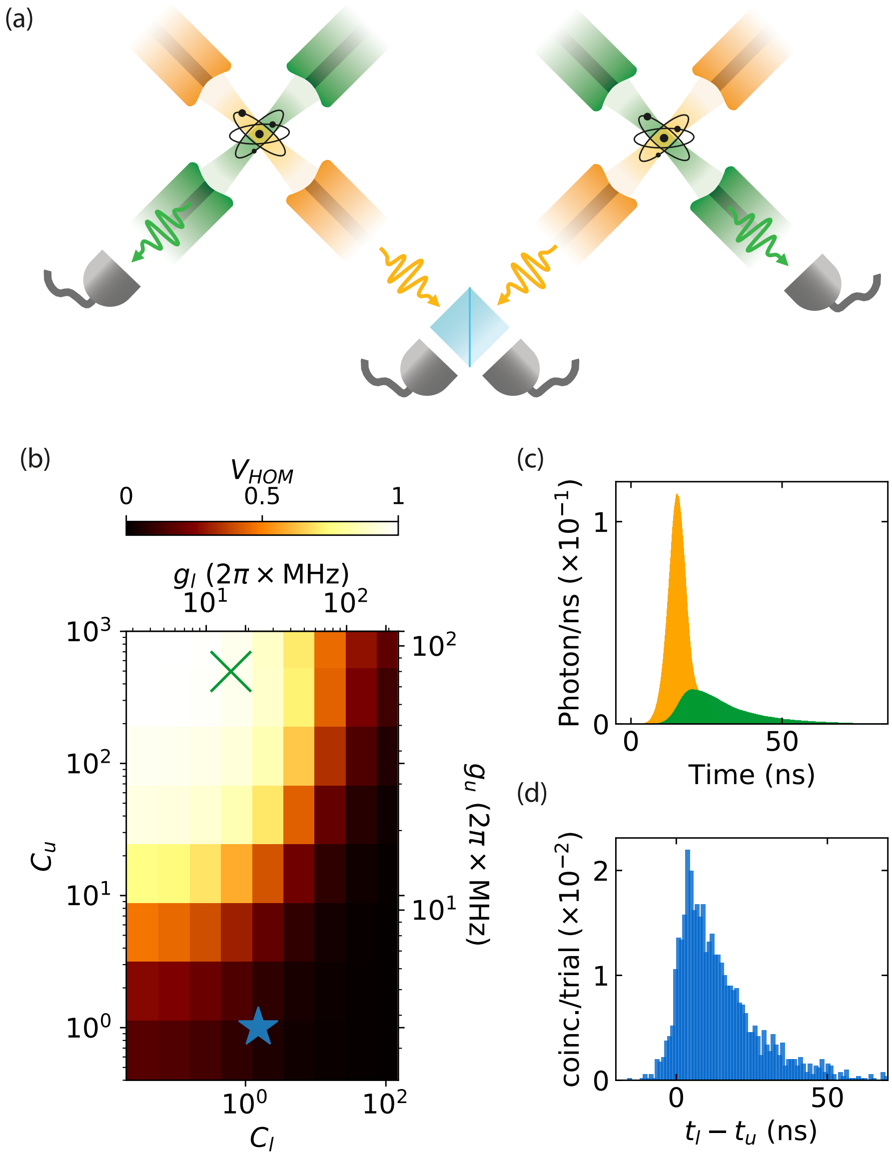


**Figure 4.8.:** (a) Two identical sources generate a photon pair each. The upper photons generated by each source interfere on a 50/50 beam splitter. (b) Simulated Hong-Ou-Mandel interference visibility ($V_{\mathrm{HOM}}$) for the upper photons as a function of the atom-cavity coupling rate of the upper ($g_u$) and lower ($g_l$) cavity. The atom-cavity cooperativities $C_u$ and $C_l$ corresponding to each coupling rate $g_u$ and $g_l$ are also indicated. The blue star marks our experimental parameters $(g_u, g_l) = (6, 21.9)$ $\times 2\pi$MHz, for which $V_{\mathrm{HOM}} = 0.08$. Plot (c) and (d) show, respectively, the simulated intensity profiles and cross-correlation for the cavity parameters marked by the green cross in the main plot $(g_u, g_l) = (80,\ 15)$ $\times 2\pi$MHz. For these parameters $V_{\mathrm{HOM}} = 0.95$.

In this situation, the system exhibits dynamics that are analogous to STIRAP. Since modifying experimentally the cavity parameters to be in the strong-coupling regime involves a noticeable overhead, such as designing, manufacturing, and installing new fiber cavities, which goes beyond the present experimental capabilities, this study is limited to simulating theoretically the regime of interest. Our cavities are designed with a mirror that has a high transmission leading to a relatively big $\kappa$, however one could in the future reach the strong coupling regime by using shorter cavities with lower transmission mirrors [90], and using a stronger upper transition such as the $5^2P_{3/2} - 4^2D_{5/2}$ in $^{87}$Rb.

Differently from section 4.1, we start by considering an atom having only three ladder levels $|e\rangle$, $|i\rangle$, and $|g\rangle$ and being coupled to the two photonic cavity modes interacting with the upper ($u$) and lower ($l$) transition [151]. For the moment, we do not consider any driving beam. The interaction picture Hamiltonian can be written

$$\hat{H} = \left(g_l \hat{a}_l^\dagger \hat{\sigma}_{gi} + g_u \hat{a}_u^\dagger \hat{\sigma}_{ie} + \text{H.c.}\right) + \Delta_l \hat{a}_l^\dagger \hat{a}_l + \Delta_u \hat{a}_u^\dagger \hat{a}_u \tag{4.3}$$

where H.c. stands for the Hermitian conjugate. For the simple case in which $\Delta_l = -\Delta_u = \Delta$, the Hamiltonian has three eigenstates and eigenenergies that resemble those of driven three-level atom in a single cavity (see Eq. 2.14). We have

$$E_0 = 0 \tag{4.4}$$

$$E_+ = \frac{\Delta}{2} + \sqrt{g_u^2 + g_l^2 + (\frac{\Delta}{2})^2} \tag{4.5}$$

$$E_- = \frac{\Delta}{2} - \sqrt{g_u^2 + g_l^2 + (\frac{\Delta}{2})^2} \tag{4.6}$$

referenced to the energy of state $|e\rangle$. The eigenstate corresponding to eigenenergy $E_0$ is

$$|\Psi_0\rangle = \left(-g_l\,|e{,}0_u{,}0_l\rangle + g_u\,|g{,}1_u{,}1_l\rangle\right)/\sqrt{g_l^2 + g_u^2} \tag{4.7}$$

where $|g{,}1_u{,}1_l\rangle$ is the state with an atom in the ground state $|g\rangle$ and 1 photon in each of the two cavities. The state described by Eq. 4.7 is similar to the dark state in STIRAP and does not include a component with population in state $|i\rangle$.

One can excite the state in Eq. 4.7 by initializing the atom in an auxiliary ground state $|g_0\rangle$ and using an optical drive beam that couples states $|g_0\rangle$ and $|\Psi_0\rangle$. If the temporal evolution of the drive is slow enough, its optical frequency spectrum will only couple to $|\Psi_0\rangle$ and not to the other eigenstates of the Hamiltonian $\hat{H}$. For $g_l \ll g_u$ and the decay rates $\kappa$ and $\gamma$ small compared to the system's time scale, one can directly go from state $|g_0{,}0_u{,}0_l\rangle$ to $|g{,}1_u{,}1_l\rangle$.

In order to study what happens to state $|i\rangle$, we fix the cavity detunings to $\Delta = 0$. To better illustrate the analogy with STIRAP, one can initially consider the scenario where atomic and cavity decays are negligible compared to the atom-cavity coupling rates. The results for a pulse with a duration of 2µs and $(g_u, g_l) = (10, 1) \times 2\pi$ MHz are shown in Fig. 4.9. Here one observes that the atomic population evolves from $|g_0\rangle$ to $|g\rangle$ (Fig. 4.9a) while generating a single photon in each of the two cavity modes (Fig. 4.9b).

We now consider some experimentally feasible parameters by taking into account the decay rates, in order to simulate the cavity fields. We consider a continuous drive with Rabi

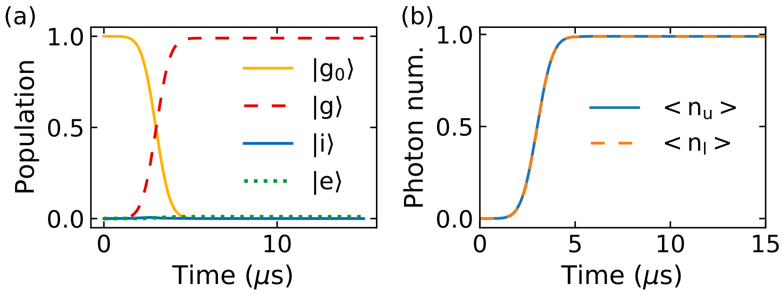


**Figure 4.9.:** Temporal evolution of the atomic populations and the cavity photon numbers. (a) The atom initially in the $|g_0\rangle$ is coupled to the state of the atom-cavities system $|\Psi_0\rangle$ by a Gaussian pulsed drive with duration 2 µs for $(g_u, g_l) = (10, 1) \times 2\pi\text{MHz}$. While the atom is quasi-adiabatically transferred from $|g_0\rangle$ to $|g\rangle$, a photon is output by each cavity (b). For this scenario, the cavities are resonant to the atomic transitions and atomic and cavity decays rates are small in comparison to the atom-photon coupling rates $(\gamma_u, \gamma_l, \kappa_u, \kappa_l) = (10^{-3}, 10^{-2}, 10^{-10}, 10^{-10}) \times 2\pi\text{MHz}$.

frequency $\Omega_D$ on the $|g\rangle - |e\rangle$ transition, as depicted in Fig. 4.10a, such that laser excitation and photon emission result in a cycling process between the atomic states. This is modeled by adding to the Hamiltonian of Eq. 4.3 the additional term

$$V = \Omega_D(\hat{\sigma}_{ge} + \hat{\sigma}_{ge}^\dagger) - \Delta_D(\hat{\sigma}_{ee} + \hat{a}_u^\dagger \hat{a}_u) \tag{4.8}$$

where $\Delta_D$ is the drive detuning. When we look at the steady-state photon number occupation in the upper cavity as a function of the $\Delta_D$ we can see, for $g_l = 0$, two photon-generation peaks which correspond to the normal-mode splitting of the $|i\rangle - |e\rangle$ transition (see Fig. 4.10b dashed blue line). However, when $g_l > 0$, a photon emission peak occurs at $\Delta_D = 0$ (Fig. 4.10b black solid line) due to the presence of the dark state in Eq. 4.7. A similar photon emission peak can be observed in the lower cavity for $\Delta_D = 0$ when both cavities are coupled to the atom (see Fig. 4.10c). One can also observe in Fig. 4.10d that for $\Delta_D = 0$ the population in state $|i\rangle$ is almost zero, much smaller than the population that results when the drive is resonant with one of the normal modes.

This observation generalizes a previous finding for one cavity [152, 153] instead of our two cavities, namely that a laser-driven cycling scheme involving a cavity vacuum can generate photons without populating the excited state $|e\rangle$, something we also find in our system. One consequence of the small population in state $|i\rangle$ is that the decay rate of the $|i\rangle - |g\rangle$ transition ($\gamma_l$) is not a limiting parameter for the spectrum of the photon. One can then obtain the photon spectrum by calculating the Fourier Transform of the $g^{(1)}$ function of the corresponding field [154]. As illustrated in Fig. 4.10e, one can observe that for $g_u = 10 \times 2\pi\text{MHz}$ the spectrum of the lower cavity photon is narrower than both the cavity and the transition linewidth, while for $g_u = 0$ the spectrum would be significantly broader.

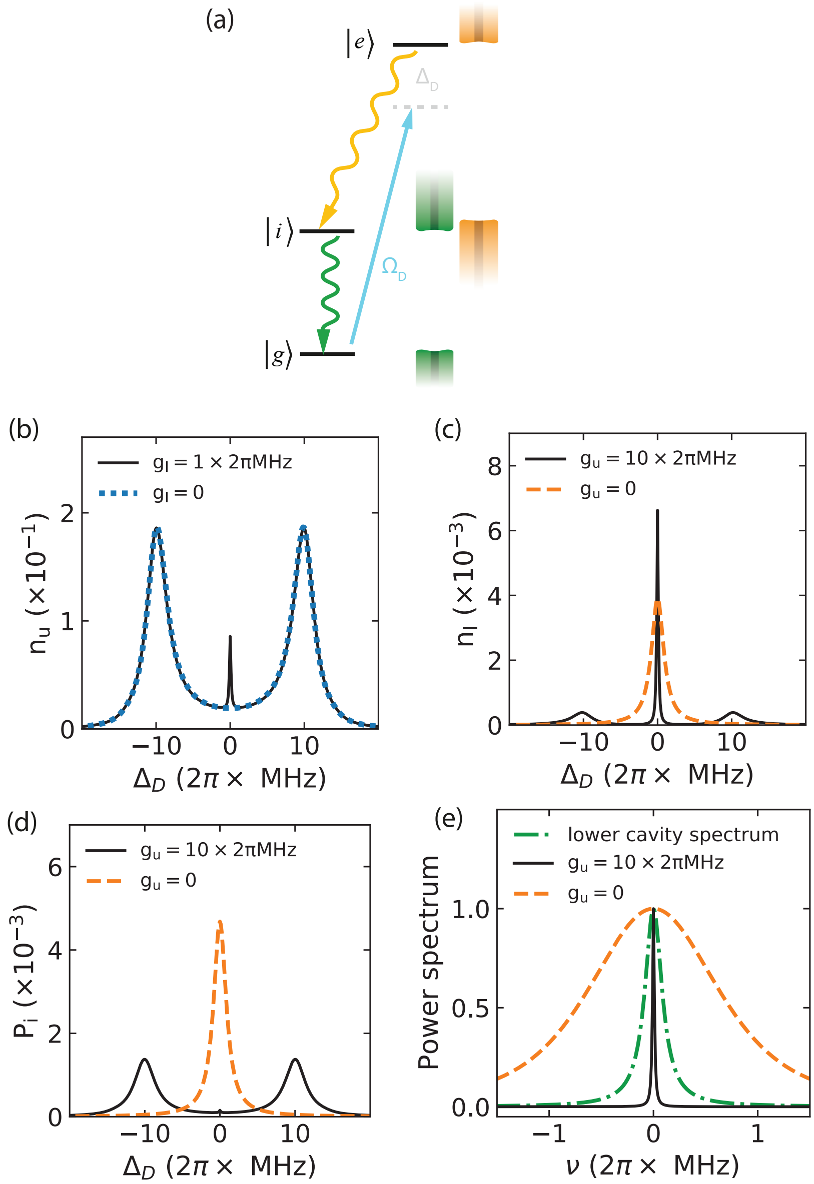


**Figure 4.10.:** (a) Simulation with a driving field coupling the transition $|g\rangle - |e\rangle$. (b) Upper cavity photon number, (c) lower cavity photon number and (d) the $|i\rangle$ state population as a function of the drive field frequency detuning. For comparison, the upper (lower) cavity photon numbers are also shown in the situation that the coupling in the other cavity $g_l$ $(g_u)$ is null. (e) Spectrum of the lower cavity photon field (black solid line $(g_u, g_l) = (10, 1) \times 2\pi$MHz, orange dashed line $(g_u, g_l) = (0, 1) \times 2\pi$MHz) and of the lower empty cavity (green dash-dotted). The other parameters used for this simulation are $(\kappa_u, \kappa_l, \gamma_u, \gamma_l, \Omega_D) = (0.01,\ 0.1,\ 1,\ 2,\ 0.1)\times 2\pi$MHz.

The atom-cavity parameters for the upper transition chosen in this simulation place the system in the strong-coupling regime. As one can see from Fig. 4.11, reaching this regime is mandatory to observe the effects just described. In particular, the high-cooperativity regime (i.e. $C = g^2/(2\kappa\gamma) > 1$ but not necessarily $g > (\kappa, \gamma)$) is not sufficient. The more the cavity losses $\kappa_u$ approach the atom-cavity coupling rate $g_u$, the harder it is to resolve the spectral line at $\Delta_D = 0$. Tracing a line at $\kappa_u = g_u$ then allows us to identify two regimes. For $\kappa_u > g_u$ the lower-cavity photon number and the intermediate-state population spectra are nearly the same. In this scenario, the cavity losses erase the coherent transfer from $|e\rangle$ to $|g\rangle$. As $\kappa_u$ decreases, the $n_l$ spectrum splits into three peaks, displaying the three eigenstates of the Hamiltonian in eq. 4.3. For $g_u \gg \kappa_u$ and $\Delta_D = 0$ photons are generated in the lower cavity while the intermediate state remains unpopulated.

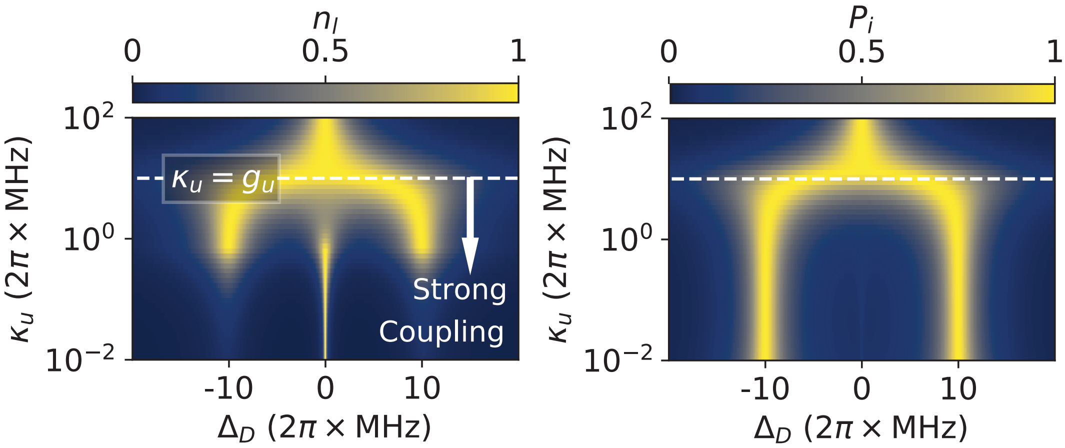


**Figure 4.11.:** Normalized lower cavity photon number $n_l$ and $|i\rangle$ state population $P_i$ as a function of the upper cavity linewidth and the drive field detuning.

# 5. Heralded generation of atom-photon entanglement

The contents of this chapter were first published in [86]:

**Source of heralded atom-photon entanglement for quantum networking**
G. Chiarella, T. Trank, L. Zukka, P. Farrera and G. Rempe
*Arxiv* 2510.15765 (2025).

Having established a source capable of generating correlated photon pairs via a cascaded atomic excitation scheme, we now turn to a more advanced application of this platform: the heralded generation of entanglement between a single atom and a single photon. This capability is relevant for quantum networking scenarios that rely on efficient and well-timed entanglement distribution between stationary and flying qubits.

The challenge in such open-system architectures lies not only in preparing entangled states with high fidelity, but also in ensuring that one can identify—herald—the successful creation of such states. While losses and imperfections are hard to eliminate in practical quantum communication systems, heralding provides a powerful strategy for mitigating their effects. By considering only the successful generation events, one can effectively filter out failed attempts and, by doing so, reduce quantum state infidelities.

In this context, heralding the creation of atom-photon entanglement directly at the source, rather than at a distant receiver, offers several advantages for the implementation of robust and scalable quantum network protocols. First and foremost, it enables local and real-time feedback, allowing the sender node to detect a failed attempt immediately and to reset or reinitialize the system without waiting for information to propagate back from the receiver. This eliminates unnecessary idle times and has the potential to improve the overall communication rate, especially important in long-distance quantum links where round-trip delays can be substantial.

Second, performing the heralding locally circumvents the inconvenience of obtaining the herald signal at the end of the transmission line. This ensures a higher detection efficiency and a higher signal-to-noise ratio, both of which are essential for maintaining the fidelity of the quantum information being transmitted.

Third, heralding at the source can provide accurate and deterministic timing information about the exact moment when the atom-photon entangled state is created. This timing information can be shared with the remote node and used to precisely gate the single-photon detectors at the receiving end. Such temporal filtering suppresses background noise, including dark counts and stray photons, which are otherwise indistinguishable from entangled photons. This becomes particularly important in low-count-rate scenarios.

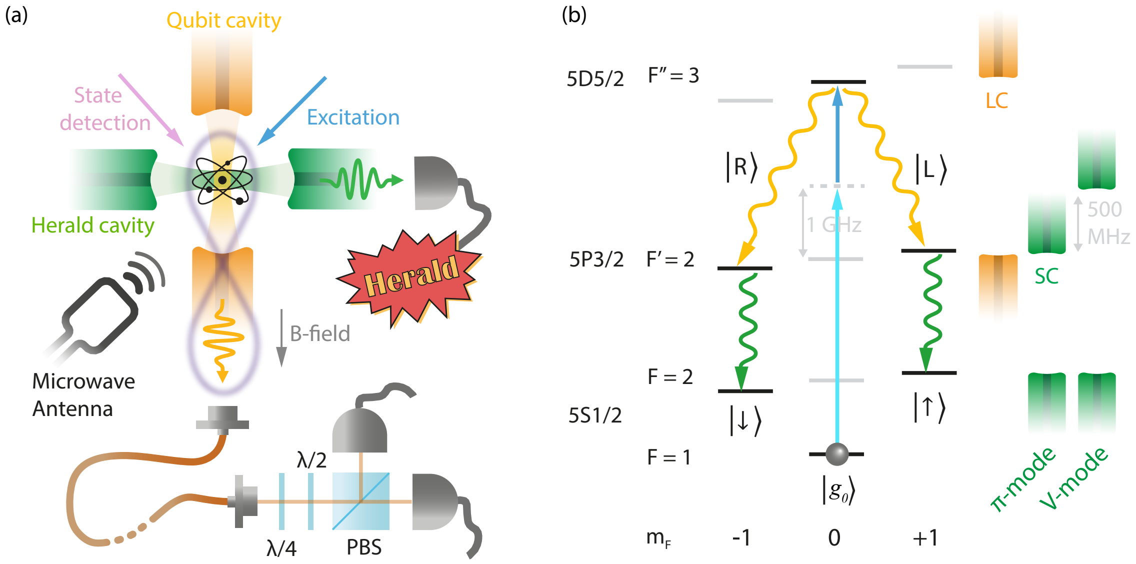


**Figure 5.1.:** (a) A $Rb^{87}$ atom coupled to two crossed fiber cavities emits a photon entangled with the atom and another photon that heralds the entangled state generation. Photon polarization and atomic qubit measurements are subsequently performed. The photonic state is read out by a polarization resolving detection setup, constituted by retardation waveplates, a polarizing beam splitter and two SNSPDs. The atomic state is read out by employing a sequence of MW pulses and fluorescence state detection. (b) After optically pumping the atom into state $|5^2S_{1/2}, F = 1, m_F = 0\rangle$, it is optically excited to state $|5^2D_{5/2}, F = 3, m_F = 0\rangle$ by two laser pulses. A polarization qubit is generated in the qubit cavity, and the cavity-enhanced emission of a $\pi$ polarized photon in the herald cavity brings the atom into a superposition of two well-defined spin states.

In this chapter, we demonstrate how our system enables such heralded generation of atom-photon entanglement, building on the photon pair source described previously. We show how detecting one of the photons projects the atom into a quantum state entangled with the remaining photon. We describe the methods used to read out the atomic and photonic state, characterize the generated entangled state, and measure its coherence time. Finally, we use the herald timing information to reduce photon qubit measurement noise and show the potential of our source for noise-limited long-distance quantum networks.

## 5.1. Experimental approach

Fig. 5.1a depicts the elements utilized to realize and measure the desired heralded atom-photon entanglement. After trapping and cooling a $Rb^{87}$ atom in the center of the fiber cavities, a static magnetic field is applied along the LC-axis, such that the quantization axis is defined in that direction. Subsequently, the atom is optically pumped into state $|g_0\rangle = |5^2S_{1/2}, F = 1, m_F = 0\rangle$ and two laser pulses induce a two-photon excitation to state $|5^2D_{5/2}, F = 3, m_F = 0\rangle$. At this stage, single photons in a superposition of right circular $|R\rangle$ and left circular $|L\rangle$ polarizations are emitted into the LC while the atom decays via $|5^2D_{5/2}, F = 3, m_F = 0\rangle \to |5^2P_{3/2}, F = 2, m_F = -1\rangle$ and

$\left|5^2D_{5/2}, F=3, m_F=0\right\rangle \rightarrow \left|5^2P_{3/2}, F=2, m_F=1\right\rangle$, respectively - see 5.1b. For this reason, this cavity is designated as the "qubit cavity". At the same time, the SC couples the lower transition $\left|5^2P_{3/2}, F=2\right\rangle \rightarrow \left|5^2S_{1/2}, F=2\right\rangle$ via the $\pi$-mode, mediating only the transitions $m_F=-1 \rightarrow |\downarrow\rangle = \left|5^2S_{1/2}, F=2, m_F=-1\right\rangle$ and $m_F=1 \rightarrow |\uparrow\rangle = \left|5^2S_{1/2}, F=2, m_F=1\right\rangle$. Since the detection of a photon emitted in this resonator projects the atom-photon state onto a superposition of $|L,\uparrow\rangle$ and $|R,\downarrow\rangle$, this cavity is designated as the "herald cavity". The atom-photon state resulting from a successful detection of a herald photon can be expressed as

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\left(|R,\downarrow\rangle + |L,\uparrow\rangle\right) \tag{5.1}$$

which is a maximally entangled state. Because of the SC birefringence, the other herald cavity eigenmode (the V mode) is 500 MHz blue detuned from the lower transition.

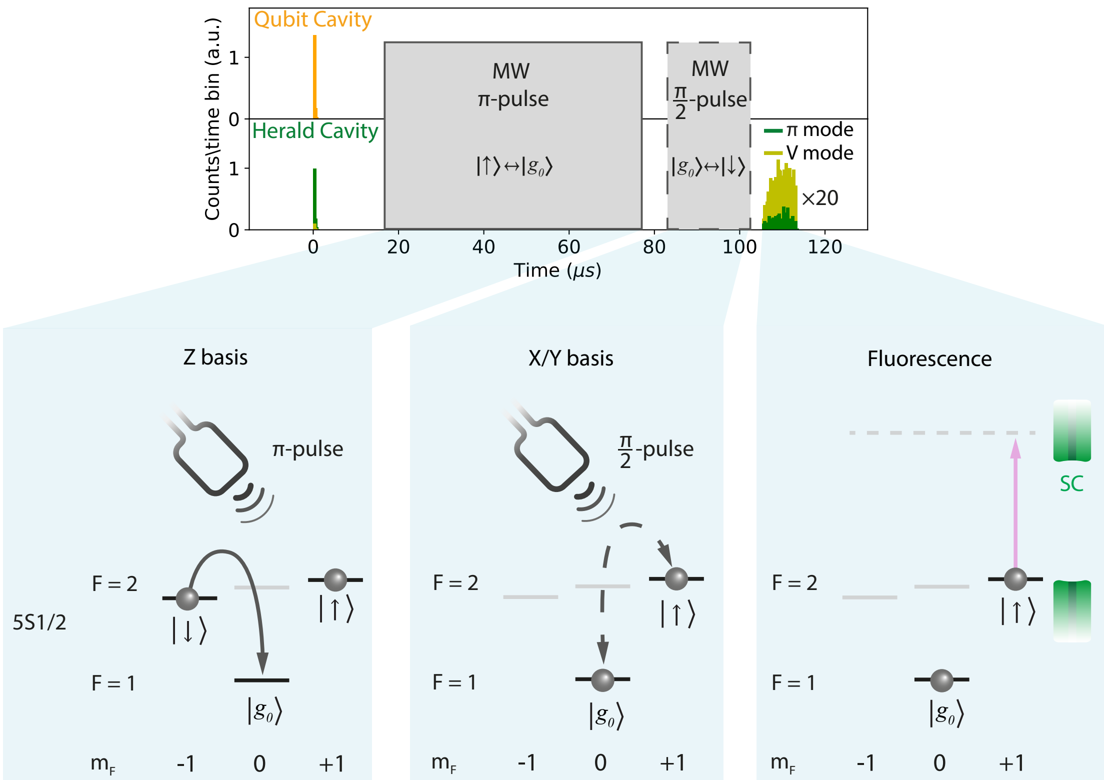


**Figure 5.2.:** Time histogram showing the photons detected in the qubit (upper plot) and herald (lower plot) cavities during the entanglement generation and measurement periods. The grey area represents the MW pulses used for the atomic qubit measurement. At first, a MW $\pi$-pulse transfers population from $|\uparrow\rangle$ to $|g_0\rangle$, enabling Z-basis readout via fluorescence detection in $F=2$. For X and Y basis measurements, an additional MW $\pi/2$-pulse with a controlled phase is applied before detection. The relative phase between the pulses defines the measurement basis, with X and Y differing by $\pi/2$. The fluorescence counts are magnified by a factor 20. The complete atomic qubit readout sequence is executed within 115 µs.

In order to analyze the entangled state, different qubit tomography techniques are employed. The single photon polarization qubit is analyzed with a combination of polarization wave plates, a polarizing beam-splitter, and single-photon detectors, allowing to perform polarization qubit measurements in an arbitrary basis (Fig. 5.1a). To perform atomic qubit measurements in an arbitrary basis, we use a sequence that employs MW pulses and fluorescence state detection - shown in Fig. 5.2. Firstly, the atomic population in $|\uparrow\rangle$ is transferred to $|g_0\rangle$ with a MW $\pi$-pulse. By directly detecting the population in F=2, the atomic state is read out in the computational ($Z$) basis. In order to read out the atomic state in the linear bases ($X$ and $Y$), a MW $\pi/2$-pulse is applied on the $|g_0\rangle - |\downarrow\rangle$ transition after the first MW pulse. The relative phase of the second pulse with respect to the first defines the arbitrary linear basis. The $X$ and $Y$ bases, in particular, are distinguished by a relative phase difference of $\pi/2$ - the atomic state bases calibration is based on the atomic state Larmor precession and is described in detail in section 5.2. The measurement finishes by performing cavity-enhanced fluorescence atomic state detection, as detailed in section 3.9.

The complete atomic qubit readout sequence, including microwave rotations, delay intervals, and fluorescence-based state detection, is executed within 115 µs following the generation of the atom-photon entangled state. The $\pi$ MW rotations on the $|g_0\rangle \leftrightarrow |\uparrow\rangle$ transition are performed with a fidelity of $\mathcal{F} = 0.95(1)$ and $\pi/2$ rotations in the $|g_0\rangle \leftrightarrow |\downarrow\rangle$ transition with a fidelity of $\mathcal{F} = 0.96(1)$ (these values include optical pumping and state detection errors). In our experiment, we choose an atomic state detection duration of $7.5\mu$s, corresponding to a fidelity of $\mathcal{F} = 0.989(2)$.

## 5.2. Atomic readout basis

Since the magnetic field splits the energy of atomic levels $|\uparrow\rangle$ and $|\downarrow\rangle$, the atomic state experiences a Larmor precession. After the measurement of the photonic qubit projects the atomic state in a superposition of $|\uparrow\rangle$ and $|\downarrow\rangle$, it will precess with a frequency given by the energy difference of the atomic levels, namely $\nu_L = |E_\uparrow - E_\downarrow|/h = 200\,\text{kHz}$ [155, 156]. Since the atomic linear basis is defined by the phase of the second MW pulse relative to the first one, we can make use of the atomic free phase evolution to precisely calibrate the pulse phase. By doing so, we can calibrate this to read out the atom in two orthogonal linear bases (e.g., $X$ and $Y$). Combined with three orthogonal photonic bases, this gives us the tools to perform a full atom-photon state tomography.

In order to observe this precession, we measure the correlation parameter, which is expressed as a function of the atom-photon state measurement probability P:

$$C_{ap} = P(\uparrow_a, D_p) + P(\downarrow_a, A_p) - P(\uparrow_a, A_p) - P(\downarrow_a, D_p) \tag{5.2}$$

where ($\uparrow_a$,$\downarrow_a$) denote the atomic spin state and ($D_p$,$A_p$) denote the diagonal and antidiagonal polarization state of the photon. This measurement is performed by applying two MW pulses and fluorescence state readout, as described in Fig. 5.2, after the successful detection of a photon pair.

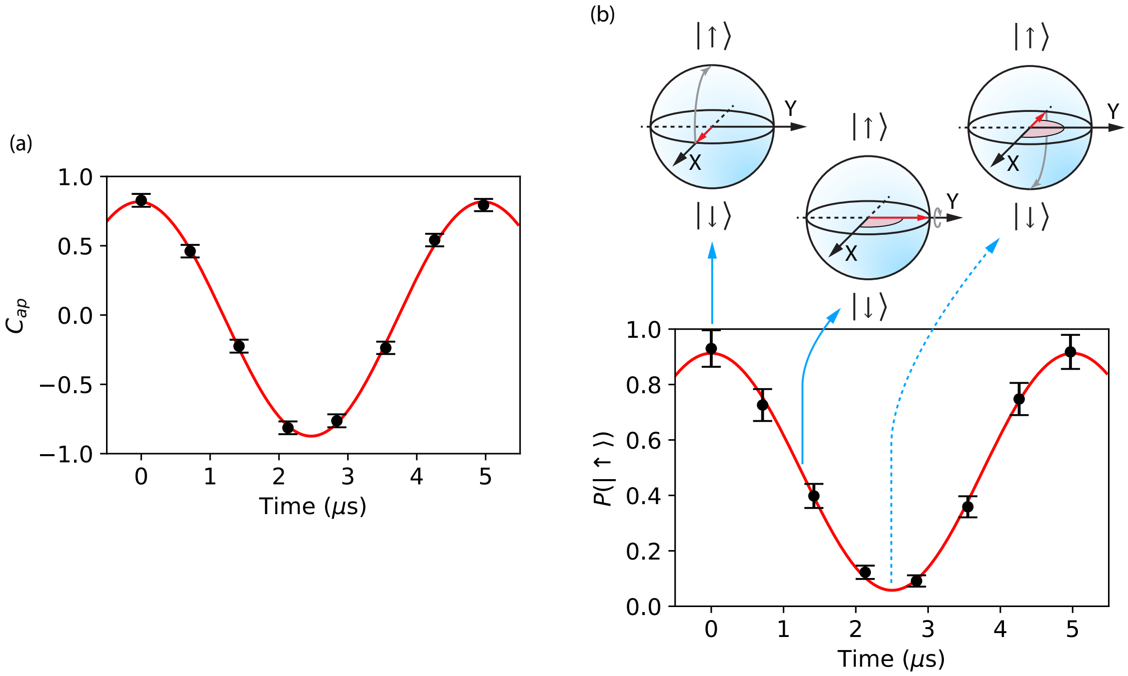


**Figure 5.3.:** (a) Atom-photon state correlation parameter $C_{ap}$ as a function of the MW frequency switch-time. The correlation oscillates sinusoidally with a frequency of 200 kHz, corresponding to the atomic states' energy difference. The fitted amplitude is 0.85(1). (b) Probability of measuring the state $|\uparrow\rangle$ after measuring a $D$ polarized photon. The Bloch spheres help visualize the atomic state evolution after the photon detection. The red arrow represents the Bloch vector and the grey arrow indicates its rotation after the MW $\pi/2$-pulse. At the switch-off time of 0 µs one measures the maximum correlation, defining therefore the atomic $X$ basis. After 1.25 µs the atomic phase has evolved of $\pi/2$, therefore the subsequent rotation leaves the state unchanged. This is the time of maximum correlation in the case in which the photon is measured in the $HV$ basis, defining therefore the atomic $Y$ basis. After 5 µs the atom has completed one revolution on the Bloch sphere's equatorial plane.

After the MW $\pi$ pulse $|\downarrow\rangle \rightarrow |g_0\rangle$ is finished, the atomic state evolves with a frequency which is 200 kHz greater than the MW frequency applied. In the time between the end of the MW pulse and the instant at which the MW frequency is switched to the $|g_0\rangle \leftrightarrow |\uparrow\rangle$ transition (i.e., rotating now at the same frequency of the atom), the atom accumulates a finite phase. By measuring $C_{ap}$ as a function of the MW frequency switch-time, it is possible to observe the atomic precession, shown in Fig. 5.3a. The period of oscillation of 5 µs matches the expected value given by the splitting of the two atomic spin states. Fixing the MW frequency switch-time is therefore equivalent to fixing the phase of the second MW pulse relative to the first one. As marked by the solid arrows in Fig. 5.3b, the switch-time 0 µs defines the $X$ basis, whereas the switch-time 1.25 µs defines the $Y$ basis, since it corresponds to a phase difference of $\pi/2$ relative to the $X$ basis.

It is to be noticed that the photonic basis of Eq. 5.2 has been arbitrarily set to $DA$. One can define the correlation parameter when the qubit photon is measured in the $HV$ basis

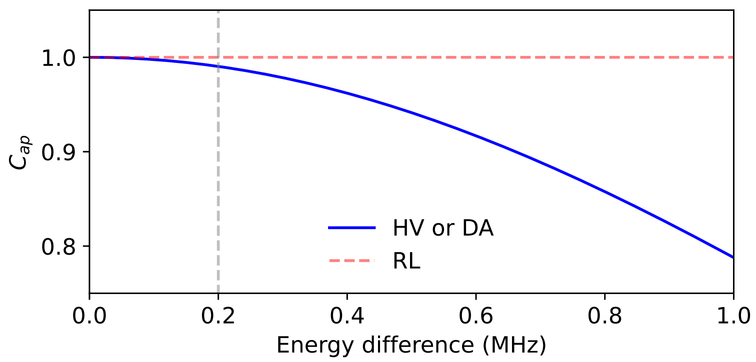


**Figure 5.4.:** Correlation parameter as a function of the qubit states' energy difference. The red dashed line represents the correlation between the atomic $Z$ and photonic $RL$ outcomes, which is unaffected by the states splitting. The blue line represents the correlation between the atomic $X(Y)$ and photonic $DA(HV)$ outcomes, modeled by eq. 5.3.

as well. The expected result is an oscillation with the same periodicity and amplitude, shifted by a quarter of the period (corresponding to a $\pi/2$ phase shift).

The atomic period is determined by the absolute Zeeman splitting, making this calibration susceptible to magnetic field fluctuations. For this reason, a curve measurement is repeated roughly every 5 hours, which has been empirically found to be a good compromise. The $X$ and $Y$ bases are then updated, and the subsequent experiments can be performed.

As the temporal shape of the qubit photon has a non-vanishing width, the phase evolution of the atomic qubit after detection of the photon differs from shot to shot with respect to a fixed sequence trigger. If $p(t)$ is the probability of detecting a photon at time $t$ (determined by the temporal shape of Fig. 4.7c) and $\Delta E$ is the atomic states energy difference, one can derive the probability of detecting the coincidence $\uparrow_a, H_p$ (or equivalently $\downarrow_a, V_p$) as

$$P(\uparrow_a, H_p) = \frac{1}{4}\left(1 + \int_{t_0}^{t_1} \cos(2\pi \cdot \Delta E \cdot t) \cdot p(t)\, dt\right) \tag{5.3}$$

where $t_0$ and $t_1$ determine the photon acceptance window. Fig. 5.4 shows the correlation parameter for photonic detection in the $RL$ basis (red dashed line) and in the $HV$ or $DA$ basis (blue line) as a function of the energy difference $\Delta E$. The latter one is based on the coincidence probability numerically computed via Eq. 5.3. By doing so we can estimate that the measured energy splitting (marked in Fig. 5.4with the gray dashed line) limits the correlation parameter to 99%.

When the photon temporal shape has a width comparable with the precession period, this effect deteriorates significantly the correlation and, therefore, the measured fidelity. However, this can be circumvented by experimentally implementing a mechanism that sweeps the photonic basis (e.g., with a polarization EOM) in such a way that the atomic phase shift is fully compensated. Since this does not constitute a limiting factor for the experiments presented, we have not utilized such a mechanism.

## 5.3. Experimental results

The probabilities of detecting a coincidence of an atomic and photonic state outcome are used to reconstruct the atom-photon density matrix. For this purpose, a total of 9 orthogonal 2-qubit measurement basis is utilized (3 atomic orthogonal bases × 3 photonic orthogonal bases). For a 2-qubit state, any density matrix $\hat{\rho}$ can be represented as [157]

$$\hat{\rho} = \frac{1}{4} \sum_{i_1, i_2 = 0}^{3} S_{i_1, i_2} \hat{\sigma}_{i_1} \otimes \hat{\sigma}_{i_2} \tag{5.4}$$

where $i_1$ and $i_2$ label the first and the second qubit, the $\hat{\sigma}_i$ matrices are the Pauli matrices, defined as

$$\hat{\sigma}_0 = \mathbb{I}\,, \hat{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \hat{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \hat{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{5.5}$$

and the $S_{i_1,i_2}$ are the Stokes parameters, given by [158]

$$S_{i_1,i_2} = \mathrm{Tr}\{\hat{\sigma}_{i_1} \otimes \hat{\sigma}_{i_2}\, \hat{\rho}\} \tag{5.6}$$

Due to normalization, $S_{0,0}$ will always equal one. Therefore, any 2-qubit density matrix can be uniquely represented by 15 real parameters. Physically, each of these parameters is directly linked to the outcome probability of a specific 2-qubit projective measurement. We consider the atomic and the photonic qubit as, respectively, the first and the second qubit. The outcomes given by their projective measurements are defined in table 5.1

From Eq. 5.6 we can derive each Stokes parameter as the following:

| Atom | Photon |
|---|---|
| X : $\lvert\uparrow_x\rangle = \frac{1}{\sqrt{2}}(\lvert\uparrow\rangle + \lvert\downarrow\rangle)$<br>$\lvert\downarrow_x\rangle = \frac{1}{\sqrt{2}}(\lvert\uparrow\rangle - \lvert\downarrow\rangle)$ | DA : $\lvert D\rangle = \frac{1}{\sqrt{2}}(\lvert R\rangle + \lvert L\rangle)$<br>$\lvert A\rangle = \frac{1}{\sqrt{2}}(\lvert R\rangle - \lvert L\rangle)$ |
| Y : $\lvert\uparrow_y\rangle = \frac{1}{\sqrt{2}}(\lvert\uparrow\rangle + i\lvert\downarrow\rangle)$<br>$\lvert\downarrow_y\rangle = \frac{1}{\sqrt{2}}(\lvert\uparrow\rangle - i\lvert\downarrow\rangle)$ | HV : $\lvert H\rangle = \frac{1}{\sqrt{2}}(\lvert R\rangle + i\lvert L\rangle)$<br>$\lvert V\rangle = \frac{1}{\sqrt{2}}(\lvert R\rangle - i\lvert L\rangle)$ |
| Z : $\lvert\uparrow\rangle$, $\lvert\downarrow\rangle$ | RL : $\lvert R\rangle$, $\lvert L\rangle$ |

**Table 5.1.:** Atom and photon measurement bases. The indexes $i_1$ and $i_2$ of eq. 5.4 label the bases as $i_1 = 1 \to X, 2 \to Y, 3 \to Z$ and $i_2 = 1 \to DA, 2 \to HV, 3 \to RL$.

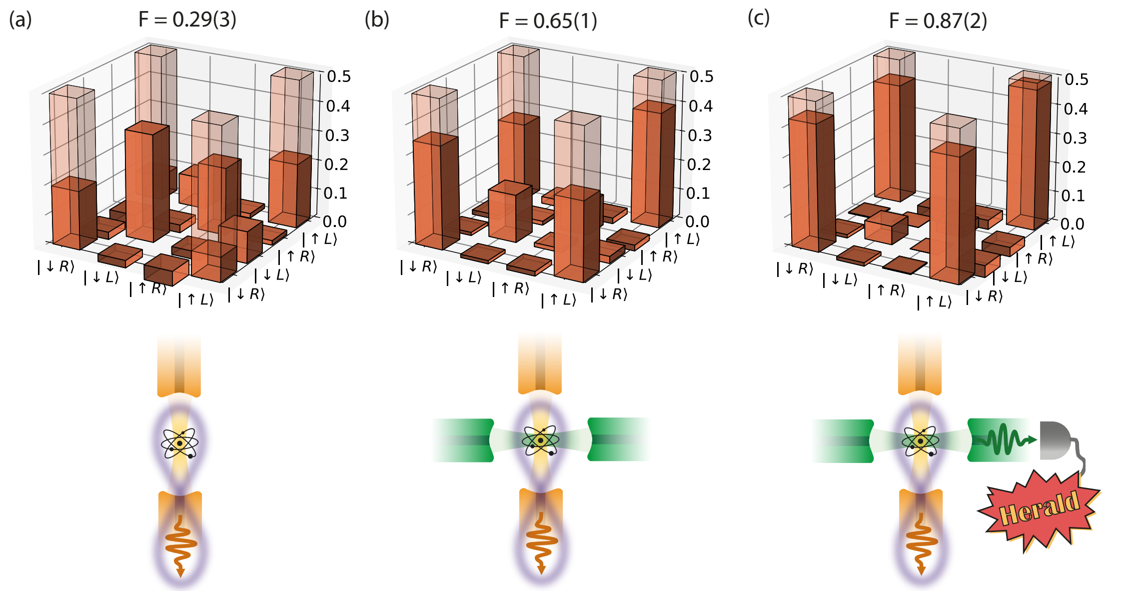


**Figure 5.5.:** Real part of the atom-photon density matrix corresponding to the situation where the herald cavity is not resonant to the lower transition (a), the herald cavity is resonant to the lower transition (b), the herald cavity is resonant to the lower transition and only events with a herald photon detection are considered (c). The shaded areas represent the ideal state in Eq. 5.1.

$$S_{1,1} = P(\uparrow_x, D) - P(\uparrow_x, A) - P(\downarrow_x, D) + P(\downarrow_x, A) \tag{5.7}$$
$$S_{1,2} = P(\uparrow_x, H) - P(\uparrow_x, V) - P(\downarrow_x, H) + P(\downarrow_x, V) \tag{5.8}$$
$$S_{1,3} = P(\uparrow_x, R) - P(\uparrow_x, L) - P(\downarrow_x, R) + P(\downarrow_x, L) \tag{5.9}$$
$$S_{2,1} = P(\uparrow_y, D) - P(\uparrow_y, A) - P(\downarrow_y, D) + P(\downarrow_y, A) \tag{5.10}$$
$$S_{2,2} = P(\uparrow_y, H) - P(\uparrow_y, V) - P(\downarrow_y, H) + P(\downarrow_y, V) \tag{5.11}$$
$$S_{2,3} = P(\uparrow_y, R) - P(\uparrow_y, L) - P(\downarrow_y, R) + P(\downarrow_y, L) \tag{5.12}$$
$$S_{3,1} = P(\uparrow, D) - P(\uparrow, A) - P(\downarrow, D) + P(\downarrow, A) \tag{5.13}$$
$$S_{3,2} = P(\uparrow, H) - P(\uparrow, V) - P(\downarrow, H) + P(\downarrow, V) \tag{5.14}$$
$$S_{3,3} = P(\uparrow, R) - P(\uparrow, L) - P(\downarrow, R) + P(\downarrow, L) \tag{5.15}$$
$$\tag{5.16}$$

Measurements are taken at each of these nine settings, directly determining the above nine 2-qubit Stokes Parameters. The six remaining required parameters are dependent upon the same measurements

$$
\begin{aligned}
S_{0,1} &= P(\uparrow_x, D) - P(\uparrow_x, A) + P(\downarrow_x, D) - P(\downarrow_x, A) && (5.17)\\
S_{0,2} &= P(\uparrow_y, H) - P(\downarrow_y, H) + P(\uparrow_y, V) - P(\downarrow_y, V) && (5.18)\\
S_{0,3} &= P(\uparrow, R) - P(\uparrow, L) + P(\downarrow, R) - P(\downarrow, L) && (5.19)\\
S_{1,0} &= P(\uparrow_x, D) + P(\uparrow_x, A) - P(\downarrow_x, D) - P(\downarrow_x, A) && (5.20)\\
S_{2,0} &= P(\uparrow_y, H) + P(\downarrow_y, H) - P(\uparrow_y, V) - P(\downarrow_y, V) && (5.21)\\
S_{3,0} &= P(\uparrow, R) + P(\uparrow, L) - P(\downarrow, R) - P(\downarrow, L) && (5.22)
\end{aligned}
$$

Through Eq. 5.4 we reconstruct the density matrix of the generated atom-photon quantum state. The closeness to the maximally entangled state of 5.1 is quantified by the fidelity, defined as

$$\mathcal{F} = \langle\Psi|\,\hat{\rho}\,|\Psi\rangle \tag{5.23}$$

Fig. 5.5 shows the density matrix of the two-qubit state for three different situations. In (a), the herald cavity is not resonant to the lower transition, and all the events with a qubit photon detection are considered. The fidelity of the atom-photon entangled state compared to an ideal Bell state is measured to be $\mathcal{F} = 0.29(3)$. This result shows no entanglement between the atom and the photon, and it is to be expected, since the atom decays in free space from the intermediate state to any ground state.

In (b), the herald cavity is resonant to the lower transition, and all the events with a qubit photon detection are considered. In this case $\mathcal{F} = 0.65(1)$, well above the fidelity bound for entangled states ($\mathcal{F} > 0.5$) [159]. The reason for this increase is that even though the atom decays to any ground state, the $\pi$-mode of the herald cavity enhances the $\pi$ atomic decay on the lower transition. However, since all qubit photon detection events are considered, regardless of the herald photon, the cases in which the atom decays to unwanted ground states are also considered.

In (c) the herald cavity is resonant with the lower transition, too, but only the events with a photon detection in both the qubit and the herald cavities are considered. We observe a further increase in the measured fidelity $\mathcal{F} = 0.87(2)$, due to the fact that the $\pi$ atomic decay on the lower transition is heralded by the detection of a herald photon. The obtained heralded entangled state fidelity is limited by the decoherence of the atomic state during the atomic qubit measurement time, including the MW rotations.

We also characterize the decrease of the entangled state fidelity as a function of the atomic-qubit measurement-delay time after the herald-detection window. Since in the time between the herald detection and the atomic-qubit measurement the atomic state freely precesses on the Bloch sphere's equator (see Fig. 5.3), for each data point we compensate for this additional rotation by recalibrating the $X$ and $Y$ basis.

We observe a Gaussian decay of the fidelity with a characteristic decay time of 206(6) µs, as shown in Fig. 5.6. By visualizing the density matrix right after the entanglement generation and after more than 800 µs, it is possible to visualize how the state has evolved in that time. Although the diagonal elements show no significant change, indicating a good

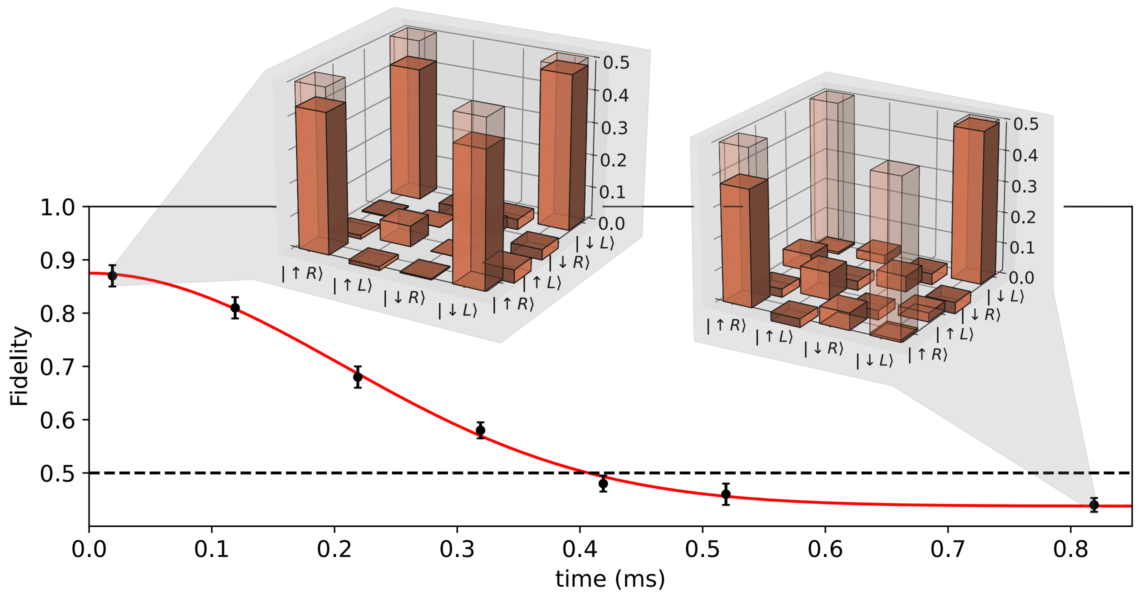


**Figure 5.6.:** Entangled state fidelity compared to an ideal Bell state as a function of the atomic qubit measurement-delay time. The dashed line represents the fidelity lower bound for an entangled state. We observe a Gaussian decay of the fidelity with a characteristic decay time of 206(6) µs. Density matrices reconstructed from the measurement after almost zero delay and after more than 800 µs are shown.

correlation in the 2-qubit computational basis, the off-diagonal elements $\langle\downarrow L|\,\hat{\rho}\,|\uparrow R\rangle$ and $\langle\uparrow R|\,\hat{\rho}\,|\downarrow L\rangle$ drop to zero. These elements are associated with the coherence of the quantum state, which is therefore completely lost. We identify the main decoherence mechanism to be the atomic qubit states' energy difference fluctuations, mainly limited by fluctuations of the magnetic field at the atomic position.

## 5.4. Application

As detailed in section 4.4, the herald photon is temporally correlated with the qubit photon (see Figs. 4.7b,c). This is a consequence of the faster photon emission into the herald cavity compared to the qubit cavity, mainly due to the stronger electric dipole moment on the $\left|5^2S_{1/2}, F=2\right\rangle \leftrightarrow \left|5^2P_{3/2}, F=2\right\rangle$ transition compared to the $\left|5^2P_{3/2}, F=2\right\rangle \leftrightarrow \left|5^2D_{5/2}, F=3\right\rangle$ transition. Due to to this temporal correlation, the detection time of the herald photon tells information about the emission time of the qubit photon. Although this property constitutes a limitation for the photon indistinguishability (see section 4.5), it can be used as a feature in the scenario depicted in Fig. 5.7a. Here, node A produces an atom-photon entangled state, heralded by the detection of the other photon at time $t_{low}$. On the other end of the 2 nodes network is node B. In this scenario, it is constituted exclusively by a polarization resolving detection setup, but it could host an heralded memory instead, as we will see later. the qubit is sent to node B via a quantum channel, e.g. an optical fiber, where, typically, measurement noise can become a limitation in situations where the signal

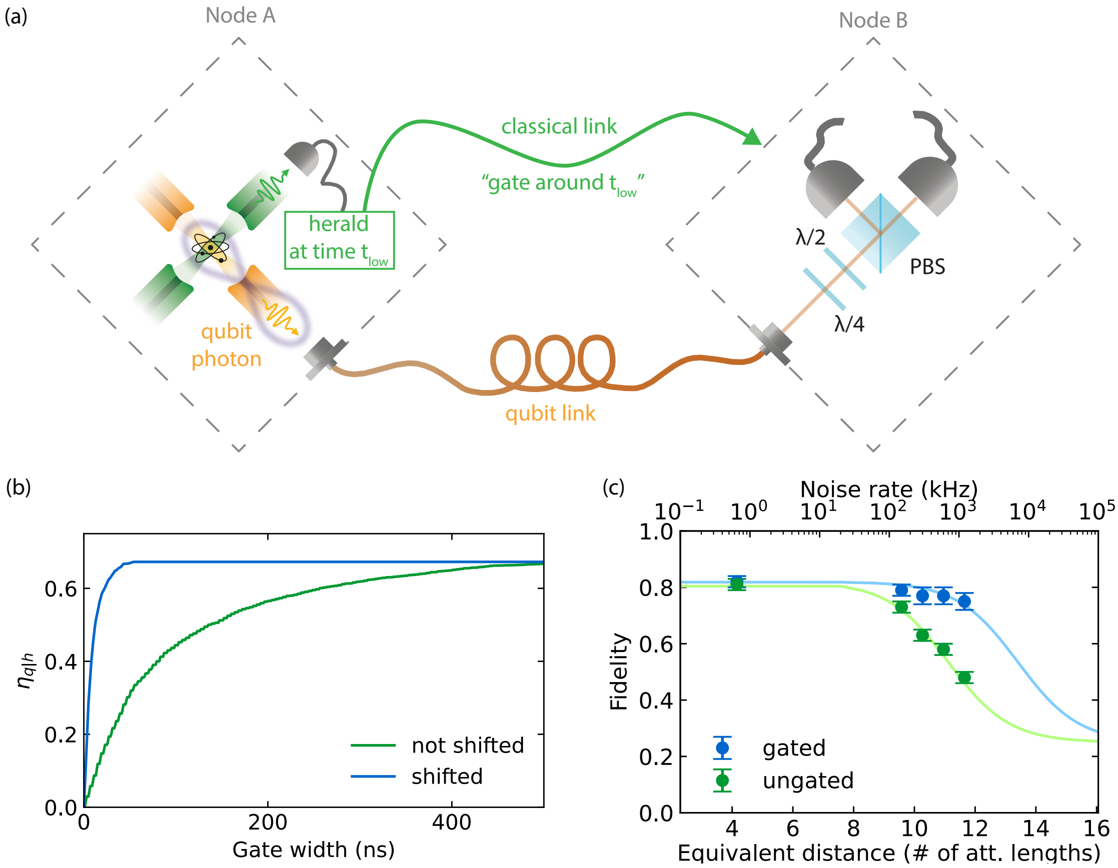


**Figure 5.7.:** Application scenario of the heralded atom-photon entanglement source. (a) At node A, a photon entangled with the atom and the herald detection time $t_{low}$ are sent via two optical links to node B, which can use the herald temporal information coming from node A to gate its detectors around $t_{low}$. A small additional delay in the quantum link, depicted by the fiber spooling, is considered to make sure that the herald temporal information arrives always first at node B. (b) Heralded qubit photon efficiency as a function of the measurement gate duration ($t_g$). The blue data uses the herald photon arrival time in order to define the gate center and the green data uses a sequence trigger. From the blue data, one can see that the efficiency starts decreasing as soon as the gate width starts being comparable with the coincidence temporal width, shown in 4.7c. (c) Entangled state fidelity as a function of the qubit measurement noise rate. For the green data points, the same photon qubit measurement gate is used for every heralded entanglement event (with gate duration $t_g = 400ns$). The blue data points indicate the situation in which the herald photon detection time information is used in order to gate the qubit photon measurement (with $t_g = 40ns$). The solid lines represent the theoretical prediction of equation 5.26 for the two scenarios.

propagates over a long distance and is significantly attenuated due to the transmission channel losses. Meanwhile, the herald detection time $t_{low}$ is communicated via a classical link to node B, which can use this information in order to shorten the detector temporal gate used in the photon qubit measurement without a decrease in the heralded efficiency. As observed in Fig. 5.7b, shifting the gate according to the herald photon detection time (blue line) allows to have higher efficiency with shorter measurement gates, compared to the situation where the gate is not temporally shifted (green line). Reducing the measurement gate reduces the amount of detection noise that is mixed with the qubit measurement signal and improves the measured entangled-state fidelity in situations where noise is a limiting factor.

In order to keep high experimental rates and short measurement times, we fix the optical length between node A and node B, and measure the entangled state fidelity as a function of the amount of noise present in the photonic qubit measurement. This is experimentally implemented by increasing the bias voltage applied to the SNSPDs until the desired dark count rates are achieved. The result of this measurement is shown in Fig. 5.7c. We observe that temporally gating (with 40ns width) the photon qubit measurement with the information received from the detection time of the herald photon improves the state fidelity (blue data), compared to the situation where the full 400ns qubit photon gate is used (green data). The bottom horizontal axis shows the equivalent distance the photon could travel in an optical fiber in order to be measured with an equivalent signal-to-noise ratio (considering a qubit measurement noise rate of 10 Hz). We observe an improvement of the entangled state fidelity from $\mathcal{F}_{q|h} = 48(2)\%$ to $\mathcal{F}_{q|h} = 75(2)\%$ in a noise-limited photon-qubit measurement situation, which is relevant in long-distance quantum communication.

This scenario can be theoretically modeled by considering an atom-photon entangled state mixed with white noise in the photonic channel. This mixture can be written in the atom-photon density matrix $\rho_{ap}$ as:

$$\hat{\rho}_{ap} = p\hat{\rho}^0_{ap} + (1-p)\hat{\rho}^0_a \otimes \frac{\mathbb{I}_p}{2} \tag{5.24}$$

where $p$ is the fraction of events in which the measurement is performed on the original atom-photon state $\hat{\rho}^0_{ap}$ and $\hat{\rho}^0_a$ is the remaining atomic state when the photonic state is traced out. When considering an entangled Bell state mixed with white noise, $\hat{\rho}^0_a = \mathrm{Tr}_p\{\hat{\rho}^0_{ap}\} = \mathbb{I}_a/2$. Therefore we can rewrite the final atom-photon state as:

$$\hat{\rho}_{ap} = p\hat{\rho}^0_{ap} + (1-p)\frac{1}{4}\mathbb{I}_a \otimes \mathbb{I}_p \tag{5.25}$$

The fidelity of this state with respect to a Bell state is

$$\mathcal{F} = p\mathcal{F}_0 + \frac{1-p}{4} \tag{5.26}$$

where $\mathcal{F}_0$ is the fidelity of $\hat{\rho}^0_{ap}$ with respect to a Bell state. This equation shows the explicit dependency of the fidelity on the signal-to-noise ratio (i.e. $p$) and it is plotted in Fig. 5.7c

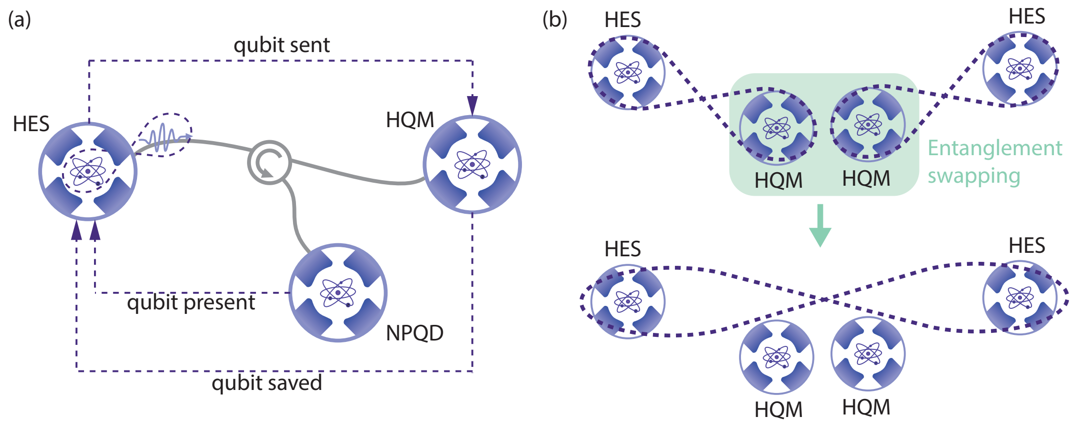


**Figure 5.8.:** Envisioned repeater architecture. (a) An entanglement-distribution link composed by a heralded entanglement source (HES), a non-destructive photonic qubit detector (NPQD)[74], and a heralded quantum memory (HQM)[84]. Whilst the HES informs the HQM about the successful generation of the entangled state, the NPQD detects the presence of the photonic qubit along the propagation path and communicates the result to the HES, which can attempt a new photon emission upon photon loss. Eventually, the HQM communicates to the HES the successful storage of a photon in the atomic memory upon the detection of a herald photon. All these components have been realized with the same crossed fiber cavities setup. In a repeater architecture, two such links are connected by an entanglement swapping operation between the HQMs (b). This operation leaves the two HESs in an entangled state.

for the full (blue solid line) and the gated (green solid line) detection window ad Node B, showing a good agreement between the theoretical model and the measured data points.

We anticipate that our source will be useful in a quantum repeater architecture which we envision as consisting of heralded atom-photon entanglement sources, non-destructive qubit trackers [74] and heralded quantum memories [84] - see Fig. 5.8. In such an architecture, herald signals provided when generating, propagating and receiving entanglement will allow one to track and quickly react on source, channel and storage errors. All the mentioned components can be realized with single atoms trapped in crossed fiber cavities, making our system highly suitable for next-generation entanglement-distribution schemes.

# 6. Summary and outlook

This thesis presents a cavity-enhanced atomic cascade platform that enables both efficient generation of photon pairs and heralded atom-photon entanglement, establishing key capabilities for modular and scalable quantum networks.

The experimental system builds upon a setup developed in 2019 and described in detail in earlier works [102, 103]. Over the course of this PhD, significant upgrades and refinements were made, particularly in the loading, trapping and manipulation of single atoms and their positioning within the cavity modes. A light-induced atom desorption (LIAD) technique was implemented for efficient atom loading, followed by precise positioning of atoms using a 852 nm standing-wave optical trap steered by a galvo scanner. A more elaborate internal state manipulation was realized via microwave radiation and multiple laser fields, and a robust cavity-enhanced fluorescence detection scheme allowed for high-fidelity atomic state readout.

Building on this robust atom–cavity interface, we explored the generation of photon pairs via cascaded atomic decay processes. In contrast to typical bulk or nonlinear crystal-based sources [60, 61], the two photons in our scheme are emitted into distinct cavity modes, allowing their spectral and temporal properties to be controlled independently via the cavity parameters. Moreover, this source has a high single photon purity and a high in-fiber efficiency of 22%, making it the most efficient in-fiber photon pair source developed sofar. A comprehensive theoretical model incorporating the full atomic level structure and experimental parameters was developed and used to guide the experimental implementation. We characterized the system's photon statistics, temporal correlations, and pair generation efficiency, finding excellent agreement with the theoretical predictions. Additionally, the strong-coupling regime of the atom–cavity interaction was explored, revealing STIRAP-like dynamics in the ladder decay, in which the atom can be coherently transported from ladder excited state to the ground state without populating the intermediate state.

The final part of this work demonstrates the heralded generation of entanglement between a single atom and a single photon. We showed that detecting the lower photon in the atomic cascade projects the atom into an entangled state with the remaining photon, effectively heralding the entanglement creation. This could be accomplished thanks to the birefringence of the lower cavity, which therefore enhances specific atomic decay channels. This process was characterized through quantum state tomography, coherence measurements, and correlation analysis. We demonstrate that the local heralding provides precise timing information that is used to suppress measurement noise through time-gated detection. These results demonstrate the potential of our system as a node in future long-distance quantum communication networks.

Looking ahead, the crossed cavity architecture opens the door to a wide range of quantum optics and quantum information experiments. One natural extension of this work is the implementation of remote atom-atom entanglement, using the heralded photon as a flying qubit to establish entanglement between distant nodes. The precise timing information provided by the local herald allows the remote node to gate its photon detector at the correct arrival time, thereby suppressing noise at the receiver. Moreover, combining our heralding techniques with recently developed multiplexed entanglement sources based on registers of intra-cavity emitters [160, 161] promises to reduce communication time and speed up repetition rates.

A particularly promising direction is the implementation of atom–photon entanglement at telecom wavelength, following the proposal by Uphoff et al. [85]. This would involve cascading through an atomic transition emitting a telecom photon into one cavity, followed by a herald photon at 795 nm into the other. Such a configuration could eliminate the need for quantum frequency conversion units [49, 162, 163, 164], which are currently a bottleneck for large-scale quantum networks. While this would require the replacement of a new telecom-compatible cavity into the current setup, the potential gains in network performance and compatibility with existing fiber infrastructure are substantial.

Another compelling direction is the investigation of an atom-photon gate via reflection onto the birefringent cavity. Although a gate between a polarization photonic qubit and a stationary atom coupled to a non-birefringent cavity has been realized in other platforms [165], the implementation with a birefringent one is yet to be proven. One of the main advantages of using a birefringent cavity is that the atomic qubit can be chosen to be in the clock states, boosting the atomic coherence time, typically limited by magnetic field fluctuations and atomic temperature. Additionally, by combining such a gate with the heralded memory realized by M. Brekenfeld [84], it is possible to implement nonlinear photon-photon interactions mediated by the single atom.

In conclusion, this thesis demonstrates that a single atom coupled to two fiber cavities constitutes a powerful and versatile platform for quantum information science. The experiments reported here showcase its capabilities for photon pair generation and heralded entanglement, and these results, together with those shown in previous works employing this platform [84, 74], provide a solid foundation for future work targeting distributed quantum networks, nonlinear quantum optics, and long-distance quantum communication.

# A. AC Stark shifts

Calculated AC Stark shifts induced by the red standing wave trap. For this calculation, the following assumptions are made: linear polarization, 852 nm wavelength, a power of 130 mW, transmission through the optical components of 95%, and a waist of 10.5 µm at the atom. The AC Stark shifts induced on the $5D_{5/2}$ levels are not calculated, since the trap is switched-off during the atomic excitation. Eq.s 3.15-3.19 are used.

| Fine level | $F$ | $m_F$ | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $5P_{3/2}$ | 3 | 2.8 | 13.2 | 19.0 | 20.9 | 19.0 | 13.2 | 2.8 |
| | 2 | | 12.5 | 13.1 | 13.3 | 13.1 | 12.5 | |
| | 1 | | | 16.4 | 4.8 | 16.4 | | |
| | 0 | | | | 12.4 | | | |
| $5S_{1/2}$ | 2 | | -27.6 | -27.6 | -27.6 | -27.6 | -27.6 | |
| | 1 | | | -27.6 | -27.6 | -27.6 | | |

**Table A.1.:** AC Stark shifts induced by the trap in frequency units (MHz).

# B. Hamiltonian derivation for multi-level and multi-cavity system

We consider the scenario represented in Fig. 4.1b. The atom has four energy levels, three of which are in a ladder configuration (ground $|g\rangle$, intermediate $|i\rangle$, and excited $|e\rangle$) supporting the two transitions that are coupled to the optical cavities. The atom has an additional ground state $|g_0\rangle$, which is coupled to the excited state $|e\rangle$ by a drive laser with Rabi frequency $\Omega_D$ and angular frequency $\omega_D$. The angular frequencies associated with the atom and the cavity field energy are $\omega_{g_0g}$, $\omega_{g_0i}$, $\omega_{g_0e}$ and $\omega_u$, $\omega_l$, respectively. Here, $g_u(g_l)$ is the coupling rate of the cavity that couples to the upper $|i\rangle - |e\rangle$ (lower $|g\rangle - |i\rangle$) transition, $\hat{a}_u$ ($\hat{a}_l$) and $\hat{a}_u^\dagger$ ($\hat{a}_l^\dagger$) are the annihilation and creation operators for the upper (lower) cavity mode, and $\hat{\sigma}_{ie}$ and $\hat{\sigma}_{gi}$ are the atomic operators which describe the transition from $|e\rangle$ to $|i\rangle$ and from $|i\rangle$ to $|g\rangle$, respectively. The atomic operator that describes the transition from $|g_0\rangle$ to $|e\rangle$ is $\hat{\sigma}_{g_0e}^\dagger$.

In general, we can write the Hamiltonian as consisting of its unperturbed energy term $\hat{H}_0$ and an interaction term $\hat{V}$, i.e. $\hat{H} = \hat{H}_0 + \hat{V}$. If we set $\hbar = 1$, we get

$$\hat{H}_0 = \omega_{g_0g}\hat{\sigma}_{g_0g}^\dagger\hat{\sigma}_{g_0g} + \omega_{g_0i}\hat{\sigma}_{g_0i}^\dagger\hat{\sigma}_{g_0i} + \omega_{g_0e}\hat{\sigma}_{g_0e}^\dagger\hat{\sigma}_{g_0e} + \omega_u\hat{a}_u^\dagger\hat{a}_u + \omega_l\hat{a}_l^\dagger\hat{a}_l \tag{B.1}$$

$$\hat{V} = g_u(\hat{a}_u^\dagger\hat{\sigma}_{ie} + \hat{a}_u\hat{\sigma}_{ie}^\dagger) + g_l(\hat{a}_l^\dagger\hat{\sigma}_{gi} + \hat{a}_l\hat{\sigma}_{gi}^\dagger) + \Omega_D(\hat{\sigma}_{g_0e} + \hat{\sigma}_{g_0e}^\dagger)(e^{i\omega_D t} + e^{-i\omega_D t}) \tag{B.2}$$

Here, we provide a general method to derive an effective, time-independent Hamiltonian.

**Interaction picture:** The transformation to the interaction picture is defined as the following [166]

$$\hat{H}^I = \hat{U}_0^\dagger\hat{V}\hat{U}_0 \tag{B.3}$$

where $\hat{U}_0$ is the propagation operator, defined as

$$\hat{U}_0 = e^{-i\hat{H}_0 t} \tag{B.4}$$

Using the unperturbed Hamiltonian of eq. B.1, we derive

$$\hat{U}_0 = e^{-i\omega_{g_0g}\hat{\sigma}_{g_0g}^\dagger\hat{\sigma}_{g_0g}t}e^{-i\omega_{g_0i}\hat{\sigma}_{g_0i}^\dagger\hat{\sigma}_{g_0i}t}e^{-i\omega_{g_0e}\hat{\sigma}_{g_0e}^\dagger\hat{\sigma}_{g_0e}t}e^{-i\omega_u\hat{a}_u^\dagger\hat{a}_u t}e^{-i\omega_l\hat{a}_l^\dagger\hat{a}_l t} \tag{B.5}$$

We consider now the Baker-Hausdorff lemma [166] for the general operators $\hat{A}$ and $\hat{B}$

$$e^{\alpha \hat{A}} \hat{B} e^{-\alpha \hat{A}} = \hat{B} + \alpha[\hat{A}, \hat{B}] + \frac{\alpha^2}{2!}[\hat{A},[\hat{A},\hat{B}]] + ... \tag{B.6}$$

By recalling the commutation rule $[\hat{a},\hat{a}^\dagger] = \mathbb{I}$, we derive

$$[\hat{a},\hat{a}^\dagger \hat{a}] = [\hat{a},\hat{a}^\dagger]\hat{a} = \hat{a} \tag{B.7}$$

By combining eq. (B.6) with eq. (B.7), one obtains

$$e^{i\omega \hat{a}^\dagger \hat{a} t} \hat{a} e^{-i\omega t \hat{a}^\dagger \hat{a} t} = \hat{a} - i\omega t \hat{a} + \frac{(i\omega)^2}{2!}\hat{a} + ... = \hat{a} e^{-i\omega t} \tag{B.8}$$

Analogously, for the atom operators we have $[\hat{\sigma},\hat{\sigma}^\dagger] = \mathbb{I}$, yielding to

$$e^{i\omega_a \hat{\sigma}^\dagger \hat{\sigma} t} \hat{\sigma} e^{-i\omega_a \hat{\sigma}^\dagger \hat{\sigma} t} = \hat{\sigma} e^{-i\omega_a t} \tag{B.9}$$

The interaction Hamiltonian is thereby obtained

$$\begin{aligned} \hat{H}^I = {} & g_u(\hat{a}_u^\dagger \hat{\sigma}_{ie} e^{i\Delta_u t} + \hat{a}_u \hat{\sigma}_{ie}^\dagger e^{-i\Delta_u t}) + g_l(\hat{a}_l^\dagger \hat{\sigma}_{gi} e^{i\Delta_l t} + \hat{a}_l \hat{\sigma}_{gi}^\dagger e^{-i\Delta_l t}) + \\ & + \Omega_D(\hat{\sigma}_{g_0 e} e^{i\Delta_D t} + \hat{\sigma}_{g_0 e}^\dagger e^{-i\Delta_D t})(\hat{\sigma}_{g_0 e} e^{-i(\omega_D + \omega_{g0e})t} + \hat{\sigma}_{g_0 e}^\dagger e^{i(\omega_D + \omega_{g0e})t}) \end{aligned} \tag{B.10}$$

where $\Delta_u = \omega_u - \omega_{ie}$, $\Delta_l = \omega_l - \omega_{gi}$ and $\Delta_D = \omega_D - \omega_{g_0 e}$. By using the rotating wave approximation (RWA) [167, 168], one obtains

$$\begin{aligned} \hat{H}^I = {} & g_u(\hat{a}_u^\dagger \hat{\sigma}_{ie} e^{i\Delta_u t} + \hat{a}_u \hat{\sigma}_{ie}^\dagger e^{-i\Delta_u t}) + g_l(\hat{a}_l^\dagger \hat{\sigma}_{gi} e^{i\Delta_l t} + \hat{a}_l \hat{\sigma}_{gi}^\dagger e^{-i\Delta_l t}) + \\ & + \Omega_D(\hat{\sigma}_{g_0 e} e^{i\Delta_D t} + \hat{\sigma}_{g_0 e}^\dagger e^{-i\Delta_D t}) \end{aligned} \tag{B.11}$$

**Rotating frame:** Under an arbitrary time dependent transformation $\hat{U} = e^{-i\hat{O}t}$ of the wavefunction $|\tilde{\psi}\rangle = U\,|\psi\rangle$, where $\hat{O}$ is an hermitian operator, the Hamiltonian $\hat{H}$ transforms as [168]

$$\hat{\tilde{H}} = e^{-i\hat{O}t} \hat{H} e^{i\hat{O}t} + \hat{O} \tag{B.12}$$

The choice of the operator $\hat{O}$ is arbitrary, and since it is convenient to transform eq. B.11 into a time-independent Hamiltonian, it pays off to choose

$$\hat{O} = \Delta_l \hat{a}_l^\dagger \hat{a}_l + (\Delta_u - \Delta_D)\hat{a}_u^\dagger \hat{a}_u - \Delta_D \hat{\sigma}_{g_0 e}^\dagger \hat{\sigma}_{g_0 e} \tag{B.13}$$

The effective time-independent Hamiltonian is then given by

$$\hat{\tilde{H}}^I = g_u(\hat{a}_u^\dagger \hat{\sigma}_{ie} + \hat{a}_u \hat{\sigma}_{ie}^\dagger) + g_l(\hat{a}_l^\dagger \hat{\sigma}_{gi} + \hat{a}_l \hat{\sigma}_{gi}^\dagger) + \Delta_u \hat{a}_u^\dagger \hat{a}_u + \Delta_l \hat{a}_l^\dagger \hat{a}_u + \\ + \Omega_D(\hat{\sigma}_{g0e}^\dagger + \hat{\sigma}_{g0e}) - \Delta_D \hat{\sigma}_{g0e}^\dagger \hat{\sigma}_{g0e} - \Delta_D \hat{a}_u^\dagger \hat{a}_u \quad \text{(B.14)}$$

# C. Modeling the $2 \leftrightarrow 3'$ Spectroscopy

We consider, for simplicity, a two-level atom with states $|g\rangle$ and $|e\rangle$ coupled to the SC. Since its modes are split by 500 MHz and mediate the photonic emission of different atomic transitions, all taking part in the spectroscopy, the presence of the SC is modeled by including two cavity couplings $g_\pi$ and $g_V$, representing the coupling constants for the $\pi$ and $V$ polarized modes, respectively. The two cavity detunings from the atomic transition are $\Delta_\pi$ and $\Delta_V$.

Here, $\hat{a}_\pi^\dagger$ ($\hat{a}_V^\dagger$) is the creation operator for the $\pi$ ($V$) cavity mode and $\hat{\sigma}_{ge}$ is the atomic operator which describes the transition from $|e\rangle$ to $|g\rangle$. Additionally, the atom is driven by an external laser with Rabi frequency $\Omega_D$ and detuning from the atomic resonance $\Delta_D$.

With the method discussed in Appendix B, we derive an effective time-independent Hamiltonian for the scenario presented

$$\begin{aligned}\hat{H}(m_F) = \Delta_\pi(m_F)\hat{a}_\pi^\dagger\hat{a}_\pi + g_\pi(m_F)(\hat{a}_\pi^\dagger\hat{\sigma}_{ge} + \hat{a}_\pi\hat{\sigma}_{ge}^\dagger) + \Delta_V(m_F)\hat{a}_V^\dagger\hat{a}_V + \\ + g_V(m_F)(\hat{a}_V^\dagger\hat{\sigma}_{ge} + \hat{a}_V\hat{\sigma}_{ge}^\dagger) + \Omega_D(\hat{\sigma}_{ge} + \hat{\sigma}_{ge}^\dagger) - \Delta_D(m_F)(\hat{\sigma}_{ge}^\dagger\hat{\sigma}_{ge} + \hat{a}_\pi^\dagger\hat{a}_\pi + \hat{a}_V^\dagger\hat{a}_V)\end{aligned} \quad (C.1)$$

where $m_F$ labels the Zeeman level in the excited manifold $5P_{3/2}, F = 3$. Since the transition strength between the excited and ground states depends on the Zeeman states considered and on the polarization, $g_\pi$ and $g_V$ depend on the Zeeman states, according to the transitions' Clebsch-Gordan coefficients [89].

The expected spectrum is obtained by simulating the steady state of the Master Equation corresponding to the Hamiltonian in Eq. C.1, as a function of the drive detuning. For each atomic Zeeman sublevel $m_F$, the steady-state intracavity photon number is computed for both cavity modes using the expressions $2\kappa\left\langle \hat{a}_\pi^\dagger\hat{a}_\pi \right\rangle$ and $2\kappa\left\langle \hat{a}_V^\dagger\hat{a}_V \right\rangle$. These contributions are then summed over all relevant $m_F$ states to produce the total spectrum. The simulation includes collapse operators accounting for both spontaneous emission into free space and photon losses from each cavity mode.

Naturally, this approach involves strong approximations. While it may not fully capture the detailed dynamics of the system, it provides useful insight into the overall spectral behavior. As parameters for the simulation we set a normalized coupling $g = 2\pi \cdot 22$ MHz $/\sqrt{1/24}$, a $\pi$-mode detuning $\Delta_\pi = -2\pi \cdot 288$ MHz and a $V$-mode detuning $\Delta_V = 2\pi \cdot 212$ MHz ( $\Delta_V - \Delta_\pi = 500$ MHz).

# D. Electric dipole matrix elements for the $5P_{3/2} \rightarrow 5D_{5/2}$ transition

The dipole matrix element that couples the two hyperfine sublevels $|F, m_F\rangle$ and $|F', m'_F\rangle$ can be calculated as [89]

$$\left\langle F, m_F | e r_q | F', m'_F \right\rangle = \left\langle F || e\boldsymbol{r} || F' \right\rangle (-1)^{F'-1+m_F} \sqrt{2F+1} \begin{pmatrix} F' & 1 & F \\ m'_F & q & -m_F \end{pmatrix} \tag{D.1}$$

where $e\boldsymbol{r}$ is the electric dipole matrix, $q$ is an index labeling the component of $\boldsymbol{r}$ in the spherical basis, the doubled bars indicate the reduced matrix element and the parentheses indicate the Wigner 3-j symbol.

The reduced matrix element can be written as

$$\left\langle F || e\boldsymbol{r} || F' \right\rangle = \left\langle J || e\boldsymbol{r} || J' \right\rangle (-1)^{F'+J+1+I} \sqrt{(2F'+1)(2J+1)} \begin{Bmatrix} J & J' & 1 \\ F' & F & I \end{Bmatrix} \tag{D.2}$$

where $J$ is the total angular momentum quantum number, and $I$ is the nuclear spin angular momentum. The reduced density matrix element $\langle J || e\boldsymbol{r} || J' \rangle$ depends only on the $J$, $L$ (orbital angular momentum) and $S$ (total spin) quantum numbers and its value for the $5P_{3/2} \rightarrow 5D_{5/2}$ transition is $\langle J = 3/2 || e\boldsymbol{r} || J' = 5/2 \rangle = 0.992 e a_0$ [169].

The hyperfine dipole matrix elements calculated via equations D.1-D.2 are given in the tables D.1-D.4 as multiples of $\langle J = 3/2 || e\boldsymbol{r} || J' = 5/2 \rangle$.

| | | $m'_F$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 |
| $F'=2$ | $\sigma^+$ | | | $\sqrt{\frac{1}{105}}$ | $\sqrt{\frac{2}{315}}$ | $\sqrt{\frac{2}{525}}$ | $\sqrt{\frac{1}{525}}$ | $\sqrt{\frac{1}{1575}}$ | | |
| | $\pi$ | | | $\sqrt{\frac{1}{315}}$ | $\sqrt{\frac{8}{1575}}$ | $\sqrt{\frac{1}{175}}$ | $\sqrt{\frac{8}{1575}}$ | $\sqrt{\frac{1}{315}}$ | | |
| | $\sigma^-$ | | | $\sqrt{\frac{1}{1575}}$ | $\sqrt{\frac{1}{525}}$ | $\sqrt{\frac{2}{525}}$ | $\sqrt{\frac{2}{315}}$ | $\sqrt{\frac{1}{105}}$ | | |
| $F'=3$ | $\sigma^+$ | | | $\sqrt{\frac{1}{30}}$ | $\sqrt{\frac{1}{18}}$ | $\sqrt{\frac{1}{15}}$ | $-\sqrt{\frac{1}{15}}$ | $-\sqrt{\frac{1}{18}}$ | $-\sqrt{\frac{1}{30}}$ | |
| | $\pi$ | | $-\sqrt{\frac{1}{10}}$ | $-\sqrt{\frac{2}{45}}$ | $-\sqrt{\frac{1}{90}}$ | | $-\sqrt{\frac{1}{90}}$ | $-\sqrt{\frac{2}{45}}$ | $-\sqrt{\frac{1}{10}}$ | |
| | $\sigma^-$ | | $-\sqrt{\frac{1}{30}}$ | $-\sqrt{\frac{1}{18}}$ | $-\sqrt{\frac{1}{15}}$ | $-\sqrt{\frac{1}{15}}$ | $\sqrt{\frac{1}{18}}$ | $\sqrt{\frac{1}{30}}$ | | |
| $F'=4$ | $\sigma^+$ | | | $\sqrt{\frac{1}{42}}$ | $\sqrt{\frac{1}{14}}$ | $\sqrt{\frac{1}{7}}$ | $\sqrt{\frac{5}{21}}$ | $\sqrt{\frac{5}{14}}$ | $\sqrt{\frac{1}{2}}$ | $\sqrt{\frac{2}{3}}$ |
| | $\pi$ | | $-\sqrt{\frac{1}{6}}$ | $-\sqrt{\frac{2}{7}}$ | $-\sqrt{\frac{5}{14}}$ | $-\sqrt{\frac{8}{21}}$ | $-\sqrt{\frac{5}{14}}$ | $-\sqrt{\frac{2}{7}}$ | $-\sqrt{\frac{1}{6}}$ | |
| | $\sigma^-$ | $\sqrt{\frac{2}{3}}$ | $\sqrt{\frac{1}{2}}$ | $\sqrt{\frac{5}{14}}$ | $\sqrt{\frac{5}{21}}$ | $\sqrt{\frac{1}{7}}$ | $\sqrt{\frac{1}{14}}$ | $\sqrt{\frac{1}{42}}$ | | |

**Table D.1.:** Hyperfine dipole matrix elements for the transitions $F = 3 \rightarrow F'$ as multiples of $\langle J = 3/2||e\boldsymbol{r}||J' = 5/2\rangle$. $\sigma^+ : m'_F = m_F + 1$, $\pi : m'_F = m_F$, $\sigma^- : m'_F = m_F - 1$.

| | | $m'_F$ | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $F'=1$ | $\sigma^+$ | | | $\sqrt{\frac{1}{50}}$ | $\sqrt{\frac{1}{100}}$ | $\sqrt{\frac{1}{300}}$ | | |
| | $\pi$ | | | $\sqrt{\frac{1}{100}}$ | $\sqrt{\frac{1}{75}}$ | $\sqrt{\frac{1}{100}}$ | | |
| | $\sigma^-$ | | | $\sqrt{\frac{1}{300}}$ | $\sqrt{\frac{1}{100}}$ | $\sqrt{\frac{1}{50}}$ | | |
| $F'=2$ | $\sigma^+$ | | | $\sqrt{\frac{7}{90}}$ | $\sqrt{\frac{7}{60}}$ | $-\sqrt{\frac{7}{60}}$ | $-\sqrt{\frac{7}{90}}$ | |
| | $\pi$ | | $-\sqrt{\frac{7}{45}}$ | $-\sqrt{\frac{7}{180}}$ | | $-\sqrt{\frac{7}{180}}$ | $-\sqrt{\frac{7}{45}}$ | |
| | $\sigma^-$ | | $-\sqrt{\frac{7}{90}}$ | $-\sqrt{\frac{7}{60}}$ | $-\sqrt{\frac{7}{60}}$ | $\sqrt{\frac{7}{90}}$ | | |
| $F'=3$ | $\sigma^+$ | | | $\sqrt{\frac{8}{225}}$ | $\sqrt{\frac{8}{75}}$ | $\sqrt{\frac{16}{75}}$ | $\sqrt{\frac{16}{45}}$ | $\sqrt{\frac{8}{15}}$ |
| | $\pi$ | | $-\sqrt{\frac{8}{45}}$ | $-\sqrt{\frac{64}{225}}$ | $-\sqrt{\frac{8}{25}}$ | $-\sqrt{\frac{64}{225}}$ | $-\sqrt{\frac{8}{45}}$ | |
| | $\sigma^-$ | $\sqrt{\frac{8}{15}}$ | $\sqrt{\frac{16}{45}}$ | $\sqrt{\frac{16}{75}}$ | $\sqrt{\frac{8}{75}}$ | $\sqrt{\frac{8}{225}}$ | | |

**Table D.2.:** Hyperfine dipole matrix elements for the transitions $F = 2 \rightarrow F'$ as multiples of $\langle J = 3/2||e\boldsymbol{r}||J' = 5/2\rangle$. $\sigma^+ : m'_F = m_F + 1$, $\pi : m'_F = m_F$, $\sigma^- : m'_F = m_F - 1$.

| | | $m'_F$ | | | | |
|---|---|---|---|---|---|---|
| | | -2 | -1 | 0 | 1 | 2 |
| $F'=1$ | $\sigma^+$ | | | $\sqrt{\frac{3}{20}}$ | $-\sqrt{\frac{3}{20}}$ | |
| | $\pi$ | | $-\sqrt{\frac{3}{20}}$ | | $-\sqrt{\frac{3}{20}}$ | |
| | $\sigma^-$ | | $-\sqrt{\frac{3}{20}}$ | $-\sqrt{\frac{3}{20}}$ | | |
| $F'=2$ | $\sigma^+$ | | | $\sqrt{\frac{7}{100}}$ | $-\sqrt{\frac{21}{100}}$ | $\sqrt{\frac{21}{50}}$ |
| | $\pi$ | | $-\sqrt{\frac{21}{100}}$ | $-\sqrt{\frac{7}{25}}$ | $-\sqrt{\frac{21}{100}}$ | |
| | $\sigma^-$ | $\sqrt{\frac{21}{50}}$ | $-\sqrt{\frac{21}{100}}$ | $\sqrt{\frac{7}{100}}$ | | |

**Table D.3.:** Hyperfine dipole matrix elements for the transitions $F = 1 \to F'$ as multiples of $\langle J = 3/2||e\boldsymbol{r}||J' = 5/2\rangle$. $\sigma^+ : m'_F = m_F + 1$, $\pi : m'_F = m_F$, $\sigma^- : m'_F = m_F - 1$.

| | | $m'_F$ | | |
|---|---|---|---|---|
| | | -1 | 0 | 1 |
| $F'=1$ | $\sigma^+$ | | | $\sqrt{\frac{1}{3}}$ |
| | $\pi$ | | $-\sqrt{\frac{1}{3}}$ | |
| | $\sigma^-$ | $\sqrt{\frac{1}{3}}$ | | |

**Table D.4.:** Hyperfine dipole matrix elements for the transitions $F = 0 \to F'$ as multiples of $\langle J = 3/2||e\boldsymbol{r}||J' = 5/2\rangle$. $\sigma^+ : m'_F = m_F + 1$, $\pi : m'_F = m_F$, $\sigma^- : m'_F = m_F - 1$.

# References


[1] Laplace, P. S. *Théorie analytique des probabilités* (1812).

[2] Kepler, J. *Astronomia Nova* (1609).

[3] Laplace, P. S. *Exposition du système du monde* (Bibliothèque nationale de France, département Réserve des livres rares, RES P-V-643, 1796).

[4] Clapeyron, B. P. E. *Mémoire sur la puissance motrice de la chaleur*. *Journal de l'École Polytechnique* **XIV**, 153–90 (1835).

[5] Maxwell, J. C. *IV. On the dynamical theory of gases*. *Philosophical Transactions of the Royal Society* **157** (1867).

[6] Boltzmann, L. *Über die mechanische Bedeutung des zweiten Hauptsatzes der Wärmetheorie*, 9–33. Cambridge Library Collection - Physical Sciences (Cambridge University Press, 2012). First published in year 1866.

[7] Einstein, A., Podolsky, B. & Rosen, N. *Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?* *Phys. Rev.* **47**, 777–780 (1935).

[8] Schrödinger, E. *Discussion of Probability Relations between Separated Systems*. *Mathematical Proceedings of the Cambridge Philosophical Society* **31**, 555–563 (1935).

[9] Schrödinger, E. *Probability relations between separated systems*. *Mathematical Proceedings of the Cambridge Philosophical Society* **32**, 446–452 (1936).

[10] Einstein, A., Born, H. & Born, M. *The Born-Einstein letters: correspondence between Albert Einstein and Max and Hedwig Born from 1916–1955, with commentaries by Max Born* (Macmillan, 1971). Pg. 158.

[11] Bell, J. S. *On the Einstein Podolsky Rosen paradox*. *Physics Physique Fizika* **1**, 195–200 (1964).

[12] Freedman, S. J. & Clauser, J. F. *Experimental Test of Local Hidden-Variable Theories*. *Phys. Rev. Lett.* **28**, 938–941 (1972).

[13] Aspect, A., Grangier, P. & Roger, G. *Experimental Tests of Realistic Local Theories via Bell's Theorem*. *Phys. Rev. Lett.* **47**, 460–463 (1981).

[14] Aspect, A., Dalibard, J. & Roger, G. *Experimental Test of Bell's Inequalities Using Time-Varying Analyzers*. *Phys. Rev. Lett.* **49**, 1804–1807 (1982).

[15] Pan, J.-W., Bouwmeester, D., Daniell, M., Weinfurter, H. & Zeilinger, A. *Experimental test of quantum nonlocality in three-photon Greenberger–Horne–Zeilinger entanglement*. *Nature* **403**, 515–519 (2000).

[16] Weihs, G., Jennewein, T., Simon, C., Weinfurter, H. & Zeilinger, A. *Violation of Bell's Inequality under Strict Einstein Locality Conditions.* *Phys. Rev. Lett.* **81**, 5039–5043 (1998).

[17] Yin, J. *et al.* *Lower Bound on the Speed of Nonlocal Correlations without Locality and Measurement Choice Loopholes.* *Phys. Rev. Lett.* **110**, 260407 (2013).

[18] Zhang, Z., You, C., na Loaiza, O. S. M., Fickler, R., de J. León-Montiel, R., Torres, J. P., Humble, T. S., Liu, S., Xia, Y. & Zhuang, Q. *Entanglement-based quantum information technology: a tutorial.* *Adv. Opt. Photon.* **16**, 60–162 (2024).

[19] Feynman, R. P. *Simulating physics with computers.* *Int. J. Theor. Phys.* **21**, 467–488 (1982).

[20] Preskill, J. *Quantum computing and the entanglement frontier.* *arXiv:1203.5813[quant-ph]* (2012).

[21] Arute, F. *et al.* *Quantum supremacy using a programmable superconducting processor.* *Nature* **574**, 505–510 (2019).

[22] Gibney, E. *Quantum cryptography: Public key distribution and coin tossing.* *Nature* **587**, 342–343 (2020).

[23] Shor, P. W. *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer.* *SIAM J. on Comput.* **26**, 1484–1509 (1997).

[24] Grover, L. K. *A fast quantum mechanical algorithm for database search.* *arXiv:quant-ph/9605043* (1996).

[25] Bennett, C. H. & Brassard, G. *Quantum cryptography: Public key distribution and coin tossing.* *Theoretical Computer Science* **560**, 7–11 (2014). Theoretical Aspects of Quantum Cryptography – celebrating 30 years of BB84.

[26] Wootters, W. K. & Zurek, W. H. *A single quantum cannot be cloned.* *Nature* **299**, 802–803 (1982).

[27] Wiesner, S. *Conjugate coding.* *SIGACT News* **15**, 78–88 (1983).

[28] Menezes, A. J., van Oorschot, P. C. & Vanstone, S. A. *Handbook of Applied Cryptography* (CRC Press, 1996).

[29] Kimble, H. J. *The quantum internet.* *Nature* **453**, 1023–1030 (2008).

[30] Wei, S.-H. *et al.* *Towards Real-World Quantum Networks: A Review.* *Laser & Photonics Reviews* **16**, 2100219 (2022). `https://onlinelibrary.wiley.com/doi/pdf/10.1002/lpor.202100219`.

[31] Wehner, S., Elkouss, D. & Hanson, R. *Quantum internet: A vision for the road ahead.* *Science* **362** (2018).

[32] Azuma, K., Economou, S. E., Elkouss, D., Hilaire, P., Jiang, L., Lo, H.-K. & Tzitrin, I. *Quantum repeaters: From quantum networks to the quantum internet.* *Rev. Mod. Phys.* **95**, 045006 (2023).

[33] Duan, L.-M., Lukin, M. D., Cirac, J. & Zoller, P. *Long-distance quantum communication with atomic ensembles and linear optics.* *Nature* **414**, 413–418 (2001).

[34] Li, Z.-D. *et al.* *Experimental quantum repeater without quantum memory.* *Nature Photonics* **13**, 644–648 (2019).

[35] Langenfeld, S., Thomas, P., Morin, O. & Rempe, G. *Quantum Repeater Node Demonstrating Unconditionally Secure Key Distribution.* *Phys. Rev. Lett.* **126**, 230506 (2021).

[36] Buhrman, H. & Röhrig, H. Distributed Quantum Computing. In *Mathematical Foundations of Computer Science 2003*, 1–20 (Springer Berlin Heidelberg, 2003).

[37] Monroe, C., Raussendorf, R., Ruthven, A., Brown, K. R., Maunz, P., Duan, L.-M. & Kim, J. *Large-scale modular quantum-computer architecture with atomic memory and photonic interconnects.* *Phys. Rev. A* **89**, 022317 (2014).

[38] Brown, K. R., Kim, J. & Monroe, C. *Co-designing a scalable quantum computer with trapped atomic ions.* *npj Quantum Inf* **2** (2016).

[39] Main, D., Drmota, P., Nadlinger, D., Ainley, E., Agrawal, A., Nichol, B. C., Srinivas, R., Araned, G. & Lucas, D. *Distributed quantum computing across an optical network link.* *Nature* **638**, 383–388 (2025).

[40] Proctor, T. J., Knott, P. A. & Dunningham, J. A. *Multiparameter Estimation in Networked Quantum Sensors.* *Phys. Rev. Lett.* **120**, 080501 (2018).

[41] Rubio, J., Knott, P. A., Proctor, T. J. & Dunningham, J. A. *Quantum sensing networks for the estimation of linear functions.* *Journal of Physics A: Mathematical and Theoretical* **53**, 344001 (2020).

[42] Hensen, B. *et al.* *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres.* *Nature* **526**, 682–686 (2015).

[43] Moehring, D. L., Maunz, P., Olmschenk, S., Younge, K. C., Matsukevich, D. N., Duan, L. M. & Monroe, C. *Entanglement of single-atom quantum bits at a distance.* *Nature* **449**, 68–71 (2007).

[44] Hofmann, J., Krug, M., Ortegel, N., Gérard, L., Weber, M., Rosenfeld, W. & Weinfurter, H. *Heralded Entanglement Between Widely Separated Atoms.* *Science* **337**, 72–75 (2012).

[45] Ritter, S., Nölleke, C., Hahn, C., Reiserer, A., Neuzner, A., Uphoff, M., Mücke, M., Figueroa, E., Bochmann, J. & Rempe, G. *An elementary quantum network of single atoms in optical cavities.* *Nature* **484**, 195–200 (2012).

[46] Pompili, M. *et al.* *Realization of a multinode quantum network of remote solid-state qubits.* *Science* **372**, 259–264 (2021). `https://www.science.org/doi/pdf/10.1126/science.abg1919`.

[47] Krutyanskiy, V. *et al.* *Entanglement of Trapped-Ion Qubits Separated by 230 Meters.* *Phys. Rev. Lett.* **130**, 050803 (2023).

[48] Knaut, C. M. *et al.* *Entanglement of nanophotonic quantum memory nodes in a telecom network*. *Nature* **629**, 573–578 (2024). 2310.01316.

[49] Liu, J.-L. *et al.* *Creation of memory–memory entanglement in a metropolitan quantum network*. *Nature* **629**, 579–585 (2024).

[50] Stolk, A. J. *et al.* *Metropolitan-scale heralded entanglement of solid-state qubits*. *Science Advances* **10**, eadp6442 (2024). https://www.science.org/doi/pdf/10.1126/sciadv.adp6442.

[51] Maunz, P., Olmschenk, S., Hayes, D., Matsukevich, D. N., Duan, L.-M. & Monroe, C. *Heralded Quantum Gate between Remote Quantum Memories*. *Phys. Rev. Lett.* **102**, 250502 (2009).

[52] Daiß, S., Langenfeld, S., Welte, S., Distante, E., Thomas, P., Hartung, L., Morin, O. & Rempe, G. *A quantum-logic gate between distant quantum-network modules*. *Science* **371**, 614–617 (2021). 2103.13095.

[53] Clauser, J. F. & Shimony, A. *Bell's theorem. Experimental tests and implications*. *Reports on Progress in Physics* **41**, 1881–1927 (1978).

[54] Gisin, N., Ribordy, G., Tittel, W. & Zbinden, H. *Quantum cryptography*. *Reviews of Modern Physics* **74**, 145–195 (2002).

[55] Zhong, H. S. *et al.* *Quantum computational advantage using photons*. *Science* **370**, 1460–1463 (2020). 2012.01625.

[56] Walther, P., Pan, J. W., Aspelmeyer, M., Ursin, R., Gasparoni, S. & Zeilinger, A. *De Broglie wavelength of a non-local four-photon state*. *Nature* **429**, 158–161 (2004).

[57] U'Ren, A. B., Silberhorn, C., Banaszek, K. & Walmsley, I. A. *Efficient Conditional Preparation of High-Fidelity Single Photon States for Fiber-Optic Quantum Networks*. *Phys. Rev. Lett.* **93**, 093601 (2004).

[58] Kocher, C. A. & Commins, E. D. *Polarization correlation of photons emitted in an atomic cascade*. *Physical Review Letters* **18**, 575–577 (1967).

[59] Srivathsan, B., Gulati, G. K., Chng, B., Maslennikov, G., Matsukevich, D. & Kurtsiefer, C. *Narrow Band Source of Transform-Limited Photon Pairs via Four-Wave Mixing in a Cold Atomic Ensemble*. *Physical Review Letters* **111**, 123602 (2013). 1302.3706.

[60] Burnham, D. C. & Weinberg, D. L. *Observation of simultaneity in parametric production of optical photon pairs*. *Physical Review Letters* **25**, 84–87 (1970).

[61] Zhang, C., Huang, Y.-F., Liu, B.-H., Li, C.-F. & Guo, G.-C. *Spontaneous Parametric Down-Conversion Sources for Multiphoton Experiments*. *Advanced Quantum Technologies* **4** (2021).

[62] Stevenson, R. M., J., Y. R., Atkinson, P., Cooper, K., Ritchie, D. A. & Shields, A. J. *A semiconductor source of triggered entangled photon pairs*. *Nature* **439**, 179–182 (2006).

[63] Huber, D., Reindl, M., Aberl, J., Rastelli, A. & Trotta, R. *Semiconductor quantum dots as an ideal source of polarization-entangled photon pairs on-demand: a review.* *Journal of Optics* **20** (2018).

[64] Rezai, M., Wrachtrup, J. & Gerhardt, I. *Polarization-entangled photon pairs from a single molecule.* *Optica* **6**, 34–40 (2019).

[65] Chiarella, G., Frank, T., Farrera, P. & Rempe, G. *Two-cavity-mediated photon-pair emission by one atom.* *Optica Quantum* **2**, 346 (2024).

[66] Reiserer, A. & Rempe, G. *Cavity-based quantum networks with single atoms and optical photons.* *Rev. Mod. Phys.* **87**, 1379–1418 (2015).

[67] Stute, A., Casabone, B., Brandstätter, B., Friebe, K., Northup, T. E. & Blatt, R. *Quantum-state transfer from an ion to a photon.* *Nature Photonics* **7**, 219–222 (2013).

[68] Körber, M., Morin, O., Langenfeld, S., Neuzner, A., Ritter, S. & Rempe, G. *Decoherence-protected memory for a single-photon qubit.* *Nature Photon.* **12**, 18–21 (2018).

[69] van Leent, T., Bock, M., Garthoff, R., Redeker, K., Zhang, W., Bauer, T., Rosenfeld, W., Becher, C. & Weinfurter, H. *Long-Distance Distribution of Atom-Photon Entanglement at Telecom Wavelength.* *Phys. Rev. Lett.* **124**, 010510 (2020).

[70] Clark, C. R. *et al.* *High-Fidelity Bell-State Preparation with Optical Qubits.* *Physical Review Letters* **127**, 130505 (2021). `2105.05828`.

[71] Gisin, N. & Thew, R. *Quantum communication.* *Nature Photon.* **1**, 165–171 (2007).

[72] Mosley, P. J., Lundeen, J. S., Smith, B. J., Wasylczyk, P., U'Ren, A. B., Silberhorn, C. & Walmsley, I. A. *Heralded Generation of Ultrafast Single Photons in Pure Quantum States.* *Phys. Rev. Lett.* **100**, 133601 (2008).

[73] Minář, J., De Riedmatten, H. & Sangouard, N. *Quantum repeaters based on heralded qubit amplifiers.* *Physical Review A - Atomic, Molecular, and Optical Physics* **85**, 1–7 (2012). `1111.5185`.

[74] Niemietz, D., Farrera, P., Langenfeld, S. & Rempe, G. *Nondestructive detection of photonic qubits.* *Nature* **591**, 570–574 (2021).

[75] Boone, K., Bourgoin, J. P., Meyer-Scott, E., Heshami, K., Jennewein, T. & Simon, C. *Entanglement over global distances via quantum repeaters with satellite links.* *Physical Review A - Atomic, Molecular, and Optical Physics* **91**, 1–5 (2015). `1410.5384`.

[76] Bouwmeester, D., Pan, J. W., Mattle, K., Eibl, M., Weinfurter, H. & Zeilinger, A. *Experimental quantum teleportation.* *Nature* **390**, 575–579 (1997).

[77] Barz, S., Cronenberg, G., Zeilinger, A. & Walther, P. *Heralded generation of entangled photon pairs.* *Nature Photonics* **4**, 553–556 (2010). `1007.2745`.

[78] Kocsis, S., Xiang, G. Y., Ralph, T. C. & Pryde, G. J. *Heralded noiseless amplification of a photon polarization qubit*. Nature Phys. **9**, 23–28 (2013).

[79] Hamel, D. R., Shalm, L. K., Hübel, H., Miller, A. J., Marsili, F., Verma, V. B., Mirin, R. P., Nam, S. W., Resch, K. J. & Jennewein, T. *Direct generation of three-photon polarization entanglement*. Nature Photonics **8**, 801–807 (2014). 1404.7131.

[80] Maring, N., Farrera, P., Kutluer, K., Mazzera, M., Heinze, G. & De Riedmatten, H. *Photonic quantum state transfer between a cold atomic gas and a crystal*. Nature **551**, 485–488 (2017). 1807.05173.

[81] Cao, H., Hansen, L. M., Giorgino, F., Carosini, L., Zahálka, P., Zilk, F., Loredo, J. C. & Walther, P. *Photonic Source of Heralded Greenberger-Horne-Zeilinger States*. Physical Review Letters **132**, 130604 (2024).

[82] Chen, S. *et al. Heralded Three-Photon Entanglement from a Single-Photon Source on a Photonic Chip*. Physical Review Letters **132**, 1–6 (2024). 2307.02189.

[83] Kalb, N., Reiserer, A., Ritter, S. & Rempe, G. *Heralded Storage of a Photonic Quantum Bit in a Single Atom*. Phys. Rev. Lett. **114**, 220501 (2015).

[84] Brekenfeld, M., Niemietz, D., Christesen, J. D. & Rempe, G. *A quantum network node with crossed optical fibre cavities*. Nature Phys. **16**, 647–651 (2020).

[85] Uphoff, M., Brekenfeld, M., Rempe, G. & Ritter, S. *An integrated quantum repeater at telecom wavelength with single atoms in optical fiber cavities*. Appl. Phys. B **122**, 46 (2016).

[86] Chiarella, G., Frank, T., Zuka, L., Farrera, P. S. & Rempe, G. A source of heralded atom-photon entanglement for quantum networking (2025). 2510.15765.

[87] Kimble, H. J. *Strong Interactions of Single Atoms and Photons in Cavity QED*. Phys. Scr. **T76**, 127–137 (1998).

[88] Loudon, R. *The Quantum Theory of Light* (Oxford University Press, 1983), 2 edn.

[89] Steck, D. A. *Rubidium 87 D Line Data*. *http://steck.us/alkalidata* (2019).

[90] Hunger, D., Steinmetz, T., Colombe, Y., Deutsch, C., Hänsch, T. W. & Reichel, J. *A fiber Fabry-Perot cavity with high finesse*. New J. Phys. **12**, 065038 (2010).

[91] Haroche, S. & Raimond, J.-M. *Exploring the Quantum: Atoms, Cavities, and Photons* (Oxford University Press, 2006).

[92] Nußmann, S. *Kühlen und Positionieren eines Atoms in einem optischen Resonator*. Ph.D. thesis, Technische Universität München, Max-Planck-Institut für Quantenoptik (2005).

[93] Shore, B. W. *The Theory of Coherent Atomic Excitation* (John Wiley & Sons, 1990).

[94] Grynberg, G., Aspect, A. & Gabre, C. *Introduction to Quantum Optics* (Cambridge University Press, 2010).

[95] Hennrich, M., Legero, T., Kuhn, A. & and, G. R. *Counter-intuitive vacuum-stimulated raman scattering*. Journal of Modern Optics **50**, 935–942 (2003). https://www.tandfonline.com/doi/pdf/10.1080/09500340308234542.

[96] Breuer, H.-P. & Petruccione, F. *The Theory of Open Quantum Systems* (Oxford University Press, 2002).

[97] Lambert, N. *et al.* QuTiP 5: The Quantum Toolbox in Python (2024). 2412.04705.

[98] Braunstein, S. L. & Mann, A. *Measurement of the Bell operator and quantum teleportation*. *Phys. Rev. A* **51**, R1727–R1730 (1995).

[99] Duan, L. M., Lukin, M. D., Cirac, J. I. & Zoller, P. *Long-distance quantum communication with atomic ensembles and linear optics*. *Nature* **414**, 413–418 (2001).

[100] Kevin A Fischer, K. G. L., Kai Müller & Vučković, J. *Dynamical modeling of pulsed two-photon interference*. *New Journal of Physics* **18** (2016).

[101] Schöll, E. *et al.* *Crux of Using the Cascaded Emission of a Three-Level Quantum Ladder System to Generate Indistinguishable Photons*. *Phys. Rev. Lett.* **125**, 233605 (2020).

[102] Brekenfeld, M. *Single atoms in crossed optical fiber cavities for a passive, heralded quantum memory*. Ph.D. thesis, Technische Universität München, Max-Planck-Institut für Quantenoptik (2023).

[103] Niemietz, D. *Nondestructive Detection of Photonic Qubits with Single Atoms in Crossed Fiber Cavities*. Ph.D. thesis, Technische Universität München, Max-Planck-Institut für Quantenoptik (2021).

[104] Steinmetz, T., Colombe, Y., Hunger, D., Hänsch, T. W., Balocchi, A., Warburton, R. J. & Reichel, J. *Stable fiber-based Fabry-Pérot cavity*. *Applied Physics Letters* **89**, 111110 (2006). https://pubs.aip.org/aip/apl/article-pdf/doi/10.1063/1.2347892/13143457/111110_1_online.pdf.

[105] Muller, A., Flagg, E. B., Lawall, J. R. & Solomon, G. S. *Ultrahigh-finesse, low-mode-volume Fabry-Perot microcavity*. *Opt. Lett.* **35**, 2293–2295 (2010).

[106] Hunger, D., Deutsch, C., Barbour, R. J., Warburton, R. J. & Reichel, J. *Laser micro-fabrication of concave, low-roughness features in silica*. *AIP Advances* **2**, 012119 (2012).

[107] Colombe, Y., Steinmetz, T., Dubois, G., Linke, F., Hunger, D. & Reichel, J. *Strong atom-field coupling for Bose-Einstein condensates in an optical cavity on a chip*. *Nature* **450**, 272–276 (2007).

[108] Miguel-Sánchez, J., Reinhard, A., Togan, E., Volz, T., Imamoglu, A., Besga, B., Reichel, J. & Estève, J. *Cavity quantum electrodynamics with charge-controlled quantum dots coupled to a fiber Fabry-Perot cavity*. *New J. Phys.* **15**, 045002 (2013).

[109] Albrecht, R., Bommer, A., Deutsch, C., Reichel, J. & Becher, C. *Coupling of a Single Nitrogen-Vacancy Center in Diamond to a Fiber-Based Microcavity*. *Phys. Rev. Lett.* **110**, 243602 (2013).

[110] Ballance, T. G., Meyer, H. M., Kobel, P., Ott, K., Reichel, J. & Köhl, M. *Cavity-induced backaction in Purcell-enhanced photon emission of a single ion in an ultraviolet fiber cavity.* Phys. Rev. A **95**, 033812 (2017).

[111] Salz, M., Herrmann, Y., Nadarajah, A., Stahl, A., Hettrich, M., Stacey, A., Prawer, S., Hunger, D. & Schmidt-Kaler, F. *Cryogenic platform for coupling color centers in diamond membranes to a fiber-based microcavity.* Appl. Phys. B **126**, 131 (2020).

[112] Ong, F. R. *et al. Probing surface charge densities on optical fibers with a trapped ion.* New J. Phys. **22**, 063018 (2020).

[113] Macha, T., Uruñuela, E., Alt, W., Ammenwerth, M., Pandey, D., Pfeifer, H. & Meschede, D. *Nonadiabatic storage of short light pulses in an atom-cavity system.* Phys. Rev. A **101**, 53406 (2020).

[114] Steiner, M., Meyer, H. M., Deutsch, C., Reichel, J. & Köhl, M. *Single Ion Coupled to an Optical Fiber Cavity.* Phys. Rev. Lett. **110**, 043003 (2013).

[115] Grinkemeyer, B., Guardado-Sanchez, E., Dimitrova, I., Shchepanovich, D., Mandopoulou, G. E., Borregaard, J., Vuletić, V. & Lukin, M. D. *Error-detected quantum operations with neutral atoms mediated by an optical cavity.* Science **387**, 1301–1305 (2025). `https://www.science.org/doi/pdf/10.1126/science.adr7075`.

[116] Ulanowski, A., Kuijpers, O., Merkel, B., Holzäpfel, A. & Reiserer, A. Cavity-enhanced spectroscopy of individual nuclear spins in a dense bath (2025). `2504.12957`.

[117] Uphoff, M., Brekenfeld, M., Rempe, G. & Ritter, S. *Frequency splitting of polarization eigenmodes in microscopic Fabry-Perot cavities.* New J. Phys. **17**, 013053 (2015).

[118] Uphoff, M. *Optical microcavities for quantum communication with single atoms.* Ph.D. thesis, Technische Universität München, Max-Planck-Institut für Quantenoptik (2017).

[119] Ott, K., Garcia, S., Kohlhaas, R., Schüppert, K., Rosenbusch, P., Long, R. & Reichel, J. *Millimeter-long fiber Fabry-Perot cavities.* Opt. Express **24**, 9839–9853 (2016).

[120] Hecht, E. *Optics* (Pearson, Boston, 2017).

[121] Kogelnik, H. & Li, T. *Laser Beams and Resonators.* Appl. Opt. **5**, 1550–1567 (1966).

[122] Siegman, A. E. *Lasers* (University Science Books, 1986).

[123] Anderson, M. H., Ensher, J. R., Matthews, M. R., Wieman, C. E. & Cornell, E. A. *Observation of Bose-Einstein Condensation in a Dilute Atomic Vapor.* Science **269**, 198–201 (1995). `https://www.science.org/doi/pdf/10.1126/science.269.5221.198`.

[124] The Nobel Prize in Physics 2001. NobelPrize.org.

[125] Corney, A. *Atomic and laser spectroscopy* (1977).

[126] Raab, E. L., Prentiss, M., Cable, A., Chu, S. & Pritchard, D. E. *Trapping of Neutral Sodium Atoms with Radiation Pressure*. *Phys. Rev. Lett.* **59**, 2631–2634 (1987).

[127] Metcalf, H. J. & van der Straten, P. *Laser Cooling and Trapping* (Springer Verlag New York, 1999).

[128] Anderson, B. P. & Kasevich, M. A. *Loading a vapor-cell magneto-optic trap using light-induced atom desorption*. *Phys. Rev. A* **63**, 023404 (2001).

[129] Atutov, S. N. *et al.* *Fast and efficient loading of a Rb magneto-optical trap using light-induced atomic desorption*. *Phys. Rev. A* **67**, 053401 (2003).

[130] Klempt, C., van Zoest, T., Henninger, T., Topic, O., Rasel, E., Ertmer, W. & Arlt, J. *Ultraviolet light-induced atom desorption for large rubidium and potassium magneto-optical traps*. *Phys. Rev. A* **73**, 013410 (2006).

[131] Aubin, S. *et al.* *Trapping Fermionic 40K and Bosonic 87Rb on a Chip*. *Journal of Low Temperature Physics* **140**, 377–396 (2005).

[132] Zhang, P., Li, G., Zhang, Y.-c., Guo, Y., Wang, J. & Zhang, T. *Light-induced atom desorption for cesium loading of a magneto-optical trap: Analysis and experimental investigations*. *Phys. Rev. A* **80**, 053420 (2009).

[133] Torralbo-Campo, L., Bruce, G. D., Smirne, G. & Cassettari, D. *Light-induced atomic desorption in a compact system for ultracold atoms*. *Scientific Reports* **5** (2015).

[134] Lett, P. D., Watts, R. N., Westbrook, C. I., Phillips, W. D., Gould, P. L. & Metcalf, H. J. *Observation of Atoms Laser Cooled below the Doppler Limit*. *Phys. Rev. Lett.* **61**, 169–172 (1988).

[135] Drever, R. W. P., Hall, J. L., Kowalsi, F. V., Hough, J., Ford, G. M., Munley, A. J. & Hard, H. *Laser phase and frequency stabilization using an optical resonator*. *Appl. Phys. B* **31**, 97–105 (1983).

[136] Frank, T. *Title to be defined*. Ph.D. thesis, Technische Universität München, Max-Planck-Institut für Quantenoptik.

[137] Kien, L., Schneeweiss, P. & Rauschenbeutel, A. *Dynamical polarizability of atoms in arbitrary light fields: General theory and application to cesium*. *The European Physical Journal D* **67** (2012).

[138] Henkel, F., Krug, M., Hofmann, J., Rosenfeld, W., Weber, M. & Weinfurter, H. *Highly Efficient State-Selective Submicrosecond Photoionization Detection of Single Atoms*. *Phys. Rev. Lett.* **105**, 253001 (2010).

[139] Myerson, A. H., Szwer, D. J., Webster, S. C., Allcock, D. T. C., Curtis, M. J., Imreh, G., Sherman, J. A., Stacey, D. N., Steane, A. M. & Lucas, D. M. *High-Fidelity Readout of Trapped-Ion Qubits*. *Phys. Rev. Lett.* **100**, 200502 (2008).

[140] Wang, J., Huang, D.-Y., Zhou, X.-L., Shen, Z.-M., He, S.-J., Huang, Q.-Y., Liu, Y.-J., Li, C.-F. & Guo, G.-C. *Ultrafast high-fidelity state readout of single neutral atom*. (2024). `2412.12584`.

[141] Chiarella, G., Frank, T., Farrera, P. & Rempe, G. Fast, efficient and lossless measurement of atom-photon entanglement. In *Quantum 2.0 Conference and Exhibition*, QW4B.3 (Optica Publishing Group, 2024).

[142] Bochmann, J. *Coherent Dynamics and State Detection of Single Atoms in a Cavity*. Ph.D. thesis, Technische Universität München, Max-Planck-Institut für Quantenoptik (2010).

[143] Allen, L. & Eberly, J. H. *Optical resonance and two-level atoms* (John Wiley and Sons, Inc.,New York, 1975).

[144] Kosugi, N., Matsuo, S., Konno, K. & Hatakenaka, N. *Theory of damped Rabi oscillations*. *Phys. Rev. B* **72**, 172509 (2005).

[145] Bergmann, K., Theuer, H. & Shore, B. W. *Coherent population transfer among quantum states of atoms and molecules*. *Rev. Mod. Phys.* **70**, 1003–1025 (1998).

[146] Grove, T. T., Sanchez-Villicana, V., Duncan, B. C., Maleki, S. & Gould, P. L. *Two-photon two-color diode laser spectroscopy of the Rb 5D5/2 state*. *Physica Scripta* **52**, 271 (1995).

[147] Bonin, K. D. & McIlrath, T. J. *Two-photon electric-dipole selection rules*. *J. Opt. Soc. Am. B* **1**, 52–55 (1984).

[148] Duncan, B. C., Sanchez-Villicana, V., Gould, P. L. & Sadeghpour, H. R. *Measurement of the* $\mathrm{Rb}(5D_{5/2})$ *photoionization cross section using trapped atoms*. *Phys. Rev. A* **63**, 043411 (2001).

[149] Gulati, G. K., Takahashi, H., Podoliak, N., Horak, P. & Keller, M. *Fiber cavities with integrated mode matching optics*. *Scientific Reports* **7** (2007).

[150] Safronova, M. S. & Safronova, U. I. *Critically evaluated theoretical energies, lifetimes, hyperfine constants, and multipole polarizabilities in* $^{87}Rb$. *Phys. Rev. A* **83**, 052508 (2011).

[151] Yoo, H. I. & Eberly, J. H. *Dynamical theory of an atom with two or three levels interacting with quantized cavity fields*. *Physics Reports* **118**, 239–337 (1985).

[152] Villas-Boas, C. J., Tolazzi, K. N., Wang, B., Ianzano, C. & Rempe, G. *Continuous Generation of Quantum Light from a Single Ground-State Atom in an Optical Cavity*. *Physical Review Letters* **124**, 93603 (2020). `1906.11449`.

[153] Tolazzi, K. N., Wang, B., Ianzano, C., Neumeier, J., Villas-Boas, C. J. & Rempe, G. *Continuous quantum light from a dark atom*. *Communications Physics* **4**, 1–5 (2021). `2103.01138`.

[154] Khintchine, A. *Korrelationstheorie der stationären stochastischen Prozesse*. *Mathematische Annalen* **109**, 604–615 (1934).

[155] Wilk, T., Webster, S., Kuhn, A. & Rempe, G. *Single-Atom Single-Photon Quantum Interface*. *Science* **317**, 488–490 (2007).

[156] Rosenfeld, W., Hocke, F., Henkel, F., Krug, M., Volz, J., Weber, M. & Weinfurter, H. *Towards Long-Distance Atom-Photon Entanglement*. *Phys. Rev. Lett.* **101**, 260403 (2008).

[157] Altepeter, J., Jeffrey, E. & Kwiat, P. Photonic State Tomography. In *Advances In Atomic, Molecular, and Optical Physics*, 105–159 (Academic Press, 2005).

[158] Nielsen, M. A. & Chuang, I. L. *Quantum Computation and Quantum Information: 10th Anniversary Edition* (Cambridge University Press, 2010).

[159] Sackett, C. A. *et al.* *Experimental entanglement of four particles*. *Nature* **404**, 256–259 (2000).

[160] Hartung, L., Seubert, M., Welte, S., Distante, E. & Rempe, G. *A quantum-network register assembled with optical tweezers in an optical cavity*. *Science (New York, N.Y.)* **385**, 179–183 (2024). `2407.09109`.

[161] Krutyanskiy, V., Canteri, M., Meraner, M., Krcmarsky, V. & Lanyon, B. P. *Multimode Ion-Photon Entanglement over 101 Kilometers*. *PRX Quantum* **5**, 1 (2024).

[162] Radnaev, A. G., Dudin, Y. O., Zhao, R., Jen, H. H., Jenkins, S. D., Kuzmich, A. & Kennedy, T. A. *A quantum memory with telecom-wavelength conversion*. *Nature Physics* **6**, 894–899 (2010).

[163] Albrecht, B., Farrera, P., Fernandez-Gonzalvo, X., Cristiani, M. & de Riedmatten, H. *A waveguide frequency converter connecting rubidium-based quantum memories to the telecom C-band*. *Nature communications* **5**, 3376 (2014). `1402.2866`.

[164] van Leent, T. *et al.* *Entangling single atoms over 33 km telecom fibre*. *Nature* **607**, 69–73 (2022). `2111.15526`.

[165] Reiserer, A., Kalb, N., Rempe, G. & Ritter, S. *A quantum gate between a flying optical photon and a single trapped atom*. *Nature* **508**, 237–240 (2014).

[166] Sakurai, J. J. & Napolitano, J. *Modern Quantum Mechanics* (Camnbridge University press, 2021).

[167] Wu, Y. & Yang, X. *Strong-Coupling Theory of Periodically Driven Two-Level Systems*. *Phys. Rev. Lett.* **98**, 013601 (2007).

[168] Scully, M. O. & Zubairy, M. S. *Quantum Optics* (Cambridge University Press, 1997).

[169] Arora, B., Safronova, M. S. & Clark, C. W. *Magic wavelengths for the $np - ns$ transitions in alkali-metal atoms*. *Phys. Rev. A* **76**, 052509 (2007).

# Software

I used the following software when writing my thesis:

| **Software** | **Use case** | **Scope** | **Comments** |
|---|---|---|---|
| Overleaf<br>texlive-full:2024.1 | Typesetting | Entire work | |
| PyCharm<br>2024.2 | Data analysis<br>and visualization | Entire work | |
| Adobe Illustrator<br>29.7 | Figures production | Entire work | |
| Grammarly<br>v1.2 | Grammar correction | Entire work | |
| ChatGPT<br>4.5 | Structure improvement | Abstract | |
| | | p. 18, par. 3 | |
| | Clarity enhancement | Introduction | |
| | | p. 13, par. 1 | |
| | | p. 14, par. 3 | Last sentence |
| | | p. 15, par. 1 | |
| | | p. 15, par. 2 | |
| | | p. 17, par. 1 | |
| | | p. 18, par. 3-4 | |
| | | p. 21, par. 1 | Second half of<br>the paragraph |
| | | p. 40, par. 2-4 | |
| | | table 3.5<br>caption | |
| | | p. 42, par. 3 | Last sentence |
| | | p. 44 par. 2 | Largely reworked<br>by the author |
| | | p. 45, par. 4-5 | |
| | | p. 50, par. 3 | |
| | | Fig. 4.3d caption | |
| | | p. 56, par. 1 | 4th sentence |
| | | p. 56, par. 2 | |
| | | p. 58, par. 2 | |
| | | Fig. 5.2 caption | |
| | | p. 70, par. 1 | 3 last sentences |

| Software | Use case | Scope | Comments |
|---|---|---|---|
| ChatGPT 4.5 | Idea generation and brainstorming | Introduction | Entirely reworked by the author |
| | | Sec. 2.4 title | |
| | | p 18, par. 5-6 | |
| | | Fig. 3.11 caption | |
| | | p. 49, par. 1-3 | |
| | | Sec. 4.1 title | |
| | | p. 67 | |
| | | p. 81 | |
| | Literature research | p. 10, par. 1 | added [28] |
| | | p. 84 | BH lemma [166] |
| | | p. 85 | RWA [168] |
| | | p. 87, par. 4 | |
| | Table formatting | Table 3.2 | Latex |
| | | Table D.4 | Latex |
| ChatAI (Gemma) 3 27B Instruct | Clarity enhancement | p. 21, par. 2-4<br>p. 22; p. 23, par. 1 and 4 | |
| | | p. 24 par. 3-4 | |
| | | Fig. 3.2 caption | |
| | Translation | Fig. 3.1a | German to English |
| Perplexity (ChatGPT 4o) | Literature research | p. 9 | |

# List of publications

- Two-cavity-mediated photon-pair emission by one atom
  G. Chiarella, T. Frank, P. Farrera, and G. Rempe
  *Optica Quantum* **2**, 346–350 (2024)

- Fast, efficient and lossless measurement of atom-photon entanglement
  G. Chiarella, T. Frank, P. Farrera, and G. Rempe
  *Quantum 2.0 Conference and Exhibition, Technical Digest Series*, Optica Publishing Group, paper QW4B.3. (2024)

- Source of heralded atom-photon entanglement for quantum networking
  G. Chiarella, T. Frank, L. Zuka, P. Farrera, and G. Rempe
  *Arxiv* 2510.15765, Accepted in *Physical Review Letters* (2025)